Geopolitical Risks and International Business

INTE 60019A · HEC Montréal

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Foreword

Every multinational now says that geopolitics matters to it. Far fewer can say *how much*, *through which channel*, or *what they would do differently if the number changed*. The gap is not one of attention — boards discuss sanctions, export controls, wars and elections every quarter — but of measurement. Geopolitical risk reaches a firm through its market access, its inputs, its finance, its people and its data, and each of those channels can be counted. Most firms do not count them.

This book is the lecture notes of a twelve-session graduate course that tries to close that gap. Its method is simple and a little stubborn: **every big idea is read, then rebuilt from its data.** A landmark article or book — Caldara and Iacoviello's geopolitical risk index, Baldwin's *Great Convergence*, Kuziemko and Werker on the price of a Security Council seat, the Global Sanctions Data Base — is paired with a practice round in which the data are collected at source and its central figure is replicated in Tableau. A famous chart is a set of decisions: which countries, which years, which deflator, which coding rule. You only see the decisions when you make them yourself.

The data will disagree with the paper. Sources are revised, codebooks change, countries are renamed and recoded. Every practice round records at least one place where the replication did not match — and asks what that disagreement means for a firm that would have relied on the original.

Each chapter ends on the same question: *what does this figure not license a multinational to conclude?* That is the habit the course is built to leave behind.

How to use it

If you want to	Go to
Prepare for next week	the session's **Before the session**
Re-read the argument after class	the session's **The lecture**
Know exactly what the practice round asks	the session's **In the practice**
Find where something was covered	the **search box**, top of the sidebar
Find a dataset, its URL and its traps	Data sources
Know how you are graded	Assessment

The search box is the reason this book exists. Slides are good for a room and bad for finding something three weeks later. Search covers every session at once, so a term you half-remember — *weaponised interdependence*, *ideal points*, *Herfindahl* — takes you to the session that introduced it, with its reading and its exercise.

The slides

Each chapter links to the decks it was built from. The slides remain the version delivered in class; the book is the same material as continuous prose, which is the better form for reading afterwards. All decks are also listed on the course site.

The twelve chapters

	Chapter	Question
1	What is geopolitical risk?	Before we can manage geopolitical risk, what exactly are we counting?
2	The international economic order	Who wrote the rules multinationals play by — and what happens to the rules when the writers' share of the world shrinks?
3	Anarchy and the balance of the major powers	In a system with no government above states, who sets the terms of trade — and how would a multinational know that the balance is shifting?
4	Trade, conflict and geoeconomic fragmentation	If states trade more with friends than with rivals, how does a multinational find out which side each of its markets is on — and how fast the lines are hardening?
5	International institutions	When a multinational relies on an international institution to manage its risk, is it relying on a rule — or on the states that buy and sell influence inside it?
6	Private authority, standards and global regulation	Most of the rules that govern a multinational's daily operations were not written by a government. Who wrote them, with whose power — and does trade spread them up or down?
7	Measuring risk: content analysis, coding, big data and composite indices	Every geopolitical risk score a multinational buys or builds is a chain of choices: which texts, which words, which coder, which weights. Can you see the chain — and defend each link?
8	Cyber geopolitics and digital threats	Is cyberspace a new battlefield for great powers — or a new way for old rivals to hurt each other, with multinationals as collateral?
9	The redefinition of economic diplomacy	When states compete for markets, investment and technology, is the multinational the object of diplomacy, its instrument — or a diplomat in its own right?
10	Economic sanctions	Sanctions have become the instrument of first resort. Are they an efficient mechanism for regulating geopolitical risk — or a source of it?
11	Space geopolitics	Satellites now carry the position, timing, connectivity and imagery a multinational runs on. When orbit is crowded, contested and dominated by one operator, who holds the risk — and what can the firm do about it?
12	Team presentations	Can you make a board act on your assessment — and say what would change your mind?

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Chapter 1

What is geopolitical risk?

Session 1 · Concepts, indicators, and the syllabus Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

Before we can manage geopolitical risk, what exactly are we counting?

Big work	Caldara, D., & Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225. doi:10.1257/aer.20191823
Target figure	The monthly GPR index since 1985, with its spikes labelled — and its historical counterpart since 1900
Data you collect	The GPR index, from the authors' page: matteoiacoviello.com/gpr.htm
Tableau skill	Connect to an Excel file · date fields · line chart · reference line · annotations
Course themes	Methodology (text-based indicators); the vocabulary of the whole course

1.1 Before the session

Three things to do before class. Budget **about two hours** — most of it is the reading. Start the Tableau licence request **today**: approval can take several days.

	Before class	Used in class for
1	Install Tableau Desktop and Tableau Prep Builder	every practice round, starting today
2	Read the paper	the lecture discussion and the replication
3	Download the GPR index	Practice round 01

1.1.1 1. Install Tableau — 20 min, plus waiting for the licence

1. Request the free academic licence through the **Tableau for Students** programme (Tableau Academic Programs) with your HEC e-mail address. You receive a product key by e-mail.
2. Install **Tableau Desktop** and, if your licence includes it, **Tableau Prep Builder**; activate with the key.
3. Create a free **Tableau Public** account — the team dashboard is published there. If the academic programme no longer offers you a licence, install the free **Tableau Public** desktop edition instead — see the fallback note in the toolkit.
4. Open Tableau Desktop, connect to *Sample – Superstore*, drag `Order Date` to Columns and `Sales` to Rows. If a line appears, you are ready.

Full instructions and troubleshooting: `TABLEAU-TOOLKIT.md`.

> If the key has not arrived by class, install the 14-day trial of Tableau Desktop. Do not arrive without a working installation: the practice round starts at minute 90.

1.1.2 2. The reading — 60–75 min

Caldara, D., & Iacoviello, M. (2022). Measuring geopolitical risk. ***American Economic Review*****, 112(4), 1194–1225.** doi:10.1257/aer.20191823 · a free working-paper version (Federal Reserve Board, *International Finance Discussion Papers* 1222) is linked from the authors' page.

Read for the **argument**, not for coverage. Annotate four things:

	What to look for
1	Their **definition** of geopolitical risk — copy it word for word
2	How the index is **built**: which newspapers, which words, which normalisation
3	The **strongest single piece of evidence** that the index measures something real
4	One **limitation** you would raise if you were the multinational's chief risk officer

Skim Sections IV–V (the macroeconomic and firm-level effects). Read Sections I–III and Figure 1 closely: **Figure 1 is today's target figure**.

1.1.3 3. The data — 10 min

Download the monthly export from the authors' page and save it, untouched, in `01-what-is-geopolitical-risk/data/raw/`. Details: `../data/README.md`.

Open it once in a spreadsheet and find: the first month of the recent index, the first month of the historical index, and the column for **one country you care about**.

1.1.4 4. Self-check — 15 min

Answer on paper. If you cannot, that is what the lecture is for.

1. In one sentence: what is the difference between **risk** and **uncertainty**?
2. The GPR index is "text-based". Name one other kind of geopolitical risk indicator and one advantage it has over text.
3. GPR is normalised so that 1985–2019 averages 100. What does a reading of 300 in March 2022 mean, in words a manager would understand?
4. Name one geopolitical event that **mattered to a multinational firm** but that you suspect barely moved the GPR index. Why might it not have?

1.1.5 Arrive with

- ☐ Tableau Desktop installed and activated (academic key, trial, or Tableau Public edition)
- ☐ A Tableau Public account
- ☐ The paper read and annotated, Figure 1 understood
- ☐ `data_gpr_export.xls` in `data/raw/`
- ☐ Your self-check answers, on paper

Session 01 overview · The lecture · The practice

1.2 The lecture

1.2.1 1. The course in fifteen minutes

The syllabus is `SYLLABUS.md`. Three things matter today:

- **The big theme.** How does a multinational firm know how much geopolitical risk it faces — and what should it do about it?
- **The rhythm.** Every session: a big article or book read before class; a lecture; a practice round in which you collect its data at source and rebuild its central figure in Tableau.
- **The grade.** 80% individual. Three quizzes (Sessions 04, 07, 10), one Risk Brief (Sessions 02–11), a team dashboard for one multinational (Session 12), a 48-hour take-home.

1.2.2 2. Risk, uncertainty, and three kinds of risk

Risk versus uncertainty. Knight (1921) separated *risk* — outcomes whose probabilities can be estimated — from *uncertainty*, where they cannot. Most geopolitical events sit uncomfortably between the two: wars and coups recur, but each is unlike the last. Everything in this course is an attempt to move a question from the second category towards the first, and to be honest about how far it moves.

	Question it answers	Typical unit	Example
Political risk	Can the government of the host country change the rules on *me*?	firm × host country	expropriation, contract renegotiation, sudden tax, licence withdrawal
Country risk	Will this country pay, convert and transfer?	country	sovereign default, capital controls, currency inconvertibility
Geopolitical risk	Will relations *between* states and political actors move against me?	the international system, or a dyad	war, sanctions, export controls, bloc formation, state-sponsored cyber operations

Kobrin (1979) made the classic distinction between **macro** risk, which hits every foreign firm in a country, and **micro** risk, which hits one sector or one firm. Geopolitical risk adds a third layer: risk that originates *outside* the host country — in Washington, Beijing, Brussels or Moscow — and reaches the firm through its suppliers, customers, banks or networks.

1.2.3 3. Six families of indicators

Family	What it counts	Examples you will use	Strength	Blind spot
Event-based	Things that happened, coded from reports	UCDP, cyber-incident data (S08), sanctions episodes (S10)	Concrete, datable	Only what was reported and coded
Text-based	How much the press or firms talk about a risk	**GPR (today)**, EPU, firm-level political risk (S07)	Timely, long, high frequency	Measures attention, not exposure
Expert-based	Scores assigned by analysts or surveys	World Governance Indicators, commercial country-risk ratings	Covers latent concepts	Opaque, sticky, perception-driven
Structural	Slow-moving features of the system	Material capabilities (S03), trade dependence (S04, S11)	Explains *why*	Rarely tells you *when*
Market-based	Prices that embed risk	sovereign CDS spreads, implied volatility	Forward-looking, daily	Mixes risk with risk appetite
Firm-level	Risk as perceived by firms themselves	earnings-call text (S07), disclosed exposures	Speaks to the decision	Self-reported, listed firms only

A good risk assessment — the team project — **triangulates** across at least three families. The quarrel between them is informative: when the text says calm and the market says danger, someone knows something.

1.2.4 4. The GPR index: how it is built

Caldara and Iacoviello (2022) define geopolitical risk as *the threat, realisation and escalation of adverse events associated with wars, terrorism and any tensions among states and political actors that affect the peaceful course of international relations.*

The construction, in four steps:

1. **Corpus.** The electronic archives of ten leading newspapers (US, UK and Canadian) for the recent index from 1985; three US newspapers for the historical index from 1900.
2. **Dictionary.** Articles are counted if they contain words from eight categories — war threats, peace threats, military build-ups, nuclear threats, terror threats, beginning of war, escalation of war, terror acts — usually in combination with words that pin the article to geopolitics.
3. **Share.** For each month, the number of GPR articles divided by the total number of articles. Dividing by the total removes the effect of newspapers growing or shrinking.
4. **Normalisation.** The share is rescaled so that the 1985–2019 average equals 100.

Threats versus acts. The first five categories form **GPR Threats**, the last three **GPR Acts**. The distinction matters to a firm: the paper finds that threats, not only realised acts, depress investment.

Validation. The authors audit samples of articles by hand, compare the index with historical narrative, and show it spikes at the events it should — the World Wars, Korea, the Cuban missile crisis, the Gulf War, 9/11, the 2003 invasion of Iraq, the 2022 Russian invasion of Ukraine.

Country-specific indices. For several dozen countries, the same counts are restricted to articles that also mention the country — a measure of how much the *anglophone press* associates that country with geopolitical tension.

1.2.5 5. Why a firm should care

The real-options channel. When an investment is costly to reverse and the future is uncertain, waiting has value (Bernanke, 1983; Bloom, 2009). A GPR spike raises the value of waiting: firms postpone hiring, capital expenditure and entry. Caldara and Iacoviello find that higher GPR predicts lower investment and employment, and that the effect is stronger in industries more exposed to geopolitical risk.

Five channels from the system to the P&L, which return all term:

Channel	Example	Session
Market access	tariffs, bloc-based trade diversion	04
Inputs	a critical service from one satellite operator	11
Finance and payments	sanctions on banks, asset freezes	10
Assets and people	conflict, expropriation, evacuation	03, 05
Data and operations	state-sponsored cyber operations	08

1.2.6 6. What GPR does not tell you

- It measures **attention in the anglophone press**, not the risk faced by a subsidiary in Abidjan or a supplier in Shenzhen.
- It is **global**: one number for the world. The country indices help, but only for countries the newspapers write about.
- **Attention fades** even while risk persists: a frozen conflict may be a constant risk and a falling index.
- It is **not a forecast**. It rises with an event; it does not tell you the next one.

That list is the template for "the catch" — the last step of every practice round.

1.2.7 7. The Tableau skill of the day

Connect to `data_gpr_export.xls` → check that `month` is a **date** → drag `month` to Columns as a **continuous month** → `GPR` to Rows → add a **constant reference line** at 100 → **annotate** the spikes. The step-by-step version is the practice brief: `../02-practice/README.md`.

1.2.8 References

Bernanke, B. S. (1983). Irreversibility, uncertainty, and cyclical investment. *Quarterly Journal of Economics*, 98(1), 85–106.

Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica*, 77(3), 623–685.

Caldara, D., & Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225.

Knight, F. H. (1921). *Risk, uncertainty and profit*. Houghton Mifflin.

Kobrin, S. J. (1979). Political risk: A review and reconsideration. *Journal of International Business Studies*, 10(1), 67–80.

1.3 In the practice

> **The target.** Caldara & Iacoviello (2022), Figure 1: the monthly GPR index since 1985, its spikes labelled with the events behind them. Then its historical counterpart since 1900.

Teams are not yet formed: today you work **in pairs**. Everyone builds their own workbook.

1.3.1 1 · Collect — 15 min

If you did the pre-session, `data_gpr_export.xls` is already in `data/raw/`. If not, download it now from matteoiacoviello.com/gpr.htm. Details: `../data/README.md`.

Start your **provenance log** — copy `templates/provenance-log.csv` into `data/` and fill one line:

file	source	url	version / release	accessed	licence	notes
data_gpr_export.xls	Caldara & Iacoviello	matteoiacoviello.com/gpr.htm	*the "last updated" date on the page*	*today*	free, cite AER 2022	*anything odd*

1.3.2 2 · Prepare — 15 min

1. Tableau Desktop → **Connect** → **Microsoft Excel** → `data_gpr_export.xls`.
2. Drag the data sheet (`Sheet1`) to the canvas. Look at the grid at the bottom. The last two columns, `var_name` and `var_label`, are the authors' notes, not data.
3. Check the type icon of `month`. It must be a **date** (calendar). If it is text (`Abc`) or a date-time, click the icon and change it to *Date*.
4. Check `GPR`, `GPRT`, `GPRA`, `GPRH` are **numbers** (`#`). Empty cells in early years of the recent index are expected: it starts in 1985.
5. Write in your log what Tableau got wrong on its own, if anything.

1.3.3 3 · Replicate — 30 min

1.3.3.1 Sheet 1 · "GPR since 1985"

1. `month` → **Columns**. Right-click the pill → choose the *continuous* **Month** (green pill, the second "Month" in the menu — the one showing *May 2015*, not *May*).
2. `GPR` → **Rows**. Keep the aggregation `SUM` — there is one row per month, so the sum is the value.
3. Filter `month` to January 1985 onward.
4. **Analytics** pane → drag **Reference Line** onto the view → *Table* → **Constant 100**, label "1985–2019 average = 100".
5. Find the five highest peaks. Right-click each → **Annotate** → **Point** and write the event: Gulf War (1991), 9/11 (2001), Iraq invasion (2003), Paris attacks (2015), Russia invades Ukraine

   (2022) — but **check the months yourself**; do not trust this list.

6. Title the sheet the way the paper would, and put the source in a caption: *Source: Caldara & Iacoviello (2022), matteoiacoviello.com, accessed ...*

1.3.3.2 Sheet 2 · "GPR since 1900"

Duplicate Sheet 1, replace `GPR` with `GPRH`, remove the 1985 filter. Relabel the reference line: `GPRH` is normalised to its **1900–2019** average = 100, not 1985–2019. Annotate the two World Wars, Korea and the Cuban missile crisis.

1.3.3.3 Sheet 3 · "Threats and acts"

Put `GPRT` and `GPRA` on the same axis: drag `GPRA` onto the `GPRT` axis until the double-ruler icon appears (**Measure Values**). Colour by **Measure Names**.

1.3.4 4 · Compare and break — 20 min

Put your Sheet 1 beside Figure 1 of the paper.

1. **Where do they differ?** Your series runs further: the index is updated monthly, and the authors also revise it. Are past values exactly the paper's? Write down one difference.
2. **Threats versus acts.** Find one episode where threats rose long before acts did. What would a firm watching only *acts* have missed?
3. **Break one decision.** Replace the monthly view by a **yearly average** (change the pill to continuous *Year*, aggregation `AVG`). Which spikes disappear? What does that tell you about the frequency at which a risk dashboard should run?
4. **Extend.** Add a `GPRC_` country index for a country of your choice (Sheet 4). It is a *percent of articles*, not an index to 100: use a dual axis. Is the country index correlated with the global one? When does it diverge?

1.3.5 5 · The catch — 10 min

Build **one slide** (a Tableau dashboard exported as an image is fine) with three panels:

Original	Ours	The catch
Figure 1 of the paper	Your Sheet 1	One sentence: what this figure does **not** license a multinational to conclude

A pair drawn at random presents in two minutes.

1.3.6 Hand in

Upload to ZoneCours before leaving: your `.twbx` (packaged workbook, *File → Export Packaged*) and your provenance log. Today's round is **formative**: it is commented on, not marked, and it counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

1.3.7 What loses credit, from Session 02 onwards

- A chart with no source caption
- A provenance log with no version or access date
- "The data matched exactly" — it rarely does; look harder
- A catch that is a limitation of the *paper* rather than of what the *firm* may conclude

Chapter 2

The international economic order

Session 2 · The system in which multinational firms operate Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **Who wrote the rules multinationals play by — and what happens to the rules when the writers' share of the world shrinks?**

Big work	Baldwin, R. (2016). *The Great Convergence: Information Technology and the New Globalization*. Belknap Press of Harvard University Press.
Target figure	The **G7 share of world GDP**, 1970–today — the "great convergence" that begins around 1990
Data you collect	World Bank, World Development Indicators: GDP in current US$ (`NY.GDP.MKTP.CD`), GDP at PPP (`NY.GDP.MKTP.PP.CD`), trade in % of GDP (`NE.TRD.GNFS.ZS`) — DataBank WDI
Tableau skill	**Pivot** wide years to long · filter on ISO codes, not names · area chart · **percent of total**
Course themes	The system of international economic order in which multinational firms operate

2.1 Before the session

Four things to do before class. Budget **about two hours**.

	Before class	Used in class for
1	Read Baldwin	the lecture discussion and the replication
2	Collect the WDI extract	Practice round 02
3	Shortlist three multinationals	team formation, first ten minutes of the practice
4	Self-check	the lecture

2.1.1 1. The reading — 60–75 min

Baldwin, R. (2016). *The Great Convergence: Information Technology and the New Globalization*. Belknap Press of Harvard University Press.

Read the **Introduction** and **Part I** (the long history of globalisation in four phases) and **Part II** (the new globalisation and the "three cascading constraints"). Skim Part IV on policy. The book is on reserve at the HEC library; the e-book is available through the library catalogue.

Annotate four things:

	What to look for
1	Baldwin's **three costs** of moving goods, ideas and people — and which one fell in each unbundling
2	Why the **first unbundling** produced divergence and the **second** convergence
3	The **figure** showing the G7 share of world income — what is on the axis, what is measured
4	One claim a **G7 multinational's strategy team** would contest, and why

The target figure is the G7 share of world GDP. Note its units (current US dollars? constant? PPP?), its start and end years, and its source. You will need all four.

2.1.2 2. The data — 15 min

Build a custom extract in World Bank DataBank: the G7, the "Industrialising Six", World; three series; 1970 to the latest year. Step-by-step: `../data/README.md`.

Open it once and find: the year the G7's GDP first appears for every member, how missing values are written, and the last year with a value for *World*.

2.1.3 3. Your team's firm — 15 min

Teams of three or four form in the first ten minutes of the practice round. Arrive with a **shortlist of three multinationals** you would want to follow all term. A good candidate:

- publishes an annual report or 10-K with **revenue by country or region**;
- has operations, suppliers or customers in **at least three of** the US, the EU, China, Russia, the Gulf, India;
- is exposed to at least one of the course's channels: sanctions, export controls, cyber, standards.

No two teams may adopt the same firm. Registration on ZoneCours is first come, first served.

2.1.4 4. Self-check — 15 min

Answer on paper.

1. What did the Bretton Woods conference of 1944 create, and what did the GATT add in 1947?
2. In one sentence each: what was unbundled in Baldwin's first unbundling, and what in the second?
3. GDP shares can be measured at market exchange rates or at purchasing-power parity. Which will show the G7 with a **larger** share, and why?
4. Rodrik's trilemma says you can have at most two of three things. Name the three.

2.1.5 Arrive with

- ☐ Baldwin read (Introduction, Parts I–II), the target figure understood
- ☐ Your WDI extract in `data/raw/`, logged
- ☐ A shortlist of three multinationals
- ☐ Your self-check answers, on paper
- ☐ If you are giving a Risk Brief today: your `.twbx` on ZoneCours before class

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Session 02 overview · The lecture · The practice

2.2 The lecture

The session opens with four Risk Briefs (20 minutes); the lecture takes the next seventy.

2.2.1 1. What an "economic order" is

An **international order** is a set of rules, institutions and expectations that governs relations between states, backed by some distribution of power. An **economic order** is the part of it that governs trade, money and investment — the part a multinational lives inside without noticing, until it changes.

Three questions describe any economic order:

Question	What to look at
Who wrote the rules?	Which states designed the institutions, and in whose interest
What do the rules say?	Openness to trade, to capital, to investment; how disputes are settled
What holds them in place?	Hegemonic power, shared interest, institutions with their own momentum

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2.2.2 2. The post-1945 order, in four layers

Bretton Woods (1944). Forty-four allied states create the **International Monetary Fund** (stable exchange rates, balance-of-payments support) and the **International Bank for Reconstruction and Development**, the core of the World Bank. Fixed but adjustable exchange rates anchored on the dollar, the dollar on gold.

The GATT (1947). A provisional agreement — the planned International Trade Organization never came into being — that cuts tariffs in successive rounds on the principle of **most-favoured-nation** treatment and **national treatment**.

Embedded liberalism. Ruggie (1982) named the bargain: openness abroad, made politically sustainable by social protection and macroeconomic autonomy at home. Capital controls were normal; free trade was managed.

The neoliberal turn and the WTO (1995). The dollar's link to gold ends in 1971; capital controls are dismantled through the 1980s and 1990s; the Uruguay Round creates the **World Trade Organization**, with a binding dispute-settlement mechanism and rules reaching into services, intellectual property (TRIPS) and investment measures (TRIMs). China joins in 2001.

The liberal order as a political project. Ikenberry (2011, *Liberal Leviathan*) argues the order was American-led but **easy to join and hard to overturn**: open, rule-based, institutional enough that rising states gain more by joining than by challenging it. Session 03 sets the realist reply beside it.

2.2.3 3. Baldwin's three unbundlings

Baldwin organises globalisation around **three costs**: of moving **goods**, moving **ideas**, and moving **people**.

Phase	What became cheap	What it unbundled	Consequence
First unbundling (steam, from the 19th century)	moving **goods**	production from consumption: you could make things far from where they were bought	Industry **clustered** in the North, where ideas stayed cheap to move locally. **Great divergence** between North and South
Second unbundling (ICT, from about 1990)	moving **ideas**	the factory itself: stages of production could be offshored while coordinated from headquarters	Northern know-how moved South **inside the multinational's value chain**. **Great convergence**
Third unbundling (prospective)	moving **people** — virtually: telepresence, telerobotics	the worker from the workplace	Services trade in tasks; competition reaches occupations, not only sectors

The key point for this course. In the second unbundling, *the multinational is the vehicle of convergence*. Know-how is transferred through supply chains the firm controls. That is why geopolitics now reaches firms so directly: the supply chain is where technology, jobs and security dependence travel together.

Baldwin's evidence is a picture: the G7's share of world income, rising for over a century, then falling sharply after about 1990 — while a handful of developing economies (his "Industrialising Six": China, Korea, India, Poland, Indonesia, Thailand) gain almost all of it. That is today's target figure.

2.2.4 4. From convergence to backlash

Rodrik's trilemma (Rodrik, 2011, *The Globalization Paradox*): a country can have at most two of **deep economic integration**, **national sovereignty** and **democratic politics**.

Choose	Give up	Looks like
integration + sovereignty	democracy	the "golden straitjacket": policies dictated by markets
integration + democracy	sovereignty	global federalism: rules made above the state
sovereignty + democracy	deep integration	Bretton Woods compromise: shallow, managed openness

The trilemma reads the 2010s and 2020s: Brexit, tariff wars, industrial policy, export controls, and "de-risking" are choices of sovereignty and democracy over integration.

Slowbalisation. World trade in percent of GDP rose steeply from the 1990s to 2008 and has broadly flattened since. Whether that is a retreat from globalisation or a shift in its form — from goods to services and data — is contested. The *extend* step of the practice round lets you see the flattening and judge it.

When the writers' share shrinks. If the G7 produced around two-thirds of world GDP when it wrote the rules, and much less today, two questions follow — and they open Sessions 03 and 04: will the rules be rewritten by the new powers (power transition), and will the world split into blocs with rules of their own (fragmentation)?

2.2.5 5. What it means for the multinational

Implication	Example
Your market is no longer where the rules are written.	A European firm earns a growing share of revenue in markets whose governments did not design the WTO and do not feel bound by its spirit
Your supply chain is a geopolitical object.	The know-how the second unbundling moved is what export controls now try to stop
Multilateral protection is weaker.	The WTO's Appellate Body has been unable to hear appeals since December 2019; disputes are settled by power more often than by panels
Measurement choices are strategy choices.	At market exchange rates, the G7 still looks large; at PPP, it looks much smaller. Which one a board sees shapes where it builds

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2.2.6 6. What the figure does not license

- A falling G7 share is **relative**: G7 economies kept growing. It does not mean the G7 market shrank.
- At **current US dollars**, the share moves with the dollar: part of any year-to-year change is the exchange rate, not production.
- **Convergence is concentrated.** Most of the gain goes to a handful of countries — above all China. "The South" did not converge.
- It shows **income**, not **rule-making power**. Whether shrinking share means shrinking influence is the question of Session 03, not a reading of this chart.

2.2.7 7. The Tableau skill of the day

DataBank gives you **years as columns**. Tableau wants **one row per country-year**. Select the year columns in the data source pane → right-click → **Pivot**. Rename `Pivot Field Names` to `Year` and `Pivot Field Values` to `Value`. Change `Year` to a number or date and `Value` to a number — DataBank's `..` become nulls. Then build the G7 share with a calculated field, and check it with a **percent of total** table calculation. Full click-path: `../02-practice/README.md`.

2.2.8 References

Baldwin, R. (2016). *The great convergence: Information technology and the new globalization*. Belknap Press of Harvard University Press.

Ikenberry, G. J. (2011). *Liberal Leviathan: The origins, crisis, and transformation of the American world order*. Princeton University Press.

Rodrik, D. (2011). *The globalization paradox: Democracy and the future of the world economy*. W. W. Norton.

Ruggie, J. G. (1982). International regimes, transactions, and change: Embedded liberalism in the postwar economic order. *International Organization*, 36(2), 379–415.

2.3 In the practice

> **The target.** Baldwin (2016): the **G7 share of world GDP**, rising for a century and falling sharply after about 1990. You rebuild it from 1970 with World Bank data, then ask what changes when the dollars change.

2.3.1 0 · Form your team — 10 min

Teams of three or four, for the rest of the term. Agree on **one multinational** from your shortlists and register it on ZoneCours (first come, first served). Write the team name and firm at the top of your provenance log.

2.3.2 1 · Collect — 10 min

If you did the pre-session, your DataBank extract is in `data/raw/`. If not, build it now — `../data/README.md`. One extract per team is enough; everyone builds their own workbook from it.

Copy `templates/provenance-log.csv` into `data/` and fill one line per file. For WDI, the **version** is the "Last Updated" date DataBank shows for the database.

2.3.3 2 · Prepare — 20 min

1. **Connect → Microsoft Excel →** your extract. Drag the data sheet to the canvas.
2. If there are notes rows at the bottom ("Data from database…", "Last Updated…"), tick **Use Data Interpreter**, or filter them out later. Write which you did.
3. **Pivot the years.** Click the first year column, shift-click the last → right-click → **Pivot**. Two new columns appear.
4. Rename `Pivot Field Names` → `Year`, `Pivot Field Values` → `Value`.
5. `Year` is text like `1990 [YR1990]`. Create a calculated field from the data source pane: `Year (n)` = `INT(LEFT([Year], 4))`. (If your extract has plain `1990`, just change the type to *Number (whole)*.)
6. `Value` is text — DataBank writes missing data as `..`. Change its type to **Number (decimal)**; the dots become nulls. Check the null count: is it plausible?
7. Look at `Country Code` and `Series Code` (in the Route B CSV the second is `Indicator Code`, the years are plain `1990`, and missing values are blank). **From now on you filter on codes, never on names.**

2.3.4 3 · Replicate — 25 min

2.3.4.1 Sheet 1 · "The G7 share of world GDP"

1. Filters: `Series Code` = `NY.GDP.MKTP.CD` only.
2. Create two calculated fields:

```
G7 GDP     = IF [Country Code] IN ("CAN","FRA","DEU","ITA","JPN","GBR","USA") THEN [Value] END
World GDP  = IF [Country Code] = "WLD" THEN [Value] END
```

3. And the share:

```
G7 share   = SUM([G7 GDP]) / SUM([World GDP])
```

   Format it as a percentage (right-click → *Default Properties* → *Number Format*).

4. `Year (n)` → Columns as a **continuous dimension** (green); `G7 share` → Rows.
5. Annotate the peak year and the latest year with their values. Title the sheet with the finding, not the variable. Caption: *World Bank, WDI, NY.GDP.MKTP.CD, last updated …, accessed ….*

2.3.4.2 Sheet 2 · "Who gained?" — a stacked area of world shares

1. Two more calculated fields:

```
I6 GDP              = IF [Country Code] IN ("CHN","KOR","IND","POL","IDN","THA") THEN [Value] END
Rest of world GDP   = SUM([World GDP]) - SUM([G7 GDP]) - SUM([I6 GDP])
```

2. `Year (n)` → Columns. Drag `G7 GDP`, `I6 GDP` and `Rest of world GDP` onto Rows as **Measure Values** (drop the second and third onto the first axis).
3. **Marks → Area**. `Measure Names` → Colour.
4. Right-click `Measure Values` → **Quick Table Calculation → Percent of Total**. Then **Edit Table Calculation → Compute Using → Measure Names**, so that the three shares sum to 100% **within each year** — not across years. Check one year by hand against Sheet 1.

2.3.5 4 · Compare and break — 15 min

1. **Compare.** Put Sheet 1 beside Baldwin's figure. Where does the curve peak in yours? How far has it fallen by Baldwin's last year, and by the latest year? Write down one difference and its most likely cause (dollars, years, data revision).
2. **Missing data.** Which I6 members have no GDP value in the 1970s or 1980s? What does that do to the I6 and "rest of world" areas in those years? This is your "the data disagreed" entry.
3. **Break — change the dollars.** Duplicate Sheet 1 and switch the filter to `NY.GDP.MKTP.PP.CD`. The PPP series starts in 1990. How much smaller is the G7 at PPP? Which board decision would you use each line for?
4. **Extend — slowbalisation.** New sheet: `NE.TRD.GNFS.ZS` for `WLD` — world trade in % of GDP (an index, so check the aggregation: there is one value per year; `AVG` or `SUM` give the same only because you filtered to one country — say so). Annotate 2008. Is the line after 2008 a retreat?

2.3.6 5 · The catch — 10 min

One slide (a dashboard exported as an image):

Original	Ours	The catch
Baldwin's G7 share figure	Your Sheet 1, plus the PPP line	One sentence: what this figure does **not** license your team's multinational to conclude

A member drawn at random presents in two minutes.

2.3.7 Hand in

On ZoneCours before leaving, one per team: the `.twbx` (*File → Export Packaged Workbook*), the provenance log, and the slide. Formative: commented, not marked; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the book.

2.3.8 What loses credit

- Filtering on country *names* ("Korea, Rep." will bite you in Session 04)
- A percent of total that sums to 100% across years instead of within each year
- A share chart in current dollars with no mention of exchange rates
- A catch about Baldwin's book rather than about what your firm may conclude

Chapter 3

Anarchy and the balance of the major powers

Session 3 · The geostrategic balance that dictates international trade relations Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **In a system with no government above states, who sets the terms of trade — and how would a multinational know that the balance is shifting?**

Big works	Mearsheimer, J. J. (2001). *The Tragedy of Great Power Politics*. W. W. Norton. · Allison, G. (2017). *Destined for War: Can America and China Escape Thucydides's Trap?* Houghton Mifflin Harcourt.
Target figure	The **shares of world material capabilities** (CINC) of the major powers, 1816–2022, and the US–China crossover
Data you collect	Correlates of War, National Material Capabilities v7.0 — correlatesofwar.org · SIPRI Military Expenditure Database — sipri.org/databases/milex · WDI GDP (from Session 02)
Tableau skill	**Calculated fields** · ratios · **parameters** to choose a country · dual axis
Course themes	Inter-state anarchy and the geostrategic balance of the major powers that dictate international trade relations

3.1 Before the session

Three things to do before class. Budget **about two hours**.

	Before class	Used in class for
1	Read Mearsheimer and Allison	the lecture discussion and the replication
2	Collect NMC v7.0 and SIPRI, read the NMC codebook	Practice round 03
3	Self-check	the lecture

3.1.1 1. The reading — 60–75 min

Mearsheimer, J. J. (2001). *The Tragedy of Great Power Politics*. W. W. Norton. Chapter 1 (*Introduction*) and Chapter 2 (*Anarchy and the Struggle for Power*). The five bedrock assumptions are in Chapter 2: copy them.

Allison, G. (2017). *Destined for War: Can America and China Escape Thucydides's Trap?* Houghton Mifflin Harcourt. The Introduction and Chapter 2 (*Athens vs. Sparta*), and the appendix listing the sixteen cases.

Annotate four things:

	What to look for
1	Mearsheimer's **five assumptions** — and which one does the most work
2	How each author **measures power**: what counts, what is left out
3	Allison's **sixteen cases** — how many ended in war, and how a case was chosen
4	One claim a **multinational with operations in both the US and China** should take seriously, and one it should doubt

The target figure is the distribution of capabilities among the major powers since 1816 — the picture that underlies both books' argument that the balance is shifting.

3.1.2 2. The data — 20 min

Download and unzip **NMC v7.0**, and download the latest **SIPRI** military expenditure workbook. Details and terminal commands: `../data/README.md`.

Read the NMC codebook before opening the data. Find: the six components and their units; how CINC is computed; the last year covered. Then find the codes for China, Russia/USSR, Germany and the UK in the COW state list (a separate file — see the data README).

3.1.3 3. Self-check — 15 min

Answer on paper.

1. "Anarchy" in realist theory means what? What does it *not* mean?
2. Waltz and Mearsheimer are both realists. On what question about how much power states want do they disagree?
3. CINC averages six shares. Name the three dimensions they cover. Which dimension do you expect to flatter a large, populous, industrialising country?
4. Allison counts how many cases of a rising power challenging a ruling one, and how many ended in war? Write the numbers down — then write one reason to doubt the ratio.

3.1.4 Arrive with

- ☐ Mearsheimer ch. 1–2 and Allison's introduction and ch. 2 read, the five assumptions copied
- ☐ `NMCv7.zip` unzipped and the SIPRI workbook in `data/raw/`, both logged
- ☐ The NMC codebook read
- ☐ Your self-check answers, on paper

Session 03 overview · The lecture · The practice

3.2 The lecture

The session opens with four Risk Briefs (20 minutes); the lecture takes the next seventy.

3.2.1 1. Anarchy: the starting point

In international relations, **anarchy** means the absence of a government above states — no world authority that can make rules binding, enforce contracts or guarantee survival. It does **not** mean chaos or constant war: most of the time, most states are at peace and trade under rules.

What follows from anarchy, in the realist tradition:

- **Self-help.** Each state must ultimately ensure its own survival.
- **Uncertainty about intentions.** A state cannot know what another will do with its power tomorrow.
- **Relative gains.** Because today's partner may be tomorrow's adversary, states care not only what they gain from cooperation but how much *more* the other gains (Grieco, 1988).

Why a firm should care. Relative-gains thinking is why trade becomes a security question: a deal that makes both richer but the rival richer *faster* may be refused, restricted or reversed. It is the logic behind export controls on advanced semiconductors.

3.2.2 2. Two realisms

Defensive realism — Waltz (1979), *Theory of International Politics*. The structure of the system — anarchy plus the **distribution of capabilities** — shapes behaviour. States seek an *appropriate* amount of power: enough for security. Too much provokes balancing coalitions, so expansion is usually self-defeating. Waltz judged **bipolar** systems more stable than multipolar ones: fewer miscalculations.

The security dilemma (Jervis, 1978). What one state does to make itself secure — arming, alliances — makes others less secure, and they respond. Both end up worse off without anyone intending aggression.

Offensive realism — Mearsheimer (2001). Five bedrock assumptions:

1. The international system is **anarchic**.
2. Great powers possess some **offensive military capability**.
3. States can never be certain of others' **intentions**.
4. **Survival** is the primary goal of great powers.
5. Great powers are **rational actors**.

From these, Mearsheimer derives that great powers **maximise relative power**; the ideal is to be the **regional hegemon** and to prevent any other region from having one. His prediction for China: it will try to dominate Asia as the United States dominates the Western Hemisphere, and the United States will try to stop it. *"China cannot rise peacefully."*

	Defensive (Waltz)	Offensive (Mearsheimer)
How much power do states want?	enough for security	as much as possible, relative to others
What restrains them?	balancing by others	only the limits of their capabilities
Expected US–China outcome	managed competition	intense security competition

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3.2.3 3. Power transition and the Thucydides's Trap

Power transition theory (Organski, 1958; Organski & Kugler, 1980) reverses Waltz's emphasis: hierarchy, not balance, keeps the peace. A **dominant power** sets the international order. War is most likely when a **dissatisfied challenger** approaches **parity** with the dominant power — a threshold often operationalised as the challenger reaching around 80% of the leader's capabilities.

Allison (2017). *"It was the rise of Athens and the fear that this instilled in Sparta that made war inevitable"* (Thucydides). Allison and the Harvard Thucydides's Trap project identify **sixteen** cases in five hundred years in which a rising power challenged a ruling one; **twelve** ended in war.

The critics.

- **Case selection.** The cases are chosen on the outcome of interest; peaceful transitions (the UK to the US) may be under-counted; the definition of "rising" and "ruling" is elastic.
- **Nuclear weapons** change the expected cost of great-power war in a way no pre-1945 case can capture.
- **Interdependence.** The US and China are bound by trade and finance far more tightly than Athens and Sparta, or Britain and Germany — though Britain and Germany were also major trading partners in 1914 (Session 04 takes this up).
- **Measuring "rise".** Beckley (2018) argues gross indicators — GDP, population, CINC — overstate China's power because they ignore the costs of governing, feeding and protecting a population.

3.2.4 4. Measuring power: the CINC

The Correlates of War **Composite Index of National Capability** (Singer, Bremer & Stuckey, 1972):

1. Take **six indicators** for every state in the system each year: total population, urban population, iron and steel production, primary energy consumption, military personnel, military expenditure.
2. For each indicator, compute the state's **share of the system total** that year.
3. **Average** the six shares. CINC is a number between 0 and 1; the system sums to about 1 (not exactly: a state missing a component is averaged over the ones it has).

What it rewards. Size and heavy industry. Two of six components are population and two are smokestack industry. CINC therefore shows China passing the United States **much earlier** than military spending or GDP at market exchange rates do. One reality, three measures, three dates: the *break* step of the practice round is to see it.

What it misses. Technology, quality of forces, alliances, financial power, control of networks — the dollar, SWIFT, the cloud — which Session 08 and Session 11 show to be the instruments of coercion that reach firms most directly.

3.2.5 5. What it means for the multinational

The balance of power sets the rules of trade	Who can impose sanctions, export controls and market-access conditions depends on who holds capabilities others need
A transition changes which rules you follow	A firm in both the US and China increasingly faces two compliance regimes — and penalties for following one in the other's jurisdiction
Relative gains turn commerce into security	Products that make a rival stronger — chips, AI, batteries, aerospace — move from the trade ministry to the security council

The date of the transition depends on the measure	Board slides that say "China has overtaken the US" should say on **what**
Security dilemmas escalate without intent	A defensive move by one side (an export control, a military exercise) is read as offensive and triggers a response that reaches your supply chain

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3.2.6 6. What the figure does not license

- CINC is a measure of **material capabilities**, not of influence, intentions or likelihood of war. A crossover on the chart is not a prediction of conflict.
- It **overweights size**: population and heavy industry. It cannot see semiconductors, alliances or finance.
- It ends in **2022**: the most recent shifts are not in it.
- It describes **states**, not markets: a firm's exposure depends on where its revenue and suppliers sit, not on the system's distribution of power.

3.2.7 7. The Tableau skill of the day

Calculated fields to build a ratio (challenger CINC ÷ US CINC), a **parameter** so the viewer can choose the challenger, a **reference line** at parity and at the 80% threshold, and a **dual axis** to set CINC against military spending. Click-path: `../02-practice/README.md`.

3.2.8 References

Allison, G. (2017). *Destined for war: Can America and China escape Thucydides's trap?* Houghton Mifflin Harcourt.

Beckley, M. (2018). The power of nations: Measuring what matters. *International Security*, 43(2), 7–44.

Grieco, J. M. (1988). Anarchy and the limits of cooperation: A realist critique of the newest liberal institutionalism. *International Organization*, 42(3), 485–507.

Jervis, R. (1978). Cooperation under the security dilemma. *World Politics*, 30(2), 167–214.

Mearsheimer, J. J. (2001). *The tragedy of great power politics*. W. W. Norton.

Organski, A. F. K. (1958). *World politics*. Knopf.

Organski, A. F. K., & Kugler, J. (1980). *The war ledger*. University of Chicago Press.

Singer, J. D., Bremer, S., & Stuckey, J. (1972). Capability distribution, uncertainty, and major power war, 1820–1965. In B. Russett (Ed.), *Peace, war, and numbers* (pp. 19–48). Sage.

Waltz, K. N. (1979). *Theory of international politics*. Addison-Wesley.

3.3 In the practice

> **The target.** The shares of world material capabilities (CINC) of the major powers, 1816–2022 — the picture behind Mearsheimer's and Allison's argument — and the year China passes the United States. Then: the same transition measured two other ways, dated two other years.

Work in your team. Everyone builds their own workbook; the team hands in one.

3.3.1 1 · Collect — 15 min

NMC v7.0 and the SIPRI workbook should already be in `data/raw/`; your Session 02 WDI extract too. If not: `../data/README.md`.

Provenance log: three lines — NMC (version 7.0, and the codebook's date), SIPRI (the file name gives the release), WDI (reused; same version as Session 02).

3.3.2 2 · Prepare — 20 min

3.3.2.1 NMC

1. **Connect** → **Text file** → `NMC-70-abridged.csv` (from `NMC-v7-abridged.zip`, inside `NMCv7.zip`). Drag it to the canvas.
2. Check the columns against the **codebook**. In v7.0 they are unchanged from v6.0: `stateabb` (three letters), `ccode` (number), `year`, the six components `milex`, `milper`, `irst`, `pec`, `tpop`, `upop`, then `cinc` and `version`. Missing component values are coded **`-9`**: before you use any component, exclude them (a filter `>= 0`), or every sum is wrong.
3. `year` should be a **number (whole)**, not a date: keep it that way for now.
4. Sanity check before you build anything: new sheet, `year` → Columns (continuous), `SUM(cinc)` → Rows, no filter. Every year should sum to **about 1** — in v7.0, between 1.000 and about 1.075. Which years are furthest from 1, and why? (The codebook's CINC section explains it: the average is taken over the components a state *has*.)

3.3.2.2 SIPRI

1. **Connect** → **Microsoft Excel** → the SIPRI workbook. Use the **`Constant (2024) US$`** sheet (US$ millions, constant 2024 prices).
2. Tick **Use Data Interpreter**: the sheet has five rows of notes above the header (`Country`, an unnamed column, `Notes`, `1949` … `2025`). Region headings (`Africa`, `North Africa`, …) remain as rows with no values: filter them out.
3. **Pivot** the year columns (as in Session 02). Rename to `Year` and `Milex`. Make `Year` a whole number with the same name as NMC's year field — it will be the link.
4. `Milex` arrives as text: SIPRI writes **`...`** for unavailable data and `xxx` for a country that did not exist (the notes at the top of the sheet; the `Footnotes` sheet explains the † § ¶ marks in `Notes`). Change to **Number (decimal)** and check what became null.
5. SIPRI identifies countries by **name**, not code. Write down the exact strings for China, the United States and Russia. This is a provenance-log entry.

3.3.3 3 · Replicate — 25 min

3.3.3.1 Sheet 1 · "Shares of world capabilities, 1816–2022"

1. `year` → Columns (continuous); `SUM(cinc)` → Rows.
2. `stateabb` → Colour. Filter to the major powers: `USA`, `UKG`, `FRN`, `GMY`, `RUS`, `JPN`, `CHN`, `ITA` — **check each code in the COW state list** (`COW-country-codes.csv`; see `../data/README.md`). Russia/USSR is one continuous series (`RUS`). Germany is not: `GMY` stops in 1945 and returns in 1991, with `GFR` and `GDR` in between. `JPN` also has a gap (1946–1951). Look for them, and decide what your line does there.
3. Highlight the US and China (strong colours), grey the rest.
4. Annotate 1945, 1991, and the year China passes the US.
5. Title with the finding; caption: *Correlates of War, NMC v7.0 (Singer, Bremer & Stuckey 1972), accessed ….*

3.3.3.2 Sheet 2 · "How close is the challenger?"

1. **Create → Parameter** `Challenger`: data type *String*, *List*, **Add values from** `stateabb`. Default `CHN`.
2. Calculated fields:

```
Challenger CINC = IF [stateabb] = [Challenger] THEN [cinc] END
US CINC         = IF [stateabb] = "USA" THEN [cinc] END
Power ratio     = SUM([Challenger CINC]) / SUM([US CINC])
```

3. `year` → Columns; `Power ratio` → Rows. Right-click the parameter → **Show Parameter**.
4. **Analytics → Reference Line**: constant **1** ("parity") and constant **0.8** ("power transition zone", Organski & Kugler).
5. Switch the parameter to `GMY`, `JPN` and `RUS`. When did each enter the transition zone against the US — or against the UK, if you also make the leader a parameter?

3.3.4 4 · Compare and break — 20 min

1. **Compare.** Put Sheet 1 beside the readings' claims. In which year does China's CINC first exceed the US's in your data? Does that date surprise you, given what you read?
2. **Break — change the measure of power.** Build two more ratios, China ÷ US:

   - **military expenditure** from SIPRI (constant dollars), and
   - **GDP** from your WDI extract (current US$ and PPP).

   Put them on one sheet with the CINC ratio by **blending** on the year: on a sheet that uses NMC, drag a SIPRI field onto the view; check that the link icon is active on the year field (*Data → Edit Blend Relationships* if Tableau did not find it). **Blending links fields only by name and type**: if the year is `year` in NMC, `Year` in SIPRI and `Year (n)` in your Session 02 WDI extract, rename them to the same name (right-click → *Rename*) and make all three whole numbers before you blend, or use *Edit Blend Relationships → Custom* to pair them by hand. Use a **dual axis** (right-click the second axis → *Dual Axis → Synchronize Axis*) so both ratios share one scale, with the parity line at 1.
3. **Read it.** Three measures, three transition dates (or no transition yet). Which date belongs in a board presentation, and for which decision?
4. **Extend.** Use the NMC components: which of the six drives China's CINC? Build one component's share (e.g. iron and steel) with a percent-of-total table calculation computed across states, restarting every year.

3.3.5 5 · The catch — 10 min

One slide:

Original	Ours	The catch
The distribution of power as the readings describe it	Sheet 1, and the three-measure ratio sheet	One sentence: what this figure does **not** license your team's multinational to conclude

A member drawn at random presents in two minutes.

3.3.6 Hand in

On ZoneCours before leaving, one per team: `.twbx`, provenance log, slide. Formative.

The one rule: your log records at least one place where the data disagreed with the readings — or with each other.

Team project milestone (Session 04): a one-paragraph footprint of your firm — its five most important countries, with sources.

3.3.7 What loses credit

- Joining NMC to WDI or SIPRI on names or on COW codes treated as ISO
- `SUM` of a ratio across years or states
- "China overtook the US in …" without saying on what measure
- A catch about the theory rather than about what the firm may conclude

Chapter 4

Trade, conflict and geoeconomic fragmentation

Session 4 · Does trade keep the peace — and is the world now trading along geopolitical lines? Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **If states trade more with friends than with rivals, how does a multinational find out which side each of its markets is on — and how fast the lines are hardening?**

Big works	Gopinath, G., Gourinchas, P.-O., Presbitero, A. F., & Topalova, P. (2025). Changing global linkages: A new Cold War? *Journal of International Economics*, 153, 104042. doi:10.1016/j.jinteco.2024.104042 (IMF Working Paper WP/24/76) · Mansfield, E. D. (2021). International trade and conflict. In *The Oxford handbook of international political economy* (pp. 660–671). doi:10.1093/oxfordhb/9780198793519.013.27
Target figure	**Trade within versus between geopolitical blocs**, indexed to the period before Russia's 2022 invasion of Ukraine
Data you collect	UN General Assembly ideal points — Bailey, Strezhnev & Voeten, Harvard Dataverse doi:10.7910/DVN/LEJUQZ · IMF, International Trade in Goods by partner country (IMTS, formerly DOTS) — data.imf.org
Tableau skill	**Joins and relationships** on dyads · **groups** and **sets** for blocs · indexing to a base year
Course themes	The system of international economic order; inter-state anarchy and the balance of the major powers that dictate international trade relations

4.1 Before the session

Four things to do before class. Budget **about two and a half hours**: today opens with Quiz 1.

	Before class	Used in class for
1	Revise Sessions 01–03	**Quiz 1**, first twenty minutes
2	Read Gopinath et al. and Mansfield	the lecture and the replication
3	Collect the ideal points and the IMF extract	Practice round 04
4	Self-check	the lecture

4.1.1 1. Quiz 1 — revise 30 min

Twenty minutes, on paper, closed book. Covers Sessions 01–03: concepts, the big works and their target figures, and reading an unseen chart. Specimen: `assessments/quizzes/`.

4.1.2 2. The reading — 60–75 min

Gopinath, G., Gourinchas, P.-O., Presbitero, A. F., & Topalova, P. (2025). Changing global linkages: A new Cold War? *Journal of International Economics*, 153, 104042. doi:10.1016/j.jinteco.2024.104042 · free as IMF Working Paper WP/24/76.

Mansfield, E. D. (2021). International trade and conflict. In J. C. W. Pevehouse & L. Seabrooke (Eds.), *The Oxford handbook of international political economy* (pp. 660–671). Oxford University Press. doi:10.1093/oxfordhb/9780198793519.013.27

Read Mansfield first — it is short and gives you the debate. Then Gopinath et al.: introduction, the section that **defines the blocs**, and the figures on trade and investment between blocs.

Annotate four things:

	What to look for
1	Mansfield: the **liberal** argument that trade reduces conflict, the **realist** reply, and his own verdict
2	Gopinath et al.: **exactly how blocs are defined** — which countries, which vote or measure, which year
3	The **target figure**: trade within vs between blocs, before and after the 2022 invasion. Its base period, its frequency, its units
4	The role of **connector** countries — and one your team's firm depends on

---

4.1.3 3. The data — 20 min

Download the ideal-point file and codebook from Harvard Dataverse, and build an IMF IMTS extract of bilateral goods exports, annual, 2015–latest. Details: `../data/README.md`.

The IMF extract is the largest download so far: do it before you read, not in class.

4.1.4 4. Self-check — 15 min

Answer on paper.

1. State the liberal "commercial peace" argument in one sentence, and the realist objection in one.
2. Martin, Mayer & Thoenig (2008) distinguish bilateral and multilateral openness. Which one makes a bilateral conflict *more* likely, and why?
3. What does a UN General Assembly ideal point measure? Name one thing it does not.
4. Why might trade between two blocs fall even if no government imposes a single new barrier?

4.1.5 Arrive with

- ☐ Sessions 01–03 revised for Quiz 1
- ☐ Mansfield and Gopinath et al. read; the bloc definition written down
- ☐ Ideal points, codebook and IMF extract in `data/raw/`, logged
- ☐ Team milestone: your firm's five most important countries, with sources
- ☐ Your self-check answers, on paper

Session 04 overview · The lecture · The practice

4.2 The lecture

The session opens with Quiz 1 (20 minutes) and four Risk Briefs (20 minutes); the lecture takes the remaining fifty.

4.2.1 1. Does trade keep the peace? The liberal answer

The **commercial peace** is one of the oldest ideas in international political economy. From Montesquieu ("peace is the natural effect of trade") to Norman Angell's *The Great Illusion* (1910), the argument is that trade raises the **opportunity cost of war**: states that gain from exchange lose those gains if they fight.

Modern versions add mechanisms:

- **Opportunity cost.** Conflict disrupts trade; the more a dyad trades, the more it loses.
- **Signalling.** Interdependent states can send costly signals — threatening economic ties — short of force, which reduces miscalculation.
- **Domestic coalitions.** Trade creates exporters and importers with a stake in peaceful relations who lobby their governments.
- **Institutions.** Trade agreements create forums and dispute-settlement procedures.

Statistical studies of dyads from the 1990s onwards (notably Russett & Oneal's *Triangulating Peace*, 2001) found that dyads with more trade relative to GDP experience fewer militarised disputes.

4.2.2 2. The realist critique

Realists reply that interdependence can **cause** conflict:

- **Vulnerability.** Dependence on a rival for essential goods is a security risk. States therefore limit it — or fight to secure supplies (Waltz, 1979).
- **Relative gains** (Session 03). If trade makes the rival stronger faster, it is a threat.
- **Security externalities** (Gowa, 1994). Trade raises the economic capacity of the partner; states therefore trade more with **allies** than with adversaries.
- **Expectations of future trade** (Copeland, 2015). What matters is not today's trade but whether leaders expect it to continue. When they expect to be cut off, interdependence becomes a reason to act before the window closes.

The history the realists point to: Britain and Germany were heavily interdependent in 1914.

4.2.3 3. What the evidence says: Mansfield (2021)

Mansfield's review in the *Oxford Handbook of International Political Economy* sets out the debate and the main empirical findings. Four points to retain:

1. **The relationship runs both ways.** Conflict reduces trade, so a negative correlation between trade and conflict does not by itself show that trade causes peace.
2. **Institutions matter.** Preferential trading arrangements are associated with fewer disputes among members (Mansfield & Pevehouse, 2000).
3. **Context matters.** The effect differs with regime type, the level of development, and the distribution of power between the partners.
4. **The question has changed.** As geopolitical competition returned, research has turned from "does trade pacify?" to "how do states use trade as an instrument of conflict?" — the questions of Sessions 08, 10 and 11.

4.2.4 4. Bilateral versus multilateral openness

Martin, Mayer & Thoenig (2008), *Make Trade Not War?*, *Review of Economic Studies*, 75(3), 865–900 (doi:10.1111/j.1467-937X.2008.00492.x).

Their argument separates two effects:

	Effect on the risk of a bilateral conflict
Bilateral trade between A and B	**lowers** it: A and B have more to lose from fighting each other
Multilateral trade openness of A and B with the rest of the world	**raises** it: if A can trade with everyone else, losing B costs less

The implication is uncomfortable for the liberal view: **globalisation lowers the cost of bilateral conflict** because it provides substitutes. A world of many alternative suppliers makes it cheaper to break with any one of them. Their empirical work finds support for both effects.

4.2.5 5. Fragmentation: the new question

Friend-shoring (a term popularised in US policy debate in 2022) proposes to move supply chains towards trusted partners. **De-risking** (the European Commission's term, 2023) proposes to reduce dependence on rivals in critical goods without breaking trade overall. **De-coupling** is the end point: separate economic blocs.

Gopinath, Gourinchas, Presbitero & Topalova (2025) ask whether the data show this happening. Their design:

1. **Define blocs.** Countries are grouped into a **US-leaning** bloc, a **China-leaning** bloc and a **non-aligned** group on the basis of their **foreign-policy alignment as revealed by UN General Assembly voting**. Read Section 2 of the paper for exactly which vote or measure, and which year, they use — ours will be a simplification, and you must say how it differs.
2. **Compare trade within and between blocs** before and after Russia's invasion of Ukraine in February 2022, controlling for the usual determinants of bilateral trade.
3. **Look for connectors** — non-aligned countries whose trade with both blocs grew, through which some between-bloc trade may be rerouted.

Their main finding: after the invasion, trade **between** the two blocs fell relative to trade **within** them, by a margin of the order of ten percent or more (find the exact estimate in the paper and write it in your notes), and foreign direct investment shows a similar pattern. They are careful to say this is a fragmentation still small compared with the Cold War — but it is happening, and it is concentrated in strategic sectors.

4.2.6 6. What it means for the multinational

Classify your markets by alignment	Which of your five most important countries (the team milestone) sit in which bloc — and which are drifting?
Connectors are opportunities and risks	Mexico, Vietnam and others gain between-bloc trade — and attract scrutiny for transshipment and rules-of-origin enforcement
The cost of switching is falling for states, not for you	Martin, Mayer & Thoenig: multilateral openness makes it cheaper for *states* to break with a partner. Your supply contracts do not get cheaper to break
De-risking is sectoral	Fragmentation is concentrated in strategic goods; a firm in consumer goods and one in semiconductors face different worlds

---

4.2.7 7. What the figure does not license

- A fall in between-bloc trade is a **correlation with a date**, not proof that geopolitics caused it: sanctions on Russia, the post-COVID rebound and price changes all move trade values in the same years.
- **Bloc membership is a modelling choice.** Change the threshold or the base year and countries move between blocs — and the figure moves with them.
- **Values, not volumes.** Trade in US dollars rises with commodity prices; 2022 was a year of very high energy and food prices.
- It shows **aggregate** trade. Your firm's products may be in a sector where fragmentation is far stronger — or absent.

4.2.8 8. The Tableau skill of the day

A dyadic table (exporter × partner × year) needs **two** copies of the country table: one for the exporter's bloc, one for the partner's. In Tableau's data model, add the bloc table, relate it on `exporter = country`, then add it again and relate the second copy on `partner = country`. Blocs are **sets** defined by a condition on the ideal point, with **parameters** as thresholds. The trade series is indexed to a base year with a table calculation. Click-path: `../02-practice/README.md`.

4.2.9 References

Angell, N. (1910). *The great illusion*. Heinemann.

Bailey, M. A., Strezhnev, A., & Voeten, E. (2017). Estimating dynamic state preferences from United Nations voting data. *Journal of Conflict Resolution*, 61(2), 430–456.

Copeland, D. C. (2015). *Economic interdependence and war*. Princeton University Press.

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Mansfield, E. D., & Pevehouse, J. C. (2000). Trade blocs, trade flows, and international conflict. *International Organization*, 54(4), 775–808.

Martin, P., Mayer, T., & Thoenig, M. (2008). Make trade not war? *Review of Economic Studies*, 75(3), 865–900.

Russett, B., & Oneal, J. R. (2001). *Triangulating peace: Democracy, interdependence, and international organizations*. W. W. Norton.

4.3 In the practice

> **The target.** Gopinath, Gourinchas, Presbitero & Topalova (2025): trade **between** geopolitical blocs against trade **within** them, before and after Russia's invasion of Ukraine. You define the blocs yourself from UN General Assembly ideal points, then watch what happens to the finding when you move the line.

Work in your team. One extract per team; everyone builds their own workbook.

4.3.1 1 · Collect — 10 min

Ideal points and codebook from Dataverse; your IMF IMTS extract of bilateral goods exports, annual, 2015–latest. If not done: `../data/README.md`.

Provenance log: one line per file. For the ideal points, log **version 39** and the change to calendar-year estimation. For IMF, log the date of the extract and the exact indicator you chose.

4.3.2 2 · Prepare — 25 min

4.3.2.1 A · Classify countries into blocs (ideal points)

1. **Connect → Text file →** `IdealpointestimatesFP_2026FP.csv`. The **ISO 3-letter code**, the **year** and the **ideal point** are `iso3c`, `year` and `IdealPointFP` (see the codebook). Rename `iso3c` → `iso3` and `IdealPointFP` → `IdealPoint` (right-click → *Rename*): the steps below use those names.
2. New sheet. Filter `year` = **2021** — the last full year before the invasion. (Gopinath et al. may use a different reference: check, and log the difference.)
3. Read the ideal points of the US (`USA`) and China (`CHN`) off the data. Create two parameters, `US ref` and `CHN ref` (*Float*), and type those two values in.
4. Create a third parameter `Margin` (*Float*, start at **0.5**) and the calculated field:

```
Bloc =
IF ABS([IdealPoint] - [CHN ref]) - ABS([IdealPoint] - [US ref]) > [Margin] THEN "US-leaning"
ELSEIF ABS([IdealPoint] - [US ref]) - ABS([IdealPoint] - [CHN ref]) > [Margin] THEN "China-leaning"
ELSE "Non-aligned"
END
```

5. `iso3` → Rows, `Bloc` → Rows, `IdealPoint` → Text. Sort by ideal point. Does the classification look right for countries you know? Adjust `Margin` until you can defend it — and write down the value and why.
6. **Export the classification** as a derived file: *Worksheet → Export → Crosstab to Excel* (or *View Data → Export All* to CSV). Save it as `data/derived/blocs_2021.csv` — **not** in `raw/`. Add a provenance line: derived from the ideal points, `Margin` = …, base year 2021.

4.3.2.2 B · Relate trade to blocs — the dyad model

1. New data source: **Connect → Text file** (or Excel) → your IMF extract. Identify the exporter, partner, year and value columns — in the IMF's SDMX layout `COUNTRY`, `COUNTERPART_COUNTRY`, `TIME_PERIOD`, `OBS_VALUE` — and rename them `exporter`, `partner`, `year`, `value`. Keep only `INDICATOR` = `XG_FOB_USD`.
2. Remove aggregate partners ("World", "Advanced Economies", regions — codes starting with `G` or `TX`, e.g. `G001` World): filter them out, and log which.
3. In the data source pane, drag `blocs_2021.csv` onto the canvas: Tableau creates a **relationship**. Click the noodle → relate `exporter` = `iso3`.
4. Drag `blocs_2021.csv` onto the canvas **a second time**. Relate `partner` = `iso3`. Rename this copy **Partner blocs** and its `Bloc` field **Partner bloc**; rename the first copy's field **Exporter bloc**.
5. IMTS uses ISO3 codes, so the relationship works directly. If your download shows names instead, you need a concordance table — log it.

4.3.3 3 · Replicate — 25 min

4.3.3.1 Sheet 1 · "Trade within and between blocs"

1. The pair type:

```
Pair type =
IF [Exporter bloc] = "Non-aligned" OR [Partner bloc] = "Non-aligned" THEN "Involves non-aligned"
ELSEIF [Exporter bloc] = [Partner bloc] THEN "Within bloc"
ELSE "Between US- and China-leaning"
END
```

2. The index, 2021 = 100:

```
Trade index = SUM([value]) / WINDOW_SUM(SUM(IF [year] = 2021 THEN [value] END)) * 100
```

3. `year` → Columns as a **discrete** dimension (blue); `Trade index` → Rows; `Pair type` → Colour. Right-click `Trade index` → **Edit Table Calculation** → **Compute Using** → **year**, so each pair type is indexed to its own 2021.

4. Reference line at 100. Annotate 2022.

5. Title with the finding. Caption: both sources with versions, and *"Blocs: ideal-point distance to US vs China, 2021, margin = ..."*.

4.3.3.2 Sheet 2 · "Who sits where?" — the classification on a map

`iso3` (geographic role *Country/Region*, ISO 3166-1 alpha-3) on the map, coloured by `Bloc`. Mark your team firm's five main countries with a **group** (*Create* → *Group*, "Our markets").

4.3.4 4 · Compare and break — 20 min

1. **Compare.** Put Sheet 1 beside the paper's figure. Their data are quarterly and their estimate controls for other determinants of trade; ours is annual and raw. Does the direction match? Is the gap between "within" and "between" of a similar order? Write one difference and its likely cause.
2. **Break the bloc definition.** Move `Margin` (0.25, 1.0), re-export, refresh. Which countries change bloc? Does the gap between the lines survive? This is the most important answer of the round.
3. **Break the side of the trade.** If your extract has the mirror series (imports reported by the partner), rebuild the index with it. Same picture?
4. **Extend — connectors.** Create a **set** of non-aligned countries (*Create* → *Set*, *Top* tab: top 10 by growth in exports to *both* blocs, 2021 → latest — or pick by hand from a table). Which are they? Is any of them among your firm's markets?

4.3.5 5 · The catch — 10 min

One slide:

Original	Ours	The catch
Gopinath et al.'s within/between figure	Sheet 1, and the same chart with another `Margin`	One sentence: what this figure does **not** license your team's multinational to conclude

A member drawn at random presents in two minutes.

4.3.6 Hand in

On ZoneCours before leaving, one per team: `.twbx` (including the derived `blocs_2021.csv`), provenance log, slide. Formative.

The one rule: your log records at least one place where the data disagreed with the paper — and your slide says how your blocs differ from theirs.

4.3.7 What loses credit

- Joining on country names (Türkiye, Côte d'Ivoire, Korea…)
- Leaving "World" or regional aggregates among the partners
- An index computed across pair types instead of within each one
- A bloc threshold that is not reported on the slide

Chapter 5

International institutions

Session 5 · Arenas of power, or constraints on it? Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **When a multinational relies on an international institution to manage its risk, is it relying on a rule — or on the states that buy and sell influence inside it?**

Big work	Kuziemko, I., & Werker, E. (2006). How much is a seat on the Security Council worth? Foreign aid and bribery at the United Nations. *Journal of Political Economy*, 114(5), 905–930. doi:10.1086/507155
Target figure	US aid to elected Security Council members **in event time** — years −3 to +3 around the start of a two-year term
Data you collect	UN Security Council elected members (United Nations); US foreign aid (Greenbook, ForeignAssistance.gov or its mirror) — or the authors' replication file from AidData
Tableau skill	**LOD expressions** (`FIXED`) · years relative to an event · event-time charts · reference bands
Course themes	The impact and role of international institutions in analysing the geopolitical risks on which multinationals base their decisions

5.1 Before the session

Three things to do before class. Budget **about two hours**.

	Before class	Used in class for
1	Read the paper	the lecture discussion and the replication
2	Collect the data — route A or route B, decided as a team	Practice round 05
3	Self-check	the first ten minutes of the lecture

5.1.1 1. The reading — 60–75 min

Kuziemko, I., & Werker, E. (2006). How much is a seat on the Security Council worth? Foreign aid and bribery at the United Nations. *Journal of Political Economy*, 114(5), 905–930. doi:10.1086/507155

Read for the **argument**, not for coverage. Annotate four things:

	What to look for
1	The **research question**, in one sentence
2	The **identification idea**: why does the rotation of elected seats let the authors separate the effect of the seat from everything else about a country?
3	The **strongest single piece of evidence** — and the evidence that the effect is larger when the Council matters more
4	One **limitation** you would raise as a referee — or as a firm relying on a UN resolution

The target figure is the paper's evidence on aid **in event time**: aid in the years before, during and after a country's term, year by year. Find where the paper shows it — as a figure, and in the year-by-year estimates behind it — and write down what is on each axis.

Background, optional: Keohane, R. O. (1984). *After Hegemony: Cooperation and Discord in the World Political Economy*. Princeton University Press — chapter 6, on why states build institutions. Mearsheimer, J. J. (1994/95). The false promise of international institutions. *International Security*, 19(3), 5–49 — the realist reply.

5.1.2 2. The data — 30–45 min

Decide as a team which route you take: details in `../data/README.md`.

- **Route A.** Download the authors' replication file from AidData. Open it; write the variable list and units in your log. The link is a Google Drive file that asked for a sign-in when checked (October 2026): if it will not open, switch to route B.
- **Route B.** Download the ForeignAssistance.gov Country Summary, `us_foreign_aid_country.csv` (official site or the mirror), and **transcribe** the UN list of elected Council members, 1970–2010, one row per term. Split the alphabet between team members — it takes each of you about twenty minutes.

Everything goes, untouched, into `05-international-institutions/data/raw/`.

5.1.3 3. Self-check — 15 min

Answer on paper.

1. In one sentence each: why do states build institutions, according to a **liberal institutionalist** and according to a **realist**?
2. How many members does the Security Council have, how many are permanent, and how long is an elected member's term?
3. Why does the paper compare a country with **itself** in other years rather than with countries that never served?
4. A Security Council resolution under Chapter VII imposes sanctions on a state where your firm has a distributor. Who is bound by it — and why should the firm care how the resolution was voted?

5.1.4 Arrive with

- ☐ The paper read and annotated; the event-time figure understood
- ☐ Route A file **or** route B files in `data/raw/`, logged
- ☐ Your self-check answers, on paper
- ☐ If you present a Risk Brief today: `.twbx` and log on ZoneCours before class

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Session 05 overview · The lecture · The practice

5.2 The lecture

5.2.1 1. Why would states under anarchy build institutions at all?

Session 03 left us with a system without a government above states. Session 04 showed trade reorganising itself along geopolitical lines. Yet the same system is dense with institutions — the UN, the WTO, the IMF, the World Bank, investment treaties, export-control regimes. Why?

	Liberal institutionalism	**Realism**
Key text	Keohane (1984), *After Hegemony*	Mearsheimer (1994/95), *The False Promise of International Institutions*
Why institutions exist	They lower the **transaction costs** of cooperation, provide **information**, and lengthen the **shadow of the future** so that states can make credible commitments	They reflect the **distribution of power**; the strong build them to serve their interests
Do they constrain states?	Yes, at the margin — states comply because the institution makes reneging costly and cooperation valuable	Rarely — when interests and rules collide, power wins
What it predicts for a firm	Rules are a reasonably stable floor: dispute settlement, treaties, sanctions regimes behave predictably	Rules are only as stable as the coalition behind them; read the coalition, not the treaty

Neither is a strawman. The useful question for a firm is not which theory is "true" but **which institution, on which issue, behaves like which**. That is an empirical question — and Kuziemko and Werker give one of its sharpest answers.

5.2.2 2. The Security Council in five facts

1. **Fifteen members**: five permanent (China, France, Russia, the United Kingdom, the United States) and ten elected. Before 1966 there were six elected members.
2. **Elected members** serve **two-year terms**, five elected each year by the General Assembly, by regional group, with a two-thirds majority; they cannot be immediately re-elected.
3. **Decisions** on substantive matters need nine votes and **no veto** from a permanent member.
4. Under **Chapter VII** the Council can impose measures — sanctions, arms embargoes, the use of force — that **bind every UN member state**.

5. An elected member's vote **can be pivotal**: the P5 need nine votes, so they must court elected members.

5.2.3 3. The price of a seat — Kuziemko and Werker (2006)

The question. Do powerful states pay for the votes of elected Security Council members?

The identification idea. Countries do not choose *when* they rotate onto the Council in any way that tracks their need for aid — rotation follows regional queues and norms. So the authors compare a country **with itself** in the years before, during and after its term. If aid rises during the term and falls back afterwards, the seat — not the country's poverty, alliances or politics — is the likeliest explanation.

The findings, in the paper's own summary:

- US aid to a developing country rises substantially — by about 59% in the abstract's headline estimate — when it rotates onto the Council, and aid through the UN rises too, more modestly.
- The increase is concentrated in the **term years** and disappears after the term.
- The effect is **larger in years when the Council matters more** — measured by how much the press covered it.

Why that last result matters. It is the difference between a statistical curiosity and a mechanism. If aid followed membership whatever the Council was doing, rotation might proxy for something else. That aid tracks the **salience** of Council business is what makes "vote buying" the most plausible reading.

The follow-up literature. Dreher, Sturm and Vreeland extended the question to the World Bank and the IMF, and Vreeland and Dreher (2014), *The Political Economy of the United Nations Security Council: Money and Influence* (Cambridge University Press), brings the evidence together. The common thread: temporary members of the Council receive more favourable treatment in several channels.

5.2.4 4. What it means for multinational firms

Institutions enter a firm's risk in at least four places:

Where	Institution	What the firm relies on	What Kuziemko and Werker warn
Sanctions	UN Security Council (Chapter VII); then national and EU regimes (Session 10)	A predictable list of who is sanctioned and why	Resolutions are bargained; votes have prices; a P5 veto blocks UN sanctions — and pushes them into unilateral national regimes that diverge
Investment protection	Bilateral investment treaties; ICSID arbitration	A neutral forum if a host state expropriates	Treaties are only as strong as the host state's willingness to pay awards
Political risk insurance	MIGA (World Bank Group), national export-credit agencies	Cover against expropriation, war, transfer restrictions	The insurer's own shareholders are states — cover is a political decision too
Trade rules	WTO dispute settlement	Tariff bindings, non-discrimination	The Appellate Body has been unable to function since 2019; "rules" now depend on the coalition

The lesson for the firm. An international institution is a risk **mitigator** when the coalition that sustains it is stable, and a risk **source** when it is not. A risk dashboard should therefore track not only the rules but the coalition: who sits on the Council, how they vote, what they are being offered — which is exactly what today's practice round builds.

5.2.5 5. The Tableau skill of the day — LOD expressions

The practice needs two numbers for every country-year that do not exist in either file:

Years relative to a term. Join the term table (one row per term) to the aid table (one row per country-year) on the country code — a **join** in the physical layer, not a relationship, because a row-level calculation cannot mix fields from two related tables. Each country-year now appears once per term of that country, which is exactly what an event study needs:

```
[Event time]   =  [Year] - [Term start]
```

The country's own pre-term baseline, a *Level of Detail* expression that fixes the computation at the country × term level whatever is in the view:

```
[Baseline aid] = { FIXED [Iso3], [Term start] :
                   AVG( IF [Event time] >= -3 AND [Event time] <= -1 THEN [Aid] END ) }

[Aid relative to baseline] = [Aid] / [Baseline aid]      // row level: one value per country-year-term
```

`FIXED` computes before the view's dimension filters (context filters apply first) and ignores what is on Rows and Columns: every row of a term carries that term's baseline. The ratio is then computed row by row, and averaging it at each event year gives the **average across terms** of a ratio computed **within** each term.

5.2.6 References

Keohane, R. O. (1984). *After hegemony: Cooperation and discord in the world political economy*. Princeton University Press.

Kuziemko, I., & Werker, E. (2006). How much is a seat on the Security Council worth? Foreign aid and bribery at the United Nations. *Journal of Political Economy*, 114(5), 905–930. https://doi.org/10.1086/507155

Mearsheimer, J. J. (1994/95). The false promise of international institutions. *International Security*, 19(3), 5–49.

Vreeland, J. R., & Dreher, A. (2014). *The political economy of the United Nations Security Council: Money and influence*. Cambridge University Press.

5.3 In the practice

> **The target.** Kuziemko & Werker (2006): US aid to developing countries **in the years around a Security Council term** — flat before, higher during the two term years, back down after.

In your teams. One workbook per team; everyone builds at least one sheet.

5.3.1 1 · Collect — 15 min

You should arrive with route A or route B files in `data/raw/` (`../data/README.md`). Finish what is missing, and fill the provenance log — one line per file.

file	source	url	version / release	accessed	licence	notes
us_foreign_aid_country.csv	US State Dept (ForeignAssistance.gov, Country Summary)	foreignassistance.gov/data/ *or the mirror*	*"frozen 2024-12-19"*	*today*	free	*fiscal years; obligations; constant_amount; economic + military combined*
unsc_elected_members.xlsx	United Nations	main.un.org/securitycouncil/…	*your download date*	*today*	free	*your transcription; name changes handled*

Route A teams also need a **term table**. Open the replication file, filter on Security Council membership = 1, show country × year as a text table, export it (*Worksheet → Export → Crosstab to Excel*), and mark the first year of each two-year run as `term_start`. Ten minutes; it gives you the same term table as route B.

5.3.2 2 · Prepare — 20 min

1. **Connect** to the aid file and to `unsc_elected_members.xlsx`. In the Country Summary, filter `Transaction Type Name` = *Obligations* (a data-source filter), exclude `Fiscal Year` = `1976tq`, then make `Fiscal Year` a **number** and rename it `Year`; rename `constant_amount` to `Aid`.
2. Drag the aid table to the canvas, then **double-click it** to open the physical layer, and drag the term table next to it: **inner join** on `Country Code` = `Iso3` (or on country name only if you have checked every match — see the trap list in `TABLEAU-TOOLKIT.md`). Route A teams: use the replication file's own country, year and aid fields. > Why a join, not a relationship? The next step is a row-level calculation that uses `Year` from > one table and `Term start` from the other. Relationships do not allow that.
3. Check the row count before and after the join and write both in your log. The join should multiply rows: each country-year now appears once per term of that country.
4. Keep only aid **recipients** (drop regional aggregates such as "World", "Africa Region" — every `Country Name` ending in "Region"), and check that aid is in **constant** dollars (`constant_amount`, not `current_amount`). If it is not, say so in the log — the ratio to baseline below limits the damage, but does not remove it.

5.3.3 3 · Replicate — 25 min

5.3.3.1 Calculated fields

```
[Event time]               = [Year] - [Term start]

[Baseline aid]             = { FIXED [Iso3], [Term start] :
                               AVG( IF [Event time] >= -3 AND [Event time] <= -1
                                    THEN [Aid] END ) }

[Aid relative to baseline] = [Aid] / [Baseline aid]      // row level: one value per country-year-term
```

Route A teams: if aid is in logs in the file, use `[Log aid] - [Baseline log aid]` instead of a ratio.

5.3.3.2 Sheet 1 · "Aid in event time"

1. Filter `[Event time]` to **−3 … +3** (make it a dimension, discrete).
2. `[Event time]` → Columns; `[Aid relative to baseline]` → Rows, then change its aggregation to **AVG** (right-click the pill → *Measure* → *Average*). Because the ratio is computed row by row — one row per country-year-term — the average at each event year is the average **across terms**.
3. Mark type **Line**, then add **Circle** marks on a dual axis so each year is visible.
4. **Analytics → Reference Band** from 0 to 1 on `[Event time]`, shaded, labelled "On the Council".
5. Constant **reference line at 1.0**, labelled "Pre-term baseline".
6. Title it with the finding; caption with both sources, versions and access dates.

5.3.3.3 Sheet 2 · "How many terms?"

`[Event time]` → Columns, `COUNTD([Iso3])` → Rows. The number of terms behind each point. Is the sample balanced across event years? If not, why?

5.3.4 4 · Compare and break — 20 min

1. **Differ.** Put your Sheet 1 beside the paper's event-time evidence. Same shape? Same size? Write one difference and its most likely cause (release of the aid data, years covered, the sample of countries, constant vs current dollars, levels vs logs).
2. **Break one decision.** Replace the **mean** across terms by the **median** (a quick change of aggregation). Does the term-year bump survive? What does that tell you about whether a few large cases drive it?
3. **Break another.** Restrict to terms starting **after 1990**. Does the pattern survive the end of the Cold War?
4. **Extend — salience.** The paper finds the effect is larger when the Council matters more. Build a crude proxy: tag the years 1990–91 (Gulf War), 2002–03 (Iraq) as *high salience* with a calculated field, and colour Sheet 1 by it. What do you see — and why is two episodes far too few to conclude?

5.3.5 5 · The catch — 10 min

Original	Ours	The catch
The paper's event-time evidence	Your Sheet 1	What it does **not** license a multinational to conclude — for instance about how durable a UN sanctions resolution is, or what a host government on the Council will do

A member drawn at random presents in two minutes.

5.3.6 Hand in

Upload to ZoneCours before leaving: the `.twbx` and the provenance log. Formative: commented, not marked; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

Team project. Is any of your firm's five key countries — or its home country — on the Council now, or about to be? Add it to your footprint notes.

Chapter 6

Private authority, standards and global regulation

Session 6 · Who writes the rules a multinational actually follows? Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **Most of the rules that govern a multinational's daily operations were not written by a government. Who wrote them, with whose power — and does trade spread them up or down?**

Big work	Prakash, A., & Potoski, M. (2006). Racing to the bottom? Trade, environmental governance, and ISO 14001. *American Journal of Political Science*, 50(2), 350–364. doi:10.1111/j.1540-5907.2006.00188.x
Companion readings	Abbott & Snidal (2009), *The governance triangle* · Mattli & Büthe (2003), *Setting international standards* · Brunsson, Rasche & Seidl (2012), *The dynamics of standardization* · Green (2013), *Rethinking Private Authority*
Target figure	**The diffusion of ISO 14001** across countries — a map and small multiples over time — and each country's adoption against the adoption level of its **export markets**
Data you collect	ISO Survey of certifications (ISO 14001, ISO 9001); IMF bilateral exports (IMTS); World Bank GDP (WDI)
Tableau skill	**Maps** · geocoding by ISO code · fixing unmatched countries · the **Pages** shelf · small multiples
Course themes	The impact and role of international institutions — here private and hybrid ones — in the geopolitical risks on which multinationals base their decisions

6.1 Before the session

Three things to do before class. Budget **about two and a half hours**, and **start the ISO Survey download at least two days ahead**: it needs a free IAF CertSearch account.

	Before class	Used in class for
1	Read the paper; skim the triangle	the lecture and the replication
2	Collect three datasets	Practice round 06
3	Self-check	the first ten minutes of the lecture

6.1.1 1. The reading — 75–90 min

The big work. Prakash, A., & Potoski, M. (2006). Racing to the bottom? Trade, environmental governance, and ISO 14001. *American Journal of Political Science*, 50(2), 350–364. doi:10.1111/j.1540-5907.2006.00188.x

	What to look for
1	The **research question**: does trade drive a race to the bottom in environmental governance?
2	The **mechanism**: why would a country's adoption depend on adoption in its **export markets**?
3	The **strongest evidence** — what is the panel, and what is the key variable?
4	One **limitation** — from a referee, or from a firm deciding whether to certify

Skim, for the lecture (30 min in total — the abstract, the introduction and the main figure or table of each):

- Abbott, K. W., & Snidal, D. (2009). The governance triangle: Regulatory standards institutions and the shadow of the state. In W. Mattli & N. Woods (Eds.), *The Politics of Global Regulation* (pp. 44–88). Princeton University Press. doi:10.1515/9781400830732.44 — **draw the triangle.**
- Mattli, W., & Büthe, T. (2003). Setting international standards: Technological rationality or primacy of power? *World Politics*, 56(1), 1–42. doi:10.1353/wp.2004.0006 — **write down the three explanations they compare.**

6.1.2 2. The data — 45 min (plus the account wait)

Details and the fallback: `../data/README.md`. Split the work in your team.

1. **ISO Survey** — ISO 14001 and ISO 9001 certificates by country and year. If you cannot download the full tables, use the **transcription fallback** (35 countries × 6 years).
2. **IMF IMTS** — goods exports from your ~35 countries to all partners, same years.
3. **WDI GDP** — current US$, all countries.

Everything goes, untouched, into `06-private-authority-and-standards/data/raw/`.

6.1.3 3. Self-check — 15 min

1. Name the three corners of Abbott and Snidal's triangle, and place **one** scheme you know (e.g. the Forest Stewardship Council, the UN Global Compact, ISO 14001) somewhere inside it.
2. What is a "race to the bottom"? What is a "California effect" (Vogel, 1995)? Which does Prakash and Potoski's evidence support?
3. Why would a firm in Viet Nam certify to ISO 14001 if no Vietnamese law requires it?
4. Who sits at the table when an ISO standard is written? Why might that matter for a firm from a country that is not there?

6.1.4 Arrive with

- ☐ The paper read; the triangle drawn; Mattli and Büthe's three explanations written down
- ☐ ISO (full or transcribed), IMTS and WDI files in `data/raw/`, logged
- ☐ Your self-check, on paper

Session 06 overview · The lecture · The practice

6.2 The lecture

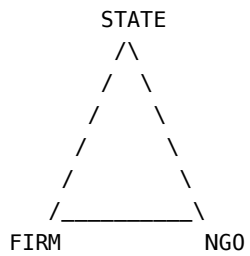
6.2.1 1. The rules a firm actually follows

Session 05 looked at intergovernmental institutions. But walk through a multinational's supply chain and count the rules it obeys: quality management (ISO 9001), environmental management (ISO 14001), information security (ISO/IEC 27001), forest and fisheries certification, labour codes of conduct, carbon accounting (the GHG Protocol), sustainability reporting standards, telecom technical specifications. **Most were not written by a government.** They are written by private or hybrid bodies, adopted voluntarily — and then made compulsory by **buyers, investors and export markets**.

That is **private authority**: the capacity of non-state actors to make rules that others treat as binding. For a multinational it is both a **risk** (a standard written elsewhere becomes a condition of market access) and a **lever** (a firm that shapes the standard shapes the market).

6.2.2 2. The governance triangle — Abbott and Snidal (2009)

Abbott and Snidal map **regulatory standards-setting (RSS) schemes** in a triangle whose corners are the three types of actor that can write rules:



- Schemes near a **corner** are run by one actor type: an industry code of conduct (firms), an activist certification (NGOs), an intergovernmental rule (states).
- Schemes in the **middle** are hybrids: firms and NGOs together (multi-stakeholder certification); states, firms and NGOs together (the UN Global Compact is often placed here).
- The **shadow of the state**: even private schemes are shaped by the possibility that states will regulate if private actors do not — and by states choosing to endorse, require or ignore them.

Why it matters for a firm. Where a scheme sits predicts how it behaves: who can change it, how legitimate NGOs and governments will find it, and whether a state may one day make it mandatory.

6.2.3 3. Standards and power — Mattli and Büthe (2003)

Who wins when an *international* product standard is written at ISO or the IEC? Mattli and Büthe compare three explanations:

Explanation	Prediction
Technological rationality	The technically best solution wins, whoever proposes it
Primacy of power	The standard favours the states with the largest markets
Institutional complementarity (their argument)	Countries whose **domestic** standard-setting system is hierarchical and coordinated — matching the structure of ISO/IEC — are more effective at shaping international standards than countries with fragmented domestic systems

Their evidence points to institutional complementarity: the coordinated European national standards bodies were better placed than the fragmented, decentralised US system. **The lesson:** influence over "technical" rules depends on how a country — and its firms — are organised to take part.

Today, that is geopolitics. The competition over telecommunications standards, over battery and charging standards, over AI and data rules, is a competition between blocs, each trying to make its domestic rules the world's. Bradford's *Brussels effect* (2020) describes how EU rules become global because firms find it cheaper to adopt the strictest rule everywhere than to run two production lines.

6.2.4 4. Three perspectives on standardization — Brunsson, Rasche and Seidl (2012)

Brunsson and colleagues organise research on standards around three aspects:

- **standardization *of* organizations** — standards that tell firms how to be organised (management system standards such as ISO 14001);
- **standardization *by* organizations** — the bodies that write standards and how they work;
- **standardization *as* organization** — standards as a way of organising markets and fields without formal hierarchy.

Their key point is that standards, usually associated with stability and sameness, are **dynamic**: they are revised, contested, translated locally, decoupled from practice. A firm "certified" to ISO 14001 may run its plants exactly as before — the certificate and the practice can diverge.

6.2.5 5. Delegated or entrepreneurial — Green (2013)

Green distinguishes two sources of private authority:

	Delegated	**Entrepreneurial**
Origin	States hand rule-making to private actors	Private actors create rules on their own, and others adopt them
Illustration (carbon governance, Green's field)	Private bodies accrediting and verifying projects under the Kyoto Protocol's Clean Development Mechanism	Voluntary carbon accounting and offset standards, such as the GHG Protocol
What a firm should watch	The state that delegated can take it back	Adoption — the rule is as strong as the number of users

---

6.2.6 6. Race to the bottom, or trading up? — Prakash and Potoski (2006)

The worry. Trade intensifies competition; governments lower environmental standards to attract investment and win exports — a race to the bottom.

The alternative. Vogel's (1995) *California effect*: exporters to strict markets adopt strict standards, and trade spreads them — trading up.

The test. ISO 14001 is a voluntary environmental management standard that goes beyond what many domestic laws require. Prakash and Potoski use a panel of 108 countries over seven years and ask whether adoption in a country depends on adoption in its **export markets**.

The finding. Trade linkages **encourage** ISO 14001 adoption **when a country's major export markets have themselves adopted it**. Trade spreads private standards upward — towards the standards of the customers.

For the firm. A private standard can become a de facto condition of access to a market without a single law changing. Monitoring "regulatory" risk means monitoring what your customers' home markets adopt.

6.2.7 7. The Tableau skill of the day — maps and Pages

- Give `iso3` the geographic role **Country/Region** (right-click → *Geographic Role*). Tableau recognises ISO 3166 codes; check the **"unknown"** indicator at the bottom right of the map and fix what it could not place (*Map → Edit Locations*).
- **Filled map** of certificates per billion US$ of GDP; colour on a sequential palette, never a rainbow.
- `Year` on the **Pages** shelf: a play button animates the diffusion.
- **Small multiples**: `Region` on Rows and `Year` on Columns with a line of the median country — the same story, without animation, on one page.

6.2.8 References

Abbott, K. W., & Snidal, D. (2009). The governance triangle: Regulatory standards institutions and the shadow of the state. In W. Mattli & N. Woods (Eds.), *The politics of global regulation* (pp. 44–88). Princeton University Press. https://doi.org/10.1515/9781400830732.44

Bradford, A. (2020). *The Brussels effect: How the European Union rules the world*. Oxford University Press.

Brunsson, N., Rasche, A., & Seidl, D. (2012). The dynamics of standardization: Three perspectives on standards in organization studies. *Organization Studies*, 33(5–6), 613–632. https://doi.org/10.1177/0170840612450120

Green, J. F. (2013). *Rethinking private authority: Agents and entrepreneurs in global environmental governance*. Princeton University Press.

Mattli, W., & Büthe, T. (2003). Setting international standards: Technological rationality or primacy of power? *World Politics*, 56(1), 1–42. https://doi.org/10.1353/wp.2004.0006

Prakash, A., & Potoski, M. (2006). Racing to the bottom? Trade, environmental governance, and ISO 14001. *American Journal of Political Science*, 50(2), 350–364. https://doi.org/10.1111/j.1540-5907.2006.00188.x

Vogel, D. (1995). *Trading up: Consumer and environmental regulation in a global economy*. Harvard University Press.

6.3 In the practice

The target. Prakash & Potoski (2006): ISO 14001 spreads across countries, and a country adopts more when its **export markets** have adopted more. Two views: the diffusion (a map and small multiples over time), and the trade channel (own adoption against export-market adoption).

In your teams. One workbook per team; everyone builds at least one sheet.

6.3.1 1 · Collect — 10 min

Your three files should already be in `data/raw/` (`../data/README.md`). Finish the provenance log:

file	source	url	version / release	accessed	licence	notes
iso_survey_1400[1].xlsx *or* iso_survey_transcribed.xlsx	ISO / IAF CertSearch	…	*survey edition*	*date*	*free / account*	*full tables or transcription; coverage changes?*
imf_imts_exports.csv	IMF	data.imf.org — IMTS	*download date*	*date*	free	*exports FOB, US$; partner codes used*
wdi_gdp.csv	World Bank	api.worldbank.org	*last updated*	*date*	CC BY 4.0	

6.3.2 2 · Prepare — 20 min

1. **ISO table.** If years are columns, **pivot** them (select the year columns → right-click → *Pivot*), rename to `Year` and `Certificates`. Add or check `iso3`. Keep 14001 and 9001 as two columns, or as a `Standard` dimension — your choice, logged.

2. **WDI GDP.** Pivot the year columns too. Keep `Country Code` (it is ISO3) and the value. From the `Metadata_Country_…` file in the same zip, bring in **Region**, and drop the aggregates (World, regions, income groups): they are the rows with an empty `Region`.

3. **Join** ISO to GDP on `iso3` **and** `Year` (both join clauses). Check the row count.

4. Calculated field:

```
[ISO 14001 per $bn GDP] = [Certificates 14001] / ([GDP current US$] / 1e9)
```

5. Log: which countries did not match, and why.

6.3.3 3 · Replicate — 30 min

6.3.3.1 Sheet 1 · "ISO 14001 spread" — a map

1. `iso3` → *Geographic Role* → *Country/Region*; double-click it. Check the **unknown** indicator, bottom right; fix with *Map* → *Edit Locations*.
2. Mark type **Map**; `[ISO 14001 per $bn GDP]` → Colour, **sequential** palette. Consider a log-like scale: the distribution is very skewed (try *Edit Colors* → *Stepped* with 5 steps).
3. `Year` → **Pages**. Press play. Title it with what you see.

6.3.3.2 Sheet 2 · "Diffusion by region" — small multiples

`Year` → Columns (continuous), `Region` → Rows, `MEDIAN([ISO 14001 per $bn GDP])` → Rows. One small line per region. Which region adopted first? Which caught up?

6.3.3.3 Sheet 3 · "The pull of export markets" — the trade channel

This is the core of Prakash and Potoski: does a country's adoption track its **export markets'** adoption?

1. New data source. Start from `imf_imts_exports.csv` (reporter × partner × year × exports). In the IMF layout reporter and partner are `COUNTRY` and `COUNTERPART_COUNTRY`, in ISO3 codes (rename them `Reporter iso3` and `Partner iso3`; `TIME_PERIOD` → `Year`, `OBS_VALUE` → `Exports`). Drop aggregate partners (codes starting with `G` or `TX`). If your download shows names instead, build a concordance. Kosovo is `KOS` at the IMF but `XKX` at the World Bank.

2. **Join** the ISO + GDP table to it **twice** (Tableau lets you add the same table again; rename the copies):

   - as **Partner**: `Partner iso3 = iso3` and `Year = Year`
   - as **Reporter**: `Reporter iso3 = iso3` and `Year = Year`

3. Calculated fields:

```
[Weighted partner adoption]  = [Exports] * [Partner ISO 14001 per $bn GDP]

[Export-market adoption]     = SUM([Weighted partner adoption]) / SUM([Exports])

[Own adoption]               = { FIXED [Reporter iso3], [Year] :
                                   AVG([Reporter ISO 14001 per $bn GDP]) }
```

   The `FIXED` matters: the reporter's own value is repeated on every partner row, so it must not be summed.

4. Scatter: `[Export-market adoption]` → Columns, `AVG([Own adoption])` → Rows, `Reporter iso3` → Detail, `Year` → Colour (or filter to one year). **Log scales** on both axes (right-click axis → *Edit Axis* → *Logarithmic*). Add a trend line (*Analytics* → *Trend Line* → *Linear*) — read it as a description, not a test: Session 09 returns to what its R^2 means.

6.3.4 4 · Compare and break — 20 min

1. **Differ.** Your scatter versus the paper's argument. The paper estimates a panel with controls; you have drawn a cloud. What can your cloud show, and what can it not?
2. **The placebo.** Rebuild Sheet 3 with **ISO 9001** — the quality standard, demanded by buyers everywhere, with no environmental content. If the export-market pull is just as strong, what does that say about *why* firms certify to 14001?
3. **Break the weighting.** Replace the export-weighted average with an **unweighted** average of partners (`AVG` instead of the weighted ratio). Does the relationship survive? Why should weighting matter for the California-effect argument?
4. **Break the normalisation.** Raw counts instead of per $bn GDP on Sheet 1. Which countries jump to the top — and why is that misleading?

6.3.5 5 · The catch — 10 min

Original	Ours	The catch
Prakash and Potoski's finding	Your Sheet 3 (and Sheet 1)	What it does **not** license a multinational to conclude — for instance that a certified supplier is environmentally better, or that adoption *caused* export success

A member drawn at random presents in two minutes.

6.3.6 Hand in

`.twbx` + provenance log on ZoneCours. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

Team project. Which private standards do your firm's buyers or home market require? Which of its key countries are below the median adoption of its export markets? Note it for the dashboard.

Chapter 7

Measuring risk: content analysis, coding, big data and composite indices

Session 7 · How a political judgement becomes a number — and what is lost on the way Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **Every geopolitical risk score a multinational buys or builds is a chain of choices: which texts, which words, which coder, which weights. Can you see the chain — and defend each link?**

Big work	Hassan, T. A., Hollander, S., van Lent, L., & Tahoun, A. (2019). Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics*, 134(4), 2135–2202. doi:10.1093/qje/qjz021
Companion works	Baker, Bloom & Davis (2016), *Measuring Economic Policy Uncertainty*, QJE, doi:10.1093/qje/qjw024 · OECD/JRC (2008), *Handbook on Constructing Composite Indicators*, doi:10.1787/9789264043466-en · Krippendorff (2018), *Content Analysis*, 4th ed.
Target figures	(a) **Average firm-level political risk over time**, against the Economic Policy Uncertainty index · (b) **your own coded index**, with its intercoder agreement · (c) **a composite exposure index** for your team's firm, with weights the reader can move
Data you collect	firmlevelrisk.com firm-quarter file; EPU (global); 40 news headlines from GDELT that **your team codes**; GPR country indices and Worldwide Governance Indicators for the composite
Tableau skill	**Tableau Prep** flows (or the data source pane) · unions · cleaning text-coded data · **normalisation** · **parameters as weights**
Course themes	Methodology: content analysis, coding, big data, statistical treatment — the Synthesis outcome

7.1 Before the session

Four things to do before class. Budget **about two and a half hours**, and **revise Sessions 04–06**: Quiz 2 opens the session.

	Before class	Used in class for
1	Read the paper; skim two companions	the lecture and practice part (a)
2	Collect the data	practice parts (a) and (c)
3	Pull 40 headlines and **code them — two coders, independently**	practice part (b)

	Before class	Used in class for
4	Self-check	the lecture

7.1.1 1. The reading — 60–75 min

The big work. Hassan, T. A., Hollander, S., van Lent, L., & Tahoun, A. (2019). Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics*, 134(4), 2135–2202. doi:10.1093/qje/qjz021

	What to look for
1	The **question**: how much political risk does an individual firm face, and does it matter?
2	The **measurement**: what text, which word pairs, how are "political" bigrams identified, and how close to "risk" or "uncertainty" must they be?
3	The **validation**: what evidence shows that PRisk measures political risk and not something else?
4	The **finding** that surprised the authors most about where the variation lies — across time, sectors, or firms?

Sections I–III closely; skim the rest.

Skim, for the lecture:

- Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636. doi:10.1093/qje/qjw024 — Section II: how the index is built, and how they audited it **by hand**.
- OECD/JRC (2008). *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing. doi:10.1787/9789264043466-en — the ten-step checklist at the start of Part 1.

7.1.2 2. The data — 30 min

Details: `../data/README.md`. Split the work.

- `firmquarter_2022q1.csv` from firmlevelrisk.com
- `Global_Policy_Uncertainty_Data.xlsx` and `All_Country_Data.xlsx` from policyuncertainty.com
- `data_gpr_export.xls` again (revised since Session 01) and the **WGI** 2025 edition
- Your Session 04 ideal-point file

7.1.3 3. Your corpus, and coding it — 45 min

One member retrieves the headlines (GDELT query in `../data/README.md`), draws **40 at random**, and shares the coding sheet. **Two other members** code all 40 — **independently, without talking to each other** — using the protocol below. The fourth member (or the first, in a team of three) is the **adjudicator** in class.

7.1.3.1 The coding protocol

Code the **headline only**. Assign **one** code: the category that best describes the headline's main event. Categories follow Caldara and Iacoviello's eight, plus a zero.

Code	Category	Assign when the headline reports…
0	Not geopolitical	anything else — domestic politics, markets, company news, disasters
1	War threats	a threat or risk of war between states
2	Peace threats	a threat to a peace process, ceasefire or agreement
3	Military build-ups	troop movements, exercises, arms build-ups, new deployments
4	Nuclear threats	nuclear tests, weapons programmes, nuclear threats
5	Terror threats	a threat or risk of a terrorist attack
6	Beginning of war	the start of a war or armed conflict between states
7	Escalation of war	an existing war intensifying
8	Terror acts	a terrorist attack that happened

Decision rules.

1. **Acts beat threats**: if a headline reports both, code the act (6, 7 or 8).
2. **Sanctions, export controls and trade wars** are *not* in the eight categories: code **0**, and write `econ` in a comment column. (That gap is one of today's discussion points.)
3. **When in doubt, code 0** and write why in the comment column.
4. **Never revise** your code after seeing your partner's.

7.1.4 4. Self-check — 15 min

1. Name the three components of the EPU index's article filter.
2. Why do Hassan et al. look for political bigrams **near** words like "risk" and "uncertainty", rather than counting political words alone?
3. Two coders agree on 80% of items. Why might that still be poor agreement?
4. Name the steps in building a composite indicator. Which one do you think is most often skipped?

7.1.5 Arrive with

- ☐ Sessions 04–06 revised — **Quiz 2**
- ☐ The paper read; the two companions skimmed
- ☐ All files in `data/raw/`, logged
- ☐ `coded/coding_sheet.xlsx` with **two independent codings** of 40 headlines
- ☐ Your self-check, on paper

Session 07 overview · The lecture · The practice

7.2 The lecture

7.2.1 1. The measurement chain

Every number in this course — GPR, a capability share, an ideal point, a sanctions count — is the end of a chain:

```
concept  →  indicator  →  source  →  coding rule  →  aggregation  →  presentation
"geopolitical    "share of      ten          the eight       monthly share,    a line with
 risk"           articles…"     newspapers   categories      base 1985–2019    spikes labelled
```

Two questions apply to every link:

- **Validity** — does the indicator measure the concept? (Is newspaper attention geopolitical risk?)
- **Reliability** — would someone else, following the same rules, get the same number? (Would a second coder put the same article in the same category?)

A measure can be perfectly reliable and invalid — a thermometer read identically by everyone that measures the wrong room. Today is about seeing every link, so you can audit a score you are sold and defend one you build.

7.2.2 2. Content analysis

Krippendorff (2018) defines content analysis as a research technique for making **replicable and valid inferences** from texts to the contexts of their use. Its steps:

Step	In the GPR index	In your coding today
Unitising — what is one unit?	an article	a headline
Sampling — which units?	all articles in ten newspapers	40 headlines drawn at random from a GDELT query
Recording / coding — which category?	a dictionary of words	two human coders and a protocol
Reducing — how are codes summarised?	monthly share	counts and shares by category
Inferring — what do we conclude?	geopolitical risk is high	your firm's countries are in the news for …

Manifest vs latent content. A dictionary sees **manifest** content — the words on the page. Human coders can see **latent** content — what an article is *about*. Dictionaries are cheap, fast and perfectly reliable; humans are slow and less reliable but can catch meaning. The best measures use both: a dictionary, **audited by humans**.

7.2.3 3. Three dictionary methods

	GPR — Caldara & Iacoviello	**EPU** — Baker, Bloom & Davis	**PRisk** — Hassan et al.
Text	newspaper articles	newspaper articles	earnings-call transcripts
Unit counted	article with GPR words	article with **E**conomy **and** **P**olicy **and** **U**ncertainty terms	political two-word combinations (bigrams) near synonyms of risk or uncertainty
How the dictionary is chosen	the authors' eight categories	the authors' term sets, audited by hand	**learned**: bigrams frequent in political texts and rare in non-political ones
Scaled by	all articles	all articles	length of the transcript
Level	world, 44 countries	country	**firm × quarter**
Validation	narrative, hand audit	large human audit of articles	correlation with EPU; firms with high PRisk lobby and donate more; cut investment and hiring

What Hassan et al. add. Political risk becomes a **firm-level** variable — measured from what managers and analysts actually discuss. Their striking finding is that most of the variation in political risk is **across firms at a given time**, not common to all firms: aggregate indices like EPU, however useful, miss most of the political risk an individual firm faces.

7.2.4 4. Reliability: how much do coders agree?

Percent agreement — the share of items both coders put in the same category — is intuitive and misleading: two coders guessing at random will often agree, especially when one category dominates.

Cohen's kappa corrects for chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \qquad p_o = \frac{\text{agreements}}{N} \qquad p_e = \sum_k \frac{n_{A,k}}{N} \cdot \frac{n_{B,k}}{N}$$

where $n_{A,k}$ is the number of items coder A put in category k.

A worked example. 40 headlines, three collapsed categories: not geopolitical (0), threat (1–5), act (6–8).

	B: 0	**B: threat**	**B: act**	**A total**
A: 0	**20**	3	1	24
A: threat	2	**8**	0	10
A: act	0	1	**5**	6
B total	22	12	6	40

- $p_o = (20 + 8 + 5)/40 = 0.825$
- $p_e = (24 \cdot 22 + 10 \cdot 12 + 6 \cdot 6)/40^2 = 684/1600 = 0.4275$
- $\kappa = (0.825 - 0.4275)/(1 - 0.4275) \approx 0.69$

An 82.5% agreement becomes a kappa of about 0.69 once chance is removed. Conventions differ: Landis and Koch's widely used labels call 0.61–0.80 "substantial"; Krippendorff, for his alpha (the generalisation to more coders, missing data and other scales), asks for at least 0.800 for firm conclusions and accepts 0.667 only for tentative ones.

What to do with disagreement. Never average it away. Read each disagreement, decide whether the protocol was unclear or a coder erred, **revise the protocol**, and adjudicate. The disagreements are where the concept is fuzzy — they are information.

7.2.5 5. Big data: what it adds, what it hides

GDELT monitors news in many languages worldwide and turns it into searchable records. Its promise is **coverage and speed**: every country, every day, without a research assistant.

Its traps:

- **Selection.** Only what is reported, by sources GDELT monitors, in languages it processes. Coverage changes over time: a rising count may be more sources, not more events.
- **Duplication.** One wire story becomes hundreds of articles.
- **Machine coding.** Automated event coding misreads irony, quotation, and historical references.
- **"N = all" is never all.** A very large biased sample is still biased — and more confidently wrong.

The rule for a risk analyst: big data for **detection** (something is happening, here, now); small, carefully coded data for **measurement** (how much, compared with what).

7.2.6 6. Composite indicators — the OECD/JRC discipline

A composite indicator combines several indicators into one number — country-risk ratings, governance indices, your team's exposure index. The OECD/JRC *Handbook* (2008) sets out ten steps; the five that decide most of the outcome are:

Step	The choice	The risk
Theoretical framework	Which dimensions of risk, and why	An index of whatever data were available
Missing data	Drop, impute, or flag	Countries silently disappear — usually the risky ones
Normalisation	Min–max, z-score, rank, distance to a reference	Different methods, different rankings
Weighting and aggregation	Equal, statistical, expert; additive or geometric	Additive aggregation lets a strength **compensate** for a weakness
Sensitivity analysis	How much do results move when choices change?	A ranking presented as fact that one weight would overturn

Min–max normalisation puts every indicator on 0–1, with the direction made consistent (higher = more risk):

$$x^* = \frac{x - \min x}{\max x - \min x} \qquad \text{or, for an indicator where higher is safer,} \qquad x^* = \frac{\max x - x}{\max x - \min x}$$

Weighted additive aggregation: $CI = \dfrac{\sum_j w_j \, x_j^*}{\sum_j w_j}$.

The point for the board. Weights are value judgements. Show them; let the reader move them; and report whether the ranking survives. A composite that hides its weights is an opinion dressed as a measurement.

7.2.7 7. The Tableau skill of the day

- **Tableau Prep** (or the data source pane): parse a quarter string into a date; aggregate firm- quarters to quarters; union or join sources on a common key.
- A **highlight table** of coder A × coder B, with grand totals: the agreement matrix, read straight off the screen.
- **Normalisation with LOD**: `{ MIN([x]) }` and `{ MAX([x]) }` — a `FIXED` expression with no dimension computes over the whole table.
- **Parameters as weights**, and `RANK()` to show how the ranking moves when the weights do.

7.2.8 References

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Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174.

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7.3 In the practice

> **The targets.** (a) Hassan et al. (2019): **average firm-level political risk over time**, against the EPU index. (b) **Your own coded index** — 40 headlines, two coders, a kappa. (c) A **composite exposure index** for your team's firm, with weights the board can move — your team's draft.

Today has three short builds instead of one long one. Keep to the clock.

Part	Minutes	The five steps, compressed
(a) Firm-level risk vs EPU	30	collect · prepare · replicate · compare
(b) Your coded index	20	agreement · adjudicate · count
(c) A composite with moving weights	30	select · normalise · weight · test
The catch	10	one slide, three panels

Data: `../data/README.md`. Provenance log as always — one line per file, including your own `coding_sheet.xlsx` ("produced by the team, protocol of Session 07").

7.3.1 (a) Firm-level political risk against EPU — 30 min

7.3.1.1 Prepare — choose a route

The firm file is **firm × quarter**; EPU is **monthly**. Both must reach a common **quarter** grain.

The quarter identifier in the firm file is a string such as `2010q1` in the vintages we have seen — **check yours**. Convert it to a date:

```
[Quarter date] = MAKEDATE( INT(LEFT([date], 4)),
                           (INT(RIGHT([date], 1)) - 1) * 3 + 1,
                           1 )
```

And for EPU (`Year` and `Month` in the current file; `Year` may arrive as text because of the citation rows at the bottom, hence the `INT`):

```
[Quarter date] = DATETRUNC('quarter', MAKEDATE(INT([Year]), INT([Month]), 1))
```

Route 1 · Tableau Prep (if your licence includes it)

1. *Input* `firmquarter_2022q1.csv` → *Clean* step: add `[Quarter date]` → *Aggregate* step: group by `[Quarter date]`; `AVG(PRisk)`, `COUNTD([gvkey])`.
2. *Input* the EPU workbook → *Clean*: add `[Quarter date]`, drop rows with no value → *Aggregate*: group by `[Quarter date]`; `AVG([GEPU_current])` (or `GEPU_ppp` — log which).
3. *Join* the two on `[Quarter date]` (inner) → *Output* as `.hyper`. Open it in Tableau Desktop.
4. Every step is a visible, saved decision: export the flow (`.tfl`) with your hand-in.

Route 2 · the data source pane (Tableau Desktop or Tableau Public edition)

1. Connect to both files. Drag the firm file to the canvas, then the EPU sheet next to it.
2. In the relationship dialog, choose **Create Relationship Calculation** on each side and enter the two `[Quarter date]` formulas above.
3. Relationships aggregate each table at its own level: `AVG(PRisk)` is averaged over firms, `AVG(GEPU_current)` over the months of the quarter — exactly what you want.

7.3.1.2 Replicate — Sheet “Firm-level political risk and EPU”

1. `[Quarter date]` → Columns as continuous **Quarter**.
2. `AVG(PRisk)` → Rows; the EPU measure → Rows; **dual axis**, *not* synchronised (different units).
3. Annotate three peaks; check what happened in those quarters.
4. Title with the finding; caption with both sources, vintages and access dates.

7.3.1.3 Compare

- Do the two series move together? Where do they part company — and which one would you trust for a firm, and which for a country?
- Add `COUNTD` of firms as a third view. Does the number of firms change over the period? Why does that matter for an average?
- Filter `hqcountrycode` (2-letter codes) to exclude `US`. How many firms are left?

7.3.2 (b) Your coded index — 20 min

Connect to `coded/coding_sheet.xlsx`.

1. **The agreement matrix.** `coder_A` → Rows, `coder_B` → Columns, `CNT(id)` → Text and Colour: a **highlight table**. *Analysis → Totals → Show Row and Column Grand Totals.*
2. **Percent agreement** in Tableau:
   `[Agree] = IF [coder_A] = [coder_B] THEN 1 ELSE 0 END` → `AVG([Agree]) = p_o`
3. **Kappa, by hand.** From the grand totals, compute $p_e = \sum_k (n_{A,k}/N)(n_{B,k}/N)$ and $\kappa = (p_o - p_e)/(1 - p_e)$. Write both on your slide. If categories are sparse, also compute kappa on three collapsed categories (0 / threat / act) — and say which you report.
4. **Adjudicate.** The adjudicator reads every disagreement and fills `final`, with a one-line reason. List the **protocol changes** you would make.
5. **The index.** From `final`: the share of headlines in each category, and the share coded geopolitical ($\neq 0$) by month. Mark `econ` comments separately: how many headlines about sanctions or export controls did the GPR categories force into 0?

7.3.3 (c) A composite exposure index with moving weights — 30 min

7.3.3.1 Select

Start from a **spine**: a sheet listing your firm's key countries plus comparators — **8 to 12 countries**, with `iso3`. Every input is **left-joined** to it, so missing values are visible rather than silently dropped.

Indicator	Source	Direction	Preparation
Country GPR	`GPRC_xxx` columns	higher = riskier	pivot the `GPRC_` columns (not `GPRHC_`; values are % of articles); extract the code with `RIGHT([Pivot Field Names], 3)`; average the last 12 months
Country EPU	`All_Country_Data.xlsx`	higher = riskier	pivot country columns; add `iso3` (the columns are country names such as `UK`, `US`, `Korea`, and China appears three times); average the last 12 months
Political stability	WGI 2025, sheet `pv`, `Governance score (0–100)`	**higher = safer**	latest year; join on `Economy (code)`
Distance in UN voting from the firm's home country	Session 04 ideal points	higher = riskier	`ABS([Ideal point] - [Home ideal point])`, latest year

Each input must be one row per country before it is joined to the spine — otherwise the min and max below are taken over months, not countries. Reduce each input first: an *Aggregate* step in Tableau Prep, or — in Desktop — a small sheet (country × the averaged value) exported with *Worksheet → Export → Crosstab* into a derived file in `data/derived/`, logged as such. Raw files are never edited.

7.3.3.2 Normalise

Min–max across the spine countries, with a `FIXED` expression over the whole table:

```
[GPR n]   = ([GPR]   - {MIN([GPR])})   / ({MAX([GPR])}   - {MIN([GPR])})
[WGI n]   = ({MAX([WGI])} - [WGI])     / ({MAX([WGI])}   - {MIN([WGI])})      // inverted
```

…and the same for EPU and voting distance.

7.3.3.3 Weight

Create four **parameters**, `w GPR`, `w EPU`, `w WGI`, `w Vote` — integers 0 to 5, default 1 — and show their controls. Missing values must not count as zero risk, so reweight over the indicators that exist:

```
[Composite] =
  ( IF NOT ISNULL([GPR n])  THEN [w GPR]  * [GPR n]  ELSE 0 END
  + IF NOT ISNULL([EPU n])  THEN [w EPU]  * [EPU n]  ELSE 0 END
  + IF NOT ISNULL([WGI n])  THEN [w WGI]  * [WGI n]  ELSE 0 END
  + IF NOT ISNULL([Vote n]) THEN [w Vote] * [Vote n] ELSE 0 END )
  /
  ( IF NOT ISNULL([GPR n])  THEN [w GPR]  ELSE 0 END
  + IF NOT ISNULL([EPU n])  THEN [w EPU]  ELSE 0 END
  + IF NOT ISNULL([WGI n])  THEN [w WGI]  ELSE 0 END
  + IF NOT ISNULL([Vote n]) THEN [w Vote] ELSE 0 END )
```

That is a **decision**, not a fix: write it in your log ("composite renormalised over available indicators").

Sheet: countries on Rows, sorted by `SUM([Composite])`, bars, with `RANK(SUM([Composite]))` as a label.

7.3.3.4 Test

Record each country's rank under **equal weights**, then under each indicator **alone** (weight 1, others 0). In a small table: how many countries change rank by more than two places? Which country is most sensitive to the weights — and why?

7.3.4 The catch — 10 min

One slide, three panels:

(a)	(b)	(c)
PRisk vs EPU: what an average of US-listed firms does **not** tell your firm	Your kappa and the protocol change you would make	Your composite's ranking — and the weight that would overturn it

A member drawn at random presents in two minutes.

7.3.5 Hand in

`.twbx` (+ `.tfl` if you used Prep), `coding_sheet.xlsx` with `final` filled, and the provenance log, on ZoneCours. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper — for (a), or for the GPR categories meeting your headlines in (b).

Team project milestone. Part (c) is your **draft composite index**. Bring it, revised, to the dashboard review in Session 10.

Chapter 8

Cyber geopolitics and digital threats

Session 8 · Restraint, rivals and weaponised networks Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **Is cyberspace a new battlefield for great powers — or a new way for old rivals to hurt each other, with multinationals as collateral?**

Big work	Valeriano, B., & Maness, R. C. (2015). *Cyber War versus Cyber Realities: Cyber Conflict in the International System*. Oxford University Press. doi:10.1093/acprof:oso/9780190204792.001.0001
Companion	Farrell, H., & Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1), 42–79. doi:10.1162/isec_a_00351
Target figure	Incidents by year, method and severity — and the **initiator × target matrix** of rival dyads
Data you collect	Dyadic Cyber Incident and Campaign Data (DCID) v2.0, doi:10.7910/DVN/CQOMYV; EuRepoC Global Dataset v1.3, doi:10.5281/zenodo.14965395
Tableau skill	Highlight tables · dyad matrices (initiator × target) · filter and highlight actions
Course themes	The evolution of cyber geopolitics and digital threats in the decision-making architecture of multinational firms

8.1 Before the session

Four things to do before class. Budget **about two hours**.

	Before class	Used in class for
1	Read Valeriano & Maness (selected chapters) and Farrell & Newman	the lecture discussion and the replication
2	Download DCID v2.0 and EuRepoC v1.3	Practice round 08
3	**Submit the GSDB request form**	Practice round 10, in two weeks
4	Self-check	the lecture

8.1.1 1. The reading — 75 min

Valeriano, B., & Maness, R. C. (2015). *Cyber War versus Cyber Realities: Cyber Conflict in the International System*. Oxford University Press. doi:10.1093/acprof:oso/9780190204792.001.0001 — available through HEC's library (Oxford Academic). Read the introduction, the chapter presenting their theory of **cyber restraint**, and the chapter describing the incident data and its patterns.

Farrell, H., & Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1), 42–79. doi:10.1162/isec_a_00351 — read in full; you will use it again in Sessions 10 and 11.

Annotate four things:

	What to look for
1	Valeriano & Maness's **claim**: who uses cyber operations against whom, how severe they are, and why
2	How their **data** are built: which dyads enter, what counts as an incident, how severity is scored
3	Farrell & Newman's two mechanisms: the **panopticon** effect and the **chokepoint** effect
4	One reason a multinational with no quarrel with anyone might still be **hit**

Today's target figure is the book's descriptive picture of incidents: their count over time, by method and by severity, and who targets whom.

8.1.2 2. The data — 15 min

Download, untouched, into `08-cyber-geopolitics/data/raw/`. Details: `../data/README.md`.

- **DCID v2.0** — Harvard Dataverse doi:10.7910/DVN/CQOMYV, file `Master_DCID_2.0_Release_public_Sep2022.tab`, downloaded **as CSV** (*Access File → Comma Separated Values (Original File Format)*). The codebook is not on Dataverse: download *Codebook 2.0* from the authors' site, ryanmaness.com/cyber-conflict-dataset.
- **EuRepoC v1.3** — Zenodo doi:10.5281/zenodo.14965395, file `eurepoc_global_dataset_1_3.csv`, plus the codebook `eurepoc_codebook_1_2.pdf` (same record).

Open the DCID codebook and find: the unit of observation, the list of methods, and the severity scale.

8.1.3 3. For Session 10 — request the Global Sanctions Data Base now — 5 min

The GSDB is free but **distributed on request**. Go to globalsanctionsdatabase.com, fill in the request form with your HEC address, and state academic use in course INTE 60019A. The files arrive by e-mail, usually within a day — but a request made the night before Session 10 is a request made too late.

> The GSDB licence forbids redistribution and commercial use. You will keep the file on your machine and hand in workbooks on ZoneCours only.

8.1.4 4. Self-check — 15 min

1. Valeriano & Maness argue for **restraint**. Restraint by whom, and why would a state hold back a weapon it has?
2. What is the difference between Farrell & Newman's **panopticon** and **chokepoint** effects? Give one example of each that involves a private firm.
3. Why is **attribution** a problem for anyone building a dataset of state cyber operations?
4. Name one multinational that suffered losses in a cyber operation it was not the target of.

8.1.5 Arrive with

- ☐ Both readings annotated
- ☐ DCID v2.0 (CSV) and EuRepoC v1.3 in `data/raw/`, with codebooks
- ☐ **The GSDB request submitted**
- ☐ Your self-check answers, on paper

Session 08 overview · The lecture · The practice

8.2 The lecture

The session opens with **four Risk Briefs** (20 min). The lecture follows.

8.2.1 1. Is cyberspace a domain of geopolitics?

Two caricatures dominate public debate. In the first, cyberspace is a revolutionary battlefield: a "cyber Pearl Harbor" can switch off a country's grid, and small states or non-state actors can humble great powers. In the second, cyber operations are simply crime and espionage with better tools, of no strategic consequence.

Both are claims about **evidence**, and this session puts them to data. The question for a multinational is narrower and more practical: *through which channels does inter-state rivalry in cyberspace reach my operations, and how should my decision architecture absorb it?*

Three kinds of activity are easily confused:

	Purpose	Typical actor	Example of the type
Espionage	Steal information — state secrets or intellectual property	intelligence services, contractors	intrusion into a defence supplier's network
Disruption and degradation	Interrupt or damage systems	military and intelligence units, proxies	denial of service against a ministry; wiper malware
Crime	Extract money	criminal groups, sometimes tolerated or used by states	ransomware against a hospital or a pipeline operator

The boundaries blur — states tolerate, hire or shelter criminal groups — and that blurring is itself a geopolitical fact.

8.2.2 2. Valeriano and Maness: cyber war versus cyber realities

Valeriano and Maness (2015) built a dataset of cyber incidents and campaigns between **rival states** — pairs of states with a history of enduring hostility — and asked what the record actually shows. Their argument, in four propositions:

1. **Restraint.** States use cyber operations far less, and far less destructively, than the "cyber war" narrative predicts. The severity of most incidents is low: defacements, denial of service, espionage — rarely physical destruction.
2. **Regional rivals.** Cyber conflict mostly follows existing rivalries — the United States and China, Russia and its neighbours, Israel and Iran, India and Pakistan, the two Koreas. Cyber does not create new conflicts; it is a tool in old ones.
3. **Why restraint?** Fear of escalation and blowback, the risk of revealing and losing a capability once used, the possibility that a weapon is reverse-engineered and turned back, and the value of keeping networks intact to spy on them.
4. **Limited coercive effect.** Cyber operations rarely change the target's behaviour. They are better at signalling and spying than at compelling.

The **Dyadic Cyber Incident and Campaign Data (DCID)** extends that work; version 2.0 covers 2000–2020. It is today's replication data.

What restraint does not mean for the firm. Low *strategic* severity is perfectly compatible with high *commercial* cost. An operation that barely registers on a scale of interstate harm can still halt a company's production for weeks.

8.2.3 3. Farrell and Newman: weaponised interdependence

Farrell and Newman (2019) look at the networks that globalisation built — finance, communications, the internet's infrastructure — and observe that they are **not flat**. They have hubs. States that have jurisdiction over a hub can use it:

Effect	Mechanism	Example of the type
Panopticon	Use access to the hub to **see** what flows through it	intelligence access to data transiting cables, data centres or cloud providers under a state's jurisdiction
Chokepoint	Use control of the hub to **deny** access to others	cutting banks out of a financial-messaging network; denying firms access to software, chips or cloud services

For a multinational, the implication is uncomfortable: **the firm itself can be the hub**. Its cloud, its payment rails, its software updates are infrastructure that its home state can watch or weaponise — and that rival states therefore treat as a target. This is the bridge from cyber (today) to sanctions (Session 10) and space (Session 11).

8.2.4 4. The attribution problem

Every cyber dataset rests on **attribution**: someone has to say who did it. Attribution is

- **slow** — months or years after the event;
- **probabilistic** — technical indicators can be faked or reused;
- **political** — governments choose when to attribute publicly, and to whom;
- **selective** — what is attributed is what someone wanted known.

So event data on cyber operations is *a record of public attributions*, not a record of operations. That is the most important caveat of today's practice round — and the heart of its "catch".

8.2.5 5. When the firm is collateral: NotPetya and its lessons

In June 2017 the **NotPetya** malware, spread through a Ukrainian accounting-software update and widely attributed by Western governments to Russian military intelligence, spread far beyond its apparent target. Shipping, pharmaceutical, food and logistics multinationals reported very large losses — Maersk, Merck and Mondelez among them.

Three lessons for the firm:

1. **Proximity is technical, not political.** Firms were hit because they ran a piece of software in Ukraine, not because anyone chose them.
2. **Insurance is not a backstop.** Insurers invoked **war exclusions** to contest claims; the litigation that followed pushed the market to write explicit exclusions for state-backed cyber operations (Lloyd's required them from 2023). A firm's cyber cover may end exactly where geopolitics begins.
3. **The board owns it.** Securities regulators increasingly require disclosure of material cyber incidents and of board oversight of cyber risk.

8.2.6 6. Digital sovereignty and the splintering of the network

States increasingly regulate **where data sits and who may access it**: data-localisation rules, restrictions on cross-border data transfers, security reviews of foreign cloud and telecoms equipment, bans on particular vendors. For a multinational this is geopolitical risk in its architecture:

- **Duplication costs** — separate data centres or cloud regions per jurisdiction;
- **Conflicting obligations** — one state's law requires disclosure that another's forbids;
- **Vendor risk** — a supplier banned in one market may be mandatory in another.

8.2.7 7. Cyber risk in the decision architecture of the firm

Who	Role in geopolitical cyber risk
Board	Sets risk appetite; oversees disclosure; asks *which state actors would want our data or our downtime?*
CISO	Turns threat intelligence into controls; owns incident response
Chief risk officer	Integrates cyber with country, sanctions and supply-chain risk — the team-project view
Insurer	Prices the risk — and draws the line at state-backed operations
Regulators	Impose disclosure, localisation and vendor rules

The question to bring to the practice round: *if the dataset shows where interstate cyber conflict concentrates, how should a firm with operations in those dyads change its architecture?*

8.2.8 8. The Tableau skill of the day

A **highlight table** (rows × columns, colour = count) turns the incident list into an **initiator × target matrix**. A **filter action** on a dashboard lets a click on one cell filter the time series and the method breakdown on the other sheets. Step by step: `../02-practice/README.md`.

8.2.9 References

Farrell, H., & Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1), 42–79. https://doi.org/10.1162/isec_a_00351

Maness, R. C., Valeriano, B., Hedgecock, K., Macias, J. M., & Jensen, B. (2022). *Dyadic Cyber Incident and Campaign Data (DCID)*, version 2.0. Harvard Dataverse. https://doi.org/10.7910/DVN/CQOMYV

Valeriano, B., & Maness, R. C. (2015). *Cyber war versus cyber realities: Cyber conflict in the international system*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780190204792.001.0001

8.3 In the practice

> **The target.** Valeriano & Maness's descriptive picture of interstate cyber conflict: incidents **by year and method**, the **distribution of severity**, and the **initiator × target matrix** of rival dyads — rebuilt from DCID v2.0, then confronted with the wider EuRepoC record.

Work in your project teams. Everyone builds their own workbook; the team hands in one slide.

8.3.1 1 · Collect — 15 min

If you did the pre-session, both files are in `data/raw/`. If not:

- **DCID v2.0** — doi:10.7910/DVN/CQOMYV → file `Master_DCID_2.0_Release_public_Sep2022.tab` → *Access File → Comma Separated Values (Original File Format)*. The codebook is not on Dataverse: *Codebook 2.0* is on ryanmaness.com/cyber-conflict-dataset.
- **EuRepoC v1.3** — doi:10.5281/zenodo.14965395 → `eurepoc_global_dataset_1_3.csv` and `eurepoc_codebook_1_2.pdf`.

Provenance log — one line per file:

file	source	url	version / release	accessed	licence	notes
Master_DCID_2.0_Release_public_Sep2022.tab	Maness, Valeriano et al.	doi:10.7910/DVN/CQOMYV	V2.0, Sep 2022	*today*	CC0, cite	*original CSV behind the .tab; 433 rows*
eurepoc_global_dataset_1_3.csv	EuRepoC	doi:10.5281/zenodo.14965395	v1.3 (6 Mar 2025)	*today*	CC BY-NC 4.0	*3,414 rows*

8.3.2 2 · Prepare — 15 min

1. **Connect** → **Text file** → the DCID CSV. Look at the grid.
2. Find the columns (checked in the September 2022 file — see `../data/README.md`): `initiator`, `StateA`, `StateB`, `Dyadpair`, `interactionstartdate`, `method`, `severity`, `targettype`. **There is no target column**, and `StateA`/`StateB` are the dyad, not initiator and target. Write the exact names in your log.
3. `initiator` is a **COW code** (16 values: 2 US, 365 Russia, 710 China, 630 Iran, 731 North Korea, 770 Pakistan, 666 Israel, 750 India, 732 South Korea…). `Dyadpair` joins the COW codes of `StateA` and `StateB` (`2365` = 2 + 365; State B's code is always the last three digits). Build names for both sides:

```
[Initiator] = IF STR([initiator]) = RIGHT(STR([Dyadpair]), 3)
              THEN TRIM([StateB]) ELSE TRIM([StateA]) END
[Target]    = IF STR([initiator]) = RIGHT(STR([Dyadpair]), 3)
              THEN TRIM([StateA]) ELSE TRIM([StateB]) END
```

   Two incidents (313 and 335) name an initiator outside the dyad — check them by hand.
4. `method` and `targettype` are **codes**. Alias them from the codebook: `method` 1 vandalism, 2 denial of service, 3 network intrusion, 4 network infiltration (4.1–4.4 are infiltration sub-types — group them, or round down with `FLOOR([method])`); `targettype` 1 private / non-state, 2 government non-military, 3 government military.
5. `interactionstartdate` is month/day/year text (`2/1/2008`). If Tableau has not read it as a date, create `Start Year` = `YEAR(DATEPARSE("M/d/yyyy", [interactionstartdate]))`.
6. Record every recode in the log.

8.3.3 3 · Replicate — 25 min

8.3.3.1 Sheet 1 · "Incidents per year, by method"

1. `Start Year` → **Columns** (discrete is fine: one bar per year).
2. `COUNTD([Cyberincidentnum])` (or the table's record count) → **Rows**.
3. Method → **Colour**. Stacked bars.
4. Title states the finding. Caption: *Source: DCID v2.0 (Maness, Valeriano et al., 2022), accessed …*

8.3.3.2 Sheet 2 · "How severe?"

1. Severity → **Columns** (discrete), count → **Rows**. A bar chart of the distribution.
2. Add a **reference line** at the median severity (*Analytics* → *Median with Quartiles* or a constant you computed).
3. Annotate the highest-severity incidents: what were they?

8.3.3.3 Sheet 3 · "Who targets whom" — the dyad matrix

1. `[Initiator]` → **Rows**, `[Target]` → **Columns** (the calculated fields from step 2.3).
2. Count → **Colour**, then *Show Me* → *highlight table*. Add count to **Label**.
3. Sort initiators by total count (right-click the pill → *Sort* → *Field* → *Count* → *Descending*).
4. Which five cells hold most incidents? Do they match the book's regional rivals?

8.3.3.4 Dashboard · link them

1. *New Dashboard* → drag Sheet 3 on the left, Sheets 1 and 2 on the right.
2. *Dashboard → Actions → Add Action → Filter…* — source sheet: Sheet 3; target sheets: 1 and 2; run on **Select**; clearing the selection **shows all values**.
3. Click the US–China cell, then the Russia–Ukraine cell. Write down how their histories differ.

8.3.4 4 · Compare and break — 20 min

1. **Differ.** Put your Sheets 1–2 beside the book's descriptive figures. DCID v2.0 runs to 2020 (one incident starts in January 2021) and has been revised since 2015. Write one difference, and its likely cause.
2. **Break the population.** Load EuRepoC as a second data source. Rebuild Sheet 1 on it — first with **all** incidents, then filtered to incidents attributed to **state** actors (field `initiator_category`; read the codebook to decide which values — `State` only, or also `State affiliated actor`). EuRepoC's `start_date` is day.month.year text: `DATEPARSE("dd.MM.yyyy", [start_date])`. How many times larger is EuRepoC? What does the difference tell you about what DCID leaves out?
3. **Restraint, tested.** In DCID, what share of incidents sits in the lowest severity levels? (*Quick Table Calculation → Percent of Total* on Sheet 2.) Does the evidence support restraint *as of 2020*? What would you need to know about 2022 onward?
4. **Extend — the firm.** Using `targettype` = 1 (*private / non-state*), isolate incidents that hit non-government targets. Which dyads hit firms most? Name a sector you would worry about.

8.3.5 5 · The catch — 15 min

One slide (a dashboard exported as an image):

Original	Ours	The catch
The book's incident picture	Your dashboard	One sentence: what this matrix does **not** license a multinational to conclude

A hint: it is a matrix of *publicly attributed* incidents between *rivals*. A member drawn at random presents in two minutes.

8.3.6 Hand in

On ZoneCours before leaving: your `.twbx` and provenance log. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the book — a dyad that is missing, a method with a different label, a severity that shifted between versions.

8.3.7 What loses credit

- Comparing DCID with *all* of EuRepoC without saying so
- Empty cells read as “no cyber conflict” rather than “no rivalry in the data”
- Codes (1, 2, 3) left on the chart instead of aliased labels
- A catch about the book rather than about what the firm may conclude

Chapter 9

The redefinition of economic diplomacy

Session 9 · Two-level games, digital diplomacy, and embassies as export promotion Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **When states compete for markets, investment and technology, is the multinational the object of diplomacy, its instrument — or a diplomat in its own right?**

Big work	Rose, A. K. (2007). The foreign service and foreign trade: Embassies as export promotion. *The World Economy*, 30(1), 22–38. doi:10.1111/j.1467-9701.2007.00870.x
Companions	Putnam, R. D. (1988). Diplomacy and domestic politics: The logic of two-level games. *International Organization*, 42(3), 427–460. doi:10.1017/S0020818300027697 · Bjola, C., & Manor, I. (2018). Revisiting Putnam's two-level game theory in the digital age: Domestic digital diplomacy and the Iran nuclear deal. *Cambridge Review of International Affairs*, 31(1), 3–32. doi:10.1080/09557571.2018.1476836
Target figure	Bilateral **exports against the number of diplomatic posts** the exporter keeps in the partner, log scale, with a trend line
Data you collect	Lowy Institute Global Diplomacy Index (`DataDownload.xlsx`); IMF bilateral goods trade (IMTS); World Bank WDI GDP
Tableau skill	Log scales · trend lines, and what their R^2 and p-value do and do not say · building the zeros a post-level file omits
Course themes	The redefinition of economic diplomacy and the implications for multinational firms

9.1 Before the session

Budget **about two hours**.

	Before class	Used in class for
1	Read Rose (2007); Putnam (1988); Bjola & Manor (2018)	the lecture and the replication
2	Download the Lowy file, IMF exports and WDI GDP	Practice round 09
3	Self-check	the lecture

9.1.1 1. The reading — 80 min

Rose, A. K. (2007). The foreign service and foreign trade: Embassies as export promotion. *The World Economy*, 30(1), 22–38. doi:10.1111/j.1467-9701.2007.00870.x — read in full; a free working-paper version (NBER) circulates under the same title. **Its central relationship — exports against foreign posts — is today's target.**

Putnam, R. D. (1988). Diplomacy and domestic politics: The logic of two-level games. *International Organization*, 42(3), 427–460. doi:10.1017/S0020818300027697 — read the first half carefully: Level I, Level II, the **win-set**.

Bjola, C., & Manor, I. (2018). Revisiting Putnam's two-level game theory in the digital age: Domestic digital diplomacy and the Iran nuclear deal. *Cambridge Review of International Affairs*, 31(1), 3–32. doi:10.1080/09557571.2018.1476836

Annotate four things:

	What to look for
1	Rose's **claim**, his **data** (which exporters, which posts, which years) and the **size** of the effect he reports
2	The **reverse-causality** objection to Rose — and how he answers it
3	Putnam's **win-set**: what enlarges it, what shrinks it
4	What Bjola & Manor say **digital** tools change at Level II

9.1.2 2. The data — 20 min

Into `09-economic-diplomacy/data/raw/`. Details: `../data/README.md`.

- **Lowy Global Diplomacy Index** — `DataDownload.xlsx` from globaldiplomacyindex.lowyinstitute.org. Open the 2023 sheet and find the columns for sending country, host country and type of post.
- **IMF IMTS** — data.imf.org: goods exports, FOB, US$, 2023, for **Canada and at least four other** Lowy countries × all partners.
- **WDI GDP** — `NY.GDP.MKTP.CD`.

9.1.3 3. Self-check — 15 min

1. In Putnam's terms, why can a negotiator with a **small** win-set sometimes obtain a better deal?
2. Rose finds that countries export more to partners where they keep more posts. Give **two** explanations other than "embassies promote exports".
3. Name one instrument of **commercial diplomacy** other than an embassy, and one firm that has used it.
4. Investment screening (CFIUS in the US, the EU screening regulation, the *Investment Canada Act* national-security review) is economic diplomacy aimed **at firms**. Name one case you know.

9.1.4 Arrive with

- ☐ Three readings annotated
- ☐ `DataDownload.xlsx`, an IMF export file and WDI GDP in `data/raw/`
- ☐ Your self-check answers, on paper
- ☐ **The GSDB file received** (for Session 10) — if not, write to the database team today

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Session 09 overview · The lecture · The practice

9.2 The lecture

The session opens with **four Risk Briefs** (20 min). The lecture follows.

9.2.1 1. What economic diplomacy was, and what it has become

Economic diplomacy is the use of the instruments of foreign policy for economic ends — and of economic instruments for foreign-policy ends. In the post-1990 order (Session 02) it was mostly the first: negotiate trade agreements, open markets, promote exports, attract investment. Diplomats served firms.

Three shifts have redefined it:

Shift	What changed	Consequence for the firm
Securitisation	Trade, investment and technology are now treated as matters of national security	Firms face screening, export controls and "trusted vendor" rules from their *own* side
From promotion to coercion	The same networks used to promote exports are used to deny access (Session 08's chokepoints; Session 10's sanctions)	Home-state support can become home-state constraint overnight
Digitalisation	Diplomacy is conducted in public, on social media, aimed at domestic and foreign publics at once	Firms are drawn into public diplomatic quarrels — boycotts, hashtags, official denunciations

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9.2.2 2. Putnam: two-level games

Putnam (1988) models international negotiation as two simultaneous games:

- **Level I** — negotiators bargain with each other to reach a tentative agreement;
- **Level II** — each negotiator's domestic constituencies decide whether to **ratify** it.

The **win-set** is the set of Level I agreements that would be ratified at Level II. Three lessons:

1. **Larger win-sets make agreement more likely** — but leave a negotiator open to being pushed.
2. **A small win-set is a bargaining asset**: "my parliament will never accept that" is credible when it is true.
3. Negotiators play **both boards at once**: they try to shape the other side's Level II and to manage their own.

For the firm. Multinationals are Level II actors in *several* countries at once. An export industry that lobbies for a trade deal enlarges the win-set at home; a firm exposed abroad may be a hostage at the other side's Level II.

9.2.3 3. Bjola and Manor: the two-level game goes digital

Bjola and Manor (2018) revisit Putnam through the negotiations over the Iran nuclear agreement (2015). Their argument: **digital diplomacy** — ministries and leaders communicating directly with publics on social media — gives negotiators a new tool to act on Level II, **their own and the other side's**. Governments used digital channels to build domestic support for ratification and to reach the counterpart's public, reshaping win-sets during the negotiation itself.

Two implications for multinationals:

- **Speed.** Domestic opinion now moves within the negotiation, not after it — so the regulatory outcome a firm plans for can change faster.
- **Exposure.** A firm's brand can be pulled into a digital campaign by either government.

9.2.4 4. Rose: embassies as export promotion

Rose (2007) asks whether the foreign service earns its keep. He combines the number of embassies and consulates that a set of exporting countries keep abroad with their bilateral exports, in a gravity framework that controls for the size of and distance between partners. His finding: exports are **higher where a country keeps more posts**, and the association survives the standard gravity controls. He reports that an additional **consulate** is associated with a further rise in exports of several percent — read the size off his tables, and note which kind of post matters most.

Three objections, which the practice round tests:

Objection	The worry	How to probe it
Reverse causality	Countries open posts *where they already trade*	Do posts predict *future* trade growth, or only follow past trade?
Omitted variables	Large, rich, close partners get more posts *and* more trade	Divide trade by partner GDP; colour by distance or region
Measurement	A post is not a budget; an honorary consulate is not an embassy	Separate embassies from consulates

Rose takes reverse causality seriously and argues that the evidence is consistent with posts having an effect, but a cross-section cannot settle it. That is the catch of the day.

9.2.5 5. The firm and the state: bargaining, screening, and diplomacy by firms

Triangular diplomacy. Stopford and Strange (1991) argued that diplomacy is no longer only state–state: it is also **state–firm** (governments bargain with multinationals over investment) and **firm–firm** (alliances). Vernon's **obsolescing bargain** (1971) adds the dynamic: once a firm's capital is sunk, its bargaining power falls.

Investment screening is economic diplomacy aimed *at* firms: the US Committee on Foreign Investment (CFIUS), the EU framework regulation on FDI screening (Regulation 2019/452), and the national-security review under the *Investment Canada Act*. Screening has widened from defence to data, critical minerals, infrastructure and technology.

Commercial diplomacy — the toolbox:

Instrument	Example of the type
Embassies and consulates	trade commissioners, market intelligence, introductions
Trade missions and state visits	ministers travelling with business delegations
Export credit and insurance	export credit agencies covering political risk
Investment promotion agencies	incentives, aftercare
Treaties	investment treaties with dispute settlement

When to lean on the home state — and when that is itself a risk: in a dispute, home-state backing helps; in a confrontation between states, being seen as an instrument of one may make the firm a target of the other.

9.2.6 6. The Tableau skill of the day

A **log scale** (*Edit Axis → Logarithmic*) for variables that span orders of magnitude; a **trend line** (*Analytics → Trend Line*) and *Describe Trend Model* for its R^2 and p-value. Neither tells you the direction of causation. Step by step: `../02-practice/README.md`.

9.2.7 References

Bjola, C., & Manor, I. (2018). Revisiting Putnam's two-level game theory in the digital age: Domestic digital diplomacy and the Iran nuclear deal. *Cambridge Review of International Affairs*, 31(1), 3–32. https://doi.org/10.1080/09557571.2018.1476836

Putnam, R. D. (1988). Diplomacy and domestic politics: The logic of two-level games. *International Organization*, 42(3), 427–460. https://doi.org/10.1017/S0020818300027697

Rose, A. K. (2007). The foreign service and foreign trade: Embassies as export promotion. *The World Economy*, 30(1), 22–38. https://doi.org/10.1111/j.1467-9701.2007.00870.x

Stopford, J. M., & Strange, S. (1991). *Rival states, rival firms: Competition for world market shares*. Cambridge University Press.

Vernon, R. (1971). *Sovereignty at bay: The multinational spread of U.S. enterprises*. Basic Books.

9.3 In the practice

> **The target.** Rose (2007)'s central relationship: for each exporter–partner pair, **exports** against the **number of diplomatic posts** the exporter keeps in the partner — on log scales, with a trend line — rebuilt with the Lowy Global Diplomacy Index (2023 data) and IMF trade.

Work in your project teams. Everyone builds their own workbook; the team hands in one slide.

9.3.1 1 · Collect — 15 min

In `data/raw/` (see `../data/README.md`):

- `DataDownload.xlsx` — Lowy Global Diplomacy Index;
- IMF IMTS — **exports of goods, FOB, US$, 2023**, for Canada and at least four other Lowy countries × all partners;
- WDI `NY.GDP.MKTP.CD` — GDP, current US$.

Provenance log — one line per file, including **which side** of the trade flow you used (exporter- reported exports) and the IMF indicator name exactly as shown.

9.3.2 2 · Prepare — 25 min

This is the hard part today. Take your time.

1. **Lowy.** Connect to `DataDownload.xlsx`, sheet **2023**. The columns for sending country, host country and post type are `COUNTRY`, `POST COUNTRY` and `TYPE OF POST`. Filter out posts that are not bilateral: `TYPE OF POST` = `Permanent Mission` (missions to the UN, EU, OECD…). Look at what `Other` holds (representative and trade offices, e.g. in Taipei) and log your choice.
2. **IMF.** Add the IMF file as a **second table** on the canvas. Keep only the five exporters and the year 2023; drop aggregate partners ("World", regional totals — codes starting with `G` or `TX`). Rename `OBS_VALUE` → `Exports`.
3. **Concordance.** Lowy has country names only; the IMF file has ISO3 codes (`COUNTRY`, `COUNTERPART_COUNTRY`). Make a small sheet `name → ISO3` for the Lowy `COUNTRY` and `POST COUNTRY` names (watch the typos listed in the data README) and add ISO3 to Lowy.
4. **Relate, don't join.** Drag the IMF table first, then the Lowy table, and create a **relationship** on `exporter ISO3 = sending ISO3` **and** `partner ISO3 = host ISO3`. A relationship keeps the export totals at their own grain; a plain join would repeat each export value once per post and inflate it.
5. **Build the zeros.** Calculated field `Posts`:

   ```
   ZN(COUNT([POST CITY]))
   ```

   (`POST CITY` is never empty in the 2023 sheet). Partners with exports but no post now get 0.
6. **Make logging possible.** Calculated field `Posts + 1` = `[Posts] + 1`. Write this decision in the log: it is the same compromise most gravity papers make, and it has consequences.
7. **Exports per partner GDP.** Add WDI GDP (relationship on partner ISO3, year 2023) and create `Exports / partner GDP (‰)` = `SUM([Exports]) / SUM([GDP]) * 1000`.

9.3.3 3 · Replicate — 20 min

9.3.3.1 Sheet 1 · "More posts, more exports"

1. `Posts + 1` → **Columns** (as a measure; aggregation `SUM` at the dyad level).
2. `Exports` → **Rows**.
3. Exporter ISO3 **and** partner ISO3 → **Detail** — one mark per pair.
4. Exporter → **Colour**.
5. Right-click each axis → *Edit Axis* → **Logarithmic**.
6. *Analytics* → **Trend Line** → *Linear*. Right-click the line → **Describe Trend Model**. Note the slope, R^2 and p-value (one line per exporter if colour splits them — try both ways).
7. Title states the finding; caption cites Lowy (2024 edition, 2023 data) and IMF IMTS, with access dates.

9.3.4 4 · Compare and break — 20 min

1. **Differ.** Rose used an earlier period and a different set of exporters. Is your slope steeper or flatter than the relationship he reports? Name one reason.
2. **Break: size.** Replace `Exports` by `Exports / partner GDP (‰)`. What happens to the slope and the R^2? What did the raw chart owe to the sheer size of partners?
3. **Break: type of post.** Filter `TYPE OF POST` to **`Embassy / High Commission`** only, then to **`Consulate / Consulate-General`** only. Which carries the relationship? Does that match Rose?
4. **Break: the zeros.** Exclude partners with `Posts = 0`. How much does the trend line move? Why should a careful analyst never drop them silently?
5. **Extend — reverse causality** (fast teams). Add the Lowy **2017** sheet and IMF exports for 2017. For each pair, compute the change in posts and the growth of exports 2017→2023. Do pairs that **gained** posts show faster export growth — or did posts open where exports had already grown? What would you need to tell the two apart?

9.3.5 5 · The catch — 10 min

Original	Ours	The catch
Rose's posts–exports relationship	Your Sheet 1	One sentence: what this trend line does **not** license a multinational — or a foreign ministry — to conclude

A member drawn at random presents in two minutes.

9.3.6 Hand in

On ZoneCours before leaving: `.twbx` and provenance log, including your concordance fixes. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

9.3.7 What loses credit

- A **join** that inflated exports, unnoticed (check one pair's total against the IMF file)
- Zeros dropped without saying so
- "$R^2 = 0.6$, so embassies explain 60% of exports" — they explain nothing; they *co-vary*
- Unmatched countries silently missing from the chart

Chapter 10

Economic sanctions

Session 10 · Do they work — and what do they do to the firms caught between sender and target? Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **Sanctions have become the instrument of first resort. Are they an efficient mechanism for regulating geopolitical risk — or a source of it?**

Big works	Hufbauer, G. C., Schott, J. J., Elliott, K. A., & Oegg, B. (2007). *Economic Sanctions Reconsidered* (3rd ed.). Peterson Institute for International Economics. · Felbermayr, G., Kirilakha, A., Syropoulos, C., Yalcin, E., & Yotov, Y. V. (2020). The Global Sanctions Data Base. *European Economic Review*, 129, 103561. doi:10.1016/j.euroecorev.2020.103561
The critique	Pape, R. A. (1997). Why economic sanctions do not work. *International Security*, 22(2), 90–136. doi:10.1162/isec.22.2.90
Target figure	**Active sanctions cases per year, by type**, 1950–today — and the **share judged successful**, by objective
Data you collect	Global Sanctions Data Base, Release 5 (by request); TIES v4.0 (Morgan, Bapat & Kobayashi, 2014)
Tableau skill	Year scaffolds and range joins · Gantt charts for episodes · stacked percent bars · quick table calculations
Course themes	The efficiency of economic sanctions as mechanisms for regulating geopolitical risks

10.1 Before the session

Budget **about two hours**, plus revision for **Quiz 3**, which opens the session.

	Before class	Used in class for
1	Read the GSDB article, Pape (1997), and Hufbauer et al. (2007), chapters as set	the lecture and the replication
2	Check the GSDB files arrived; download TIES; make the year scaffold	Practice round 10
3	Revise Sessions 07–09	Quiz 3
4	Team: bring your dashboard skeleton	the last 20 minutes of the practice round

10.1.1 1. The reading — 80 min

Felbermayr, G., Kirilakha, A., Syropoulos, C., Yalcin, E., & Yotov, Y. V. (2020). The Global Sanctions Data Base. *European Economic Review*, 129, 103561. doi:10.1016/j.euroecorev.2020.103561 — read in full. **Its descriptive figures — cases over time by type, and outcomes by objective — are today's target.**

Hufbauer, G. C., Schott, J. J., Elliott, K. A., & Oegg, B. (2007). *Economic Sanctions Reconsidered* (3rd ed.). Peterson Institute for International Economics. — chapter 1 and the chapter on how success is assessed (library reserve).

Pape, R. A. (1997). Why economic sanctions do not work. *International Security*, 22(2), 90–136. doi:10.1162/isec.22.2.90

Annotate four things:

	What to look for
1	How Hufbauer et al. **code success** — the two components they combine
2	Pape's **critique**: why he believes most of their "successes" are not
3	How the GSDB is **built**: what a case is, the types, the objectives, the outcomes
4	One way in which sanctions became a **risk for firms** that were not their target

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10.1.2 2. The data — 15 min

Into `10-economic-sanctions/data/raw/`. Details: `../data/README.md`.

- **GSDB Release 5** — the files you requested after Session 08. Open the codebook; find the type, objective and outcome variables. **If they have not arrived, tell the instructor before class.**
- **TIES v4.0** — sanctions.web.unc.edu: case file and user manual.
- **`years.csv`** — one column `Year`, 1945 to 2025.

10.1.3 3. Quiz 3 — revise Sessions 07–09

Twenty minutes, closed book. Concepts (content analysis, coding, composite indices; cyber restraint, weaponised interdependence, attribution; two-level games, digital diplomacy, commercial diplomacy), one big work, one unseen Tableau chart. Specimen: `assessments/quizzes/`.

10.1.4 4. Self-check — 15 min

1. Distinguish **comprehensive** from **targeted** ("smart") sanctions, with one example of each.
2. Why might the **threat** of sanctions succeed more often than sanctions actually imposed?
3. What are **secondary sanctions**, and why do they matter to a firm headquartered in a third country?
4. A firm leaves a sanctioned market although the law does not require it. Name two reasons.

10.1.5 Arrive with

- ☐ Three readings annotated
- ☐ GSDB files, TIES, codebooks and `years.csv` in `data/raw/`
- ☐ Revised for Quiz 3
- ☐ Team: the dashboard skeleton, on one laptop

Session 10 overview · The lecture · The practice

10.2 The lecture

The session opens with **Quiz 3** (20 min) and **four Risk Briefs** (20 min). The lecture is therefore shorter today.

10.2.1 1. The instrument of first resort

Economic sanctions are restrictions on economic relations imposed by one or more **senders** on a **target** — a state, a government, a firm or an individual — to change its behaviour, weaken it, or signal disapproval. Their use has grown markedly since the end of the Cold War; the Global Sanctions Data Base documents the rise and its changing composition.

Dimension	Categories
Sender	unilateral (one state) · plurilateral (the EU, a coalition) · multilateral (the UN Security Council)
Type	trade · financial · travel · arms · military assistance · other — the GSDB's classification
Scope	**comprehensive** (a whole economy) vs **targeted** or "smart" (named persons, firms, sectors)
Objective	change a policy · destabilise a regime · end a war or a territorial conflict · prevent war · fight terrorism · human rights · democracy

---

10.2.2 2. Do they work? The great debate

Hufbauer, Schott, Elliott and Oegg (2007) — the third edition of a study first published in 1985 — code each episode on two components: how far the **policy outcome** sought was achieved, and how far **sanctions contributed** to it. Combining the two, they judge **roughly a third** of cases at least partly successful, with success more likely when goals are modest, the target is small, weak and friendly to the sender, and sanctions are imposed quickly and decisively.

Pape (1997) re-examined the cases counted as successes and argued that most do not survive scrutiny: in many, the outcome was achieved by **force**, was not really conceded, or cannot be attributed to sanctions. His conclusion: economic sanctions rarely achieve **major** political goals. Elliott (1998) replied that the glass is half full rather than empty — the debate turns on *coding*.

The selection effect (Drezner, 2003). If targets that would concede do so when sanctions are only **threatened**, then the cases we observe as *imposed* are the hard ones. Imposed sanctions will look ineffective even if the instrument works. This is why **TIES** records threats as well as impositions — and why the practice round compares the two.

The lesson for an analyst. "Do sanctions work?" has no answer independent of how a case, an objective and a success are coded. Every number in this debate is a coding decision.

10.2.3 3. From comprehensive to smart sanctions

The humanitarian cost of the comprehensive sanctions on Iraq in the 1990s pushed senders towards **targeted** measures: asset freezes and travel bans on named individuals, sectoral restrictions, financial measures. Farrell and Newman (Session 08) explain why **financial** sanctions became the weapon of choice: the dollar-clearing system and financial-messaging networks are **chokepoints** under the jurisdiction of a few senders.

Targeted does not mean contained. Sanctions on banks, insurers and shippers propagate to every firm that depends on them.

10.2.4 4. The firm in the middle

The compliance perimeter. A firm must screen customers, suppliers, banks and vessels against sanctions lists — in the US, the Treasury's OFAC lists; in the EU, Council regulations; in Canada, regulations under the *Special Economic Measures Act* and related statutes. Ownership rules extend listings to entities owned by listed persons, so the perimeter is wider than the list.

Secondary sanctions threaten third-country firms with penalties — loss of access to the sender's market or financial system — for dealing with the target. They make a sender's foreign policy binding on firms that are not its nationals.

Overcompliance and de-risking. Because penalties are severe and the rules complex, banks and firms often withdraw from whole markets or sectors the law does not close. The *effective* scope of sanctions is set by compliance departments as much as by governments.

The exit decision: Russia 2022. Following the full-scale invasion of Ukraine, Western sanctions and public pressure led many multinationals to suspend, scale back or leave Russian operations. Exits were complicated by countermeasures — approval requirements and discounts on sales of foreign assets, and in some cases temporary state management of foreign-owned firms. The firm's options:

Option	Cost	Risk
Stay	reputational, compliance burden	new sanctions; asset seizure; home-state pressure
Suspend	idle assets	write-off if the situation persists
Sell	forced discount, approval delays	buyer later sanctioned; buy-back clauses
Walk away	full write-off	—

10.2.5 5. What a risk dashboard should track

1. **Exposure** — the firm's revenues, assets and suppliers in sanctioned or at-risk jurisdictions;
2. **Pipeline** — threats and announced measures (the TIES lesson);
3. **Propagation** — banks, insurers, logistics providers whose own exposure would cut the firm off;
4. **Counter-measures** — what the target can do to foreign firms on its soil.

10.2.6 6. The Tableau skill of the day

A **year scaffold** joined with a *range* condition (`Year ≥ begin_year` and `Year ≤ end_year`, in the GSDB's column names) turns episodes into active cases per year; a **Gantt chart** draws each episode as a bar from start to end. Step by step: `../02-practice/README.md`.

10.2.7 References

Drezner, D. W. (2003). The hidden hand of economic coercion. *International Organization*, 57(3), 643–659.

Elliott, K. A. (1998). The sanctions glass: Half full or completely empty? *International Security*, 23(1), 50–65.

Felbermayr, G., Kirilakha, A., Syropoulos, C., Yalcin, E., & Yotov, Y. V. (2020). The Global Sanctions Data Base. *European Economic Review*, 129, 103561. https://doi.org/10.1016/j.euroecorev.2020.103561

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Pape, R. A. (1997). Why economic sanctions do not work. *International Security*, 22(2), 90–136. https://doi.org/10.1162/isec.22.2.90

10.3 In the practice

> **The target.** The Global Sanctions Data Base's descriptive picture (Felbermayr et al., 2020): **active sanctions cases per year, by type**, from 1950 to today, and the **distribution of outcomes by objective** — rebuilt from Release 5, then confronted with the **threats** in TIES.

Work in your project teams. The last **20 minutes** are the team dashboard skeleton review.

> **Licence.** GSDB data may not be redistributed. Hand in on **ZoneCours only**. Do not publish this workbook to Tableau Public.

10.3.1 1 · Collect — 5 min

Everything should already be in `data/raw/`: the GSDB files as e-mailed, TIES v4.0, both codebooks, and `years.csv` (see `../data/README.md`). Fill the provenance log — for the GSDB, the **release number** and the **date of the e-mail**.

10.3.2 2 · Prepare — 20 min

1. **Connect → Text file / Excel →** the GSDB **case-level** file. Confirm in the codebook, and log, the names used below: `case_id`, `sanctioned_state`, `sanctioning_state`, `begin_year`, `end_year`, the six **type** columns, `objective`, `success`.
2. **Pivot the types.** The six types are separate 0/1 columns (`trade`, `financial`, `travel`, `arms`, `military`, `other`): select them in the grid → right-click → **Pivot**. Rename `Pivot Field Names` → `Type`, `Pivot Field Values` → `Type flag`. Filter later on `Type flag = 1`.
3. **Build the years.** Add `years.csv` to the canvas and **join** it to the GSDB table with two clauses:
   - `Year` >= `begin_year`
   - `Year` <= *join calculation* `IFNULL([end_year], 2025)` (click the right-hand field of a clause → *Create Join Calculation*). Each case now has one row per active year — and, after the pivot, one per type. (In the R5 file ongoing cases already carry `end_year` = 2025; the `IFNULL` is a guard for releases that leave it empty.)
4. **Check the grain.** Note the row count before and after. Write in the log what one row now is.
5. **Outcomes.** Outcomes are **not** indicator columns: `success` is one text field holding one code per objective, in the order of `objective` (e.g. `objective` = `human_rights,democracy`, `success` = `failed,ongoing`). Take the first objective and its outcome:

```
Objective  =  TRIM(SPLIT([objective], ",", 1))
Outcome    =  CASE TRIM(SPLIT([success], ",", 1))
              WHEN "success_total"   THEN "Success"
              WHEN "success_part"    THEN "Partial success"
              WHEN "nego_settlement" THEN "Negotiated settlement"
              WHEN "failed"          THEN "Failure"
              WHEN "ongoing"         THEN "Ongoing"
              ELSE "Not coded" END
```

   Log the decision: cases with several objectives count here under their **first** one only.

10.3.3 3 · Replicate — 25 min

10.3.3.1 Sheet 1 · "Active sanctions cases per year, by type"

1. `Year` → **Columns** (continuous).
2. `COUNTD([case_id])` → **Rows** — *count distinct*, never `SUM`, because the pivot and the scaffold multiply rows.
3. Filter `Type flag = 1`; `Type` → **Colour**; mark type **Area** or **Bar**.
4. Note in a caption: a case with several types is counted once **in each** type, so the stacked total exceeds the number of cases. Then build a version with no colour to show the true total.
5. Annotate 1990–91, 2014 and 2022. What explains the shape?

10.3.3.2 Sheet 2 · "How do cases end, by objective?"

1. Use the **case-level** grain: duplicate the data source **without** the year scaffold, or use `{FIXED [case_id] : MIN([Outcome])}`.
2. `Objective` → **Rows**, `COUNTD([case_id])` → **Columns**, `Outcome` → **Colour**.
3. *Quick Table Calculation → Percent of Total*; *Edit Table Calculation → Compute using* `Outcome` so each objective sums to 100%.
4. Exclude ongoing cases from the denominator in a second version. How much does the success share move? (That is a coding decision too.)

10.3.3.3 Sheet 3 · "One target's history" — a Gantt chart

1. Filter `sanctioned_state` to **one** target (Iran, Russia, Cuba, North Korea…) — with a calculated filter `CONTAINS([sanctioned_state], "Iran")`, because some cases list several targets.
2. `begin_year` → **Columns** (continuous); `case_id` (or `sanctioning_state` + `case_id`) → **Rows**.
3. Mark type **Gantt Bar**; calculated field `Duration` = `IFNULL([end_year], 2025) - [begin_year]` → **Size**.
4. `Type` → **Colour**. What does the target's history look like — layers added, never removed?

10.3.4 4 · Compare and break — 10 min

1. **Differ.** Put Sheet 1 beside the GSDB paper's figure. Release 5 runs years further and has added and recoded cases: write one difference.
2. **Threats vs impositions.** Open TIES. Using the manual, separate **threatened-only** cases (`imposition` = 0) from **imposed** cases (`imposition` = 1) and count each per `startyear` (1945–2005). Put the counts beside GSDB impositions for the same years. What does the gap say about Drezner's selection effect?
3. **Extend — the firm.** For your team's firm, list its top five countries. How many active GSDB cases name each as the sanctioned state in the latest year?

10.3.5 5 · The catch — 10 min

Original	Ours	The catch
The GSDB paper's figure	Your Sheet 1 (and Sheet 2)	One sentence: what these charts do **not** license a multinational to conclude about *its* sanctions exposure

A member drawn at random presents in two minutes.

10.3.6 6 · Team dashboard skeleton review — 20 min

Have your team-project workbook open. The instructor checks: four datasets from three indicator families; the composite index; at least one channel view; the watchlist with thresholds. You leave with three written comments.

10.3.7 Hand in

On **ZoneCours only**: `.twbx` and provenance log. Formative; counts towards participation.

The one rule: your log records at least one place where the data disagreed with the paper.

10.3.8 What loses credit

- `SUM` of anything after the scaffold join
- Stacked type totals presented as the number of cases
- Success shares without saying whether ongoing cases are in the denominator
- The GSDB workbook published on Tableau Public

Chapter 11

Space geopolitics

Session 11 · The orbital commons and the new space race Pre-session slides · Lecture slides · Practice slides · Session folder · Data instructions

> **Satellites now carry the position, timing, connectivity and imagery a multinational runs on. When orbit is crowded, contested and dominated by one operator, who holds the risk — and what can the firm do about it?**

Big works	ESA Space Debris Office. (2026). *ESA's annual space environment report* (Issue 10.0, 1 May 2026; GEN-DB-LOG-00288-OPS-SD). European Space Agency. PDF · Samson, V., & Brett, K. (Eds.). (2026). *Global counterspace capabilities: An open source assessment*. Secure World Foundation. swfound.org
Companion readings	Pekkanen, S. M. (2019). Governing the new space race. *AJIL Unbound*, 113, 92–97. doi:10.1017/aju.2019.16 · The Outer Space Treaty (1967) · Farrell & Newman (2019), from Session 08
Target figure	ESA's **evolution of the number of objects in geocentric orbit by object class**, 1957–2025 (Fig. 2.1), with the 2007 Chinese ASAT test, the 2009 Iridium–Cosmos collision and the 2021 Russian ASAT test annotated — and the **concentration** of who puts satellites up, then and now
Data you collect	CelesTrak **SATCAT** (`satcat.csv`, the public satellite catalogue); optionally Jonathan McDowell's **GCAT** launch log
Tableau skill	**Running sum** (table calculation) on a pivoted event table · **Herfindahl–Hirschman index with LOD expressions** · **dashboard actions** across data sources
Course themes	Inter-state anarchy and the balance of the major powers; cyber geopolitics and digital threats in the decision-making architecture of multinational firms

11.1 Before the session

> **Satellites now carry the position, timing, connectivity and imagery a multinational runs on. When orbit is crowded, contested and dominated by one operator, who holds the risk?**

Four things to do before class. Budget **about two hours**. **The download in step 3 is done once per team, not once per student** — CelesTrak blocks addresses that download the same file too often.

	Before class	Used in class for
1	Read the ESA report (selected pages)	the target figure
2	Read the SWF executive summary, Pekkanen and the Outer Space Treaty	the lecture discussion and the catch
3	One teammate downloads `satcat.csv` and shares it; everyone opens it	Practice round 11
4	Self-check	the lecture

11.1.1 1. The big work — 45 min

ESA Space Debris Office. (2026). ***ESA's annual space environment report*** **(Issue 10.0, 1 May 2026; GEN-DB-LOG-00288-OPS-SD). European Space Agency.** PDF · free

Read the **executive summary** (pp. 3–11), **section 1.1–1.2** on definitions and data sources (pp. 13–16), **section 2.1** (pp. 23–24) and **Table 3.1** (p. 63). Annotate four things:

	What to look for
1	What counts as an **object** — catalogued, asserted, and the nine object classes (Table 1.1)
2	Where the data **come from** — which catalogues, and what is beyond the report's scope
3	The **strongest single piece of evidence** that the environment is getting worse
4	One **limitation** an insurer or a satellite-service buyer would raise

Find **Figure 2.1** — *evolution of number of objects in geocentric orbit by object class* (the same chart opens the executive summary as Figure 1). It is today's target. Mark where you expect the 2007, 2009 and 2021 events to show.

11.1.2 2. The companion readings — 45 min

Samson, V., & Brett, K. (Eds.). (2026). ***Global counterspace capabilities: An open source assessment.*** **Secure World Foundation.** — the **executive summary** only (24 pages; free on swfound.org). Annotate: (a) the five categories of counterspace capability; (b) which of them are used in current military operations; (c) one capability that would hit a commercial operator first.

Pekkanen, S. M. (2019). Governing the new space race. ***AJIL Unbound*****, 113, 92–97.** doi:10.1017/aju.2019.16 · open access. Annotate her three trends and the share of space technologies she calls dual-use.

The Outer Space Treaty (1967) — *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies*, on the UNOOSA site. Read Articles I, II, IV, VI, VII and IX — two pages.

11.1.3 3. The data — 15 min

Details: `../data/README.md`.

1. **One teammate** downloads `satcat.csv` from celestrak.org/pub/satcat.csv into `data/raw/` and shares it with the team. Note the date.
2. Everyone opens it in a spreadsheet. Find the rows whose `OBJECT_NAME` is `FENGYUN 1C DEB`. What is their `LAUNCH_DATE`? When did the event that created them happen? **Write down the gap: it is the first trap of the practice round.**
3. Optional, for the extension: GCAT's `launch.tsv` and `orgs.tsv`.

11.1.4 4. Self-check — 15 min

Answer on paper. If you cannot, that is what the lecture is for.

1. Which article of the Outer Space Treaty makes orbit a **commons**, and which one makes a state answer for a **private** company's satellites?
2. What is the **Kessler** mechanism, and why does it make one destructive ASAT test a problem for every operator in that orbit, not only for the target?
3. According to the SWF, are **destructive** or **non-destructive** counterspace capabilities being used in current wars? Why does that matter more to a firm than to a defence ministry?
4. One owner has 75% of active satellites, the next two 9% and 4%, and the rest is spread thinly over a hundred owners. Roughly what is the **HHI**? Is it "highly concentrated"?
5. Name one service your team's firm buys that would stop, or degrade, without a satellite.

11.1.5 Arrive with

- ☐ ESA report annotated; Figure 2.1 located; the three events marked on it
- ☐ SWF executive summary, Pekkanen and the Treaty articles annotated
- ☐ `satcat.csv` in `data/raw/` (one download per team), the date noted
- ☐ The `FENGYUN 1C DEB` gap written down
- ☐ Provenance log started
- ☐ Your self-check answers, on paper

Session 11 overview · The lecture · The practice

11.2 The lecture

The session opens with **four Risk Briefs** (20 min). The lecture follows.

11.2.1 1. Infrastructure in orbit

Few multinationals own a satellite. Almost all of them depend on one. Three services carry most of that dependence:

Service	What the firm uses it for	Who provides it
Positioning, navigation and timing (PNT)	fleet and container tracking, aviation and shipping, precision agriculture — and above all **time**: telecom networks, power grids and trading venues synchronise their clocks to GNSS signals	four global systems, each run by a state: **GPS** (United States), **Galileo** (European Union), **BeiDou** (China), **GLONASS** (Russia)
Connectivity	ships, aircraft, mines, rigs and remote sites; back-up links for networks on the ground	geostationary operators, and since 2019 **constellations** of thousands of satellites in low Earth orbit
Earth observation	crop and commodity forecasts, insurance claims, supply-chain and site monitoring, due diligence	state agencies and a growing number of commercial imaging firms

Sessions 08 and 10 showed how networks become instruments of power. Today's network is physically above the firm, and its hubs are a handful of states and, increasingly, a handful of companies. The scale of the change is recent: ESA's 2026 report notes that the exponential growth in the number of new payloads continued in 2025, driven by a higher launch frequency, rideshare launches and the deployment of large constellations by commercial operators (ESA Space Debris Office, 2026).

11.2.2 2. The orbital commons: law and physics

The law. The **Outer Space Treaty** (1967) is the constitution of the domain, and it was written for two superpowers, not for a market:

Article	Rule	What it means for a firm
II	outer space "is not subject to national appropriation" by claim of sovereignty, use or occupation	no one can own an orbit — but whoever occupies it first, in practice, keeps it
IV	no nuclear weapons or other weapons of mass destruction in orbit; the Moon and celestial bodies for peaceful purposes only	**conventional** weapons in orbit and anti-satellite tests are **not** prohibited by the Treaty
VI	states bear international responsibility for national activities, "whether … carried on by governmental agencies or by non-governmental entities", which require "authorization and continuing supervision"	every commercial satellite has a **state behind it** — its licensing state
VII	the launching state is liable for damage caused by its objects	debris has an owner, in law, long after it has stopped working
IX	"due regard" to the interests of others; consultations before potentially harmful interference	the closest the Treaty comes to traffic rules — without an enforcer

Two further layers matter to business. The **International Telecommunication Union** coordinates radio frequencies and orbital positions through its Radio Regulations: an operator's state files a satellite network with the ITU and coordinates it with earlier filings, so priority goes, broadly, to the first to file — an incentive to file early and large, sometimes for "paper satellites" that may never fly. And **national licensing** — of launches, spectrum, remote sensing, exports — is where the Treaty's "authorization and continuing supervision" becomes a permit a firm must hold.

The physics. Kessler and Cour-Palais (1978) showed that once the density of objects in an orbital shell passes a threshold, collisions produce fragments that cause further collisions: a debris belt that grows on its own, even if no one launches again. A commons in the strict sense — no one can be excluded, and every use depletes it for others — with physics in place of an economy. Debris in low orbit re-enters as the atmosphere drags it down; the higher the orbit, the longer it stays — years at a few hundred kilometres, decades to centuries above that.

> **For the firm:** an orbit is a **shared infrastructure** whose quality is set by the least careful actor in it. A destructive test by one state raises the collision risk — and the insurance price — of every operator in that shell.

11.2.3 3. Astropolitics: three readings of the new space race

	Reading	Implication
Dolman (2002), *Astropolitik*	classical geopolitics carried into space: in the spirit of Mackinder, whoever controls low Earth orbit controls access to near-Earth space, and through it the Earth; a realist case for the United States to take and hold that control	space is a **territory to dominate** — Session 03's anarchy, with orbits as chokepoints
Pekkanen (2019), "Governing the new space race"	the new space race is driven by three intertwined trends — **democratisation** (more states and non-state actors), **commercialisation** (firms striking out on their own, not only supplying governments), **militarisation** — while about **95% of space technologies are dual-use**, and treaty-making is in decline	space is a **market** governed by national laws, contracts and a few old principles — and the firms in it are part of the security problem
Bowen (2020), *War in Space*	spacepower matters for its effects **on Earth**; Earth orbit is a "cosmic coastline" from which states support, and threaten, what happens on land, at sea and in the air	space is **infrastructure for terrestrial power** — and therefore a target when terrestrial power is contested

The three readings do not exclude one another. Read together they explain why the same constellation can be a consumer broadband business, a military communications network and a foreign-policy instrument, all at once — Pekkanen's dual use, with the strategic stakes of Dolman and Bowen.

11.2.4 4. Counterspace: the threats

The Secure World Foundation's annual open-source assessment (Samson & Brett, 2026) tracks what states can do to deceive, disrupt, deny, degrade or destroy space systems, across five categories, in thirteen countries:

Category	What it is	Reversible?	Who it hits first
Direct-ascent	a missile launched from Earth to destroy a satellite	no — and it creates debris	the target, then every operator in that shell
Co-orbital	a satellite that approaches, inspects, interferes with or attacks another	depends	the target
Electronic warfare	jamming or spoofing of signals — uplinks, downlinks, GNSS	yes	**users on the ground**: ships, aircraft, networks
Directed energy	lasers or microwaves to dazzle or damage sensors	partly	imaging satellites
Cyber	attacks on satellites, ground stations or user terminals	yes, unless hardware is destroyed	**customers** of the operator

Its central finding for this course: many states have researched and tested destructive capabilities, but **only non-destructive capabilities are actively used against satellites in current military operations**. Four states have destroyed a satellite with a missile from the ground: the United States (1985, and the low-altitude intercept of a failing satellite in 2008), China (2007), India (2019) and Russia (2021). In April 2022 the United States declared it would not conduct destructive direct-ascent tests, and the UN General Assembly called on all states to make the same commitment in December 2022.

What firms have actually suffered is the reversible kind. On **24 February 2022**, the day of the invasion of Ukraine, a cyberattack on the ground segment of Viasat's KA-SAT network disabled thousands of user modems in Ukraine and across Europe — including the remote monitoring of thousands of wind turbines in Germany; the EU, the UK and the US attributed it to Russia. GNSS jamming and spoofing around conflict zones has become routine; in 2024 Finnair suspended flights to Tartu, in Estonia, for several weeks because interference made satellite-based approaches unusable.

> **The lesson for an analyst.** Destructive tests make headlines and debris. Jamming, spoofing and cyberattacks on ground segments make **losses** — and they fall on the customers, not on the satellite.

11.2.5 5. The evidence: the debris record

ESA's Space Debris Office has published its *Space Environment Report* every year since 2017. The 2026 edition (Issue 10.0, data to the end of 2025) is today's big work. Its method is worth reading before its figures: it counts **catalogued** objects — from the US public catalogue, the Russian Vimpel catalogue and the UK's historical RAE tables — and **asserted** objects known to exist by design, cross-correlated in ESA's DISCOS database; objects too small or too erratic to be tracked are outside its scope.

Its headline numbers, at the end of 2025 (Table 3.1):

- **44,964 objects** in Earth orbit, of which **16,946 payloads**; **25,060** in low Earth orbit;
- "ever since the start of the space age there has been more space debris in orbit than operational satellites";
- **fragmentations other than collisions** — explosions, mostly of old rocket stages and batteries — are the **dominant source** of debris: on average 9.8 non-deliberate fragmentations a year over the last two decades; 2024 added over 3,000 catalogued fragments, mostly from propulsion-related break-ups.

The target figure — Figure 2.1, the number of objects in geocentric orbit by object class since 1957 — shows the cumulative result. Three events stand out, all recorded in the report:

Date	Event	Altitude	What it did
11 January 2007	China destroys its Fengyun-1C weather satellite with a ground-launched missile	about 850 km	the largest single source of catalogued debris in the public catalogue; thousands of fragments, most still in orbit
10 February 2009	the active Iridium 33 and the defunct Cosmos 2251 collide	about 780 km	the first accidental collision between two intact satellites — Kessler, observed
15 November 2021	Russia destroys Cosmos 1408 with a ground-launched missile	about 470–490 km	some 1,800 catalogued fragments; low enough that almost all re-entered within three years — ESA attributes the 2022 spike in re-entering debris to the test

What the figure does not measure. It counts objects, not risk: a 10-cm fragment and a 5-tonne rocket stage each count once, and objects too small to catalogue — the majority of the fragments able to disable a satellite — do not appear at all. And it is drawn from catalogues that themselves improve: when sensors get better, the number of objects rises without anything new happening in orbit. In the practice round, you will find a third limit — the one hidden in how the public catalogue **dates** debris.

11.2.6 6. One operator: concentration, and weaponised interdependence in orbit

The new space race has a feature the Cold War one did not: most of the active satellites in orbit belong to **one company**. SpaceX's Starlink, first launched in May 2019, grew in six years into the majority of the world's operating satellites; you will measure how large a majority in the practice round, and you will find that concentration of **who launches payloads** has returned to Cold War levels — with a firm, not a superpower, at the top.

Starlink in Ukraine shows what that means. Within days of the 2022 invasion, Starlink terminals became a backbone of Ukraine's military and civil communications. The company's own decisions about where the service would work — including, by its owner's account, a refusal to enable it for an attack near Crimea in 2022 — became questions of war policy, and governments moved to contract for the service directly.

Read it with **Farrell and Newman (2019)** from Session 08. A constellation is a **hub**: its operator sees who connects, where (a panopticon), and can switch a user, a region or a country off (a chokepoint). The two conditions for weaponisation are met when the network is **concentrated** and when a **state can direct the firm** that runs the hub — and Article VI of the Outer Space Treaty guarantees that every operator has a licensing state behind it. A firm that depends on one operator depends on that operator's state as well.

States are responding as you would expect from Session 03: by building their own. The EU's **IRIS²** secure-connectivity programme (Regulation (EU) 2023/588), China's **Qianfan** and **Guowang** constellations, and national launch programmes are all, in part, answers to dependence on one American firm. **Export controls** draw the same line from the other side: spacecraft and their components sit on the US Munitions List (Category XV of the ITAR) or, for most commercial satellites since 2014, under the Export Administration Regulations (the "9x515" entries) — so a non-US manufacturer with a US component needs US permission to sell to some customers.

11.2.7 7. What can the firm do?

Through which channel does space risk reach the P&L? The course's five channels apply:

Channel	How space risk arrives	Example
Market access	export controls and licensing on space technology; spectrum and landing rights	a satellite or terminal that cannot be sold or switched on in a market
Inputs	a service from a single operator or a single state's GNSS	connectivity for ships or mines from one constellation
Finance and payments	trading and payment systems time-stamped from GNSS; insurance capacity	a timing outage; a premium that doubles after a fragmentation event
Assets and people	ships and aircraft navigating in jammed or spoofed areas	rerouted flights; vessels misreporting their position
Data and operations	cyberattacks on ground segments and terminals; loss of imagery feeds	the KA-SAT modems; a crop forecast without images

What a dashboard should track — each with a threshold that triggers a named decision:

1. **Dependence** — the share of a critical service (connectivity, imagery, timing) supplied by one operator or one state's system; *e.g. above 50% from one operator → contract a second one*.
2. **Interference** — official warnings of GNSS interference (aviation authorities publish them) in the regions where the firm's ships, aircraft or sites operate.
3. **Environment** — destructive tests and fragmentation events in the orbital shells of the firm's providers; ESA's yearly environment indicators.
4. **Rules** — changes in export-control lists, licensing conditions and ITU filings that affect the firm's suppliers or markets.

Strategies, each with a cost and a limit:

Strategy	Example	Cost	Limit
Multi-orbit, multi-operator	geostationary + low-orbit links from two providers	price, integration	the second operator may share the first one's state
Multi-constellation receivers	GPS + Galileo (+ others) in the same receiver	small	one jammer can still blind them all locally
Terrestrial back-up	holdover clocks and fibre-delivered time; cellular and fibre links	capital	not available at sea or in remote sites
Insurance	in-orbit and third-party liability cover	premium	a thin market; capacity shrinks after large losses
Compliance by design	export-control classification of every space component bought or sold	staff, delays	rules change with the balance of power

---

11.2.8 8. The Tableau skill of the day

- **A stock from flows.** Pivot `LAUNCH_DATE` and `DECAY_DATE` into one `Event date` column; a calculated field `Change` = +1 for a launch, −1 for a decay; `SUM([Change])` by year with a **Running Total** table calculation is the number of objects in orbit.

- **HHI with LOD expressions.** Share of each owner in a year's payloads: `{FIXED [Launch year], [OWNER] : COUNT([NORAD_CAT_ID])} / {FIXED [Launch year] : COUNT([NORAD_CAT_ID])}`; the sum of squared shares, built so that it adds up correctly over rows, is the HHI.
- **Dashboard action across data sources**: *Dashboard → Actions → Add Action → Filter*, mapping `Year` in one source to `Launch year` in the other.

The step-by-step version is the practice brief: `../02-practice/README.md`.

11.2.9 References

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Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies, opened for signature 27 January 1967, 610 U.N.T.S. 205 (entered into force 10 October 1967). https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/outerspacetreaty.html

11.3 In the practice

> **The target.** ESA's *Space Environment Report 2026*, Figure 2.1: the **number of objects in geocentric orbit by object class**, from 1957 to the end of 2025, with the 2007 Chinese ASAT test, the 2009 Iridium–Cosmos collision and the 2021 Russian ASAT test annotated — rebuilt from the public catalogue. You add the **concentration** of who puts payloads into orbit, year by year, and who operates the active satellites today — and wire the views into **one dashboard with an action**.

The last practice round of the term. The last twenty minutes of Session 10 reviewed your team dashboard's skeleton. Most skeletons still lack a space-infrastructure channel and any interaction between views. Leave today with both: the dependence view you build is designed to be dropped into your team dashboard, for the satellite services your own firm relies on.

Work in your project teams. Everyone builds the workbook; the team hands in one.

11.3.1 1 · Collect — 15 min

You arrive with `satcat.csv` (one download per team, pre-session). Use these fifteen minutes to **check**, not to download again:

1. **The snapshot.** What is the latest `LAUNCH_DATE` in the file? Log it with your download date: the catalogue changes every day.
2. **The codes.** Open the format documentation and the status codes. Which `OBJECT_TYPE` values are present? Which `OPS_STATUS_CODE` values count as **active**? What does a blank status mean?
3. **The trap you found before class.** Filter `OBJECT_NAME` = `FENGYUN 1C DEB`. How many rows? Which `LAUNCH_DATE`? Write the gap to 11 January 2007 in your log.

Provenance log — one line per file, as always:

file	source	url	version / release	accessed	licence	notes
satcat.csv	CelesTrak / US Space Force public catalogue	celestrak.org/pub/satcat.csv	*latest; LAUNCH_DATE; Last-Modified*	*date*	free (usage policy)	*debris dated by parent launch*
ESA report	ESA Space Debris Office	sdo.esoc.esa.int/publications/...	*Issue 10.0, May 2026*	*date*	public	*the original: Fig. 2.1, Table 3.1*
launch.tsv *(optional)*	J. C. McDowell, GCAT	planet4589.org/space/gcat	*"Updated" line*	*date*	free	

11.3.2 2 · Prepare — 15 min

You build **two data sources from the same file**: one reshaped into events (for the stock of objects in orbit), one left as it is (for concentration).

11.3.2.1 Data source A · "SATCAT events"

1. *Connect* → *Text file* → `satcat.csv`. Check the types: `LAUNCH_DATE` and `DECAY_DATE` must be **dates** (click the type icon above the column if they arrive as strings), `NORAD_CAT_ID` a number.
2. **Data source filter** (top right of the data source page → *Filters* → *Add*): `ORBIT_CENTER` = `EA`. Objects around the Sun or the Moon never decay and would stay "in orbit" for ever.
3. **Pivot the two dates.** Select `LAUNCH_DATE` and `DECAY_DATE` in the grid → right-click → **Pivot**. Rename `Pivot Field Names` → `Event` and `Pivot Field Values` → `Event date`. Each object now has two rows: its launch and its decay (empty if it is still in space).
4. Calculated fields:

```
Change  =  IF [Event] = "LAUNCH_DATE" THEN 1 ELSE -1 END
Year    =  YEAR([Event date])
```

5. **Check the grain.** Note the row count before and after the pivot, then exclude the null `Event date` rows (decays that have not happened). What is one row now? Write it in the log.

11.3.2.2 Data source B · "SATCAT objects"

6. *Data* → *New Data Source* → the same `satcat.csv`, **not** pivoted. Data source filter `OBJECT_TYPE` = `PAY` — a **data source** filter, not a sheet filter (step 3 explains why).
7. Calculated fields:

```
Launch year  =  YEAR([LAUNCH_DATE])
Power        =  CASE [OWNER] WHEN "US" THEN "United States"
                             WHEN "PRC" THEN "China"
                             WHEN "CIS" THEN "USSR / Russia"
                             ELSE "Rest of the world" END
Active       =  [OPS_STATUS_CODE] = "+" OR [OPS_STATUS_CODE] = "P" OR [OPS_STATUS_CODE] = "B"
                OR [OPS_STATUS_CODE] = "S" OR [OPS_STATUS_CODE] = "X"
Operator     =  IF STARTSWITH([OBJECT_NAME], "STARLINK") THEN "Starlink (SpaceX)"
                ELSEIF STARTSWITH([OBJECT_NAME], "ONEWEB") THEN "OneWeb"
                ELSEIF STARTSWITH([OBJECT_NAME], "KUIPER") THEN "Kuiper (Amazon)"
                ELSEIF STARTSWITH([OBJECT_NAME], "QIANFAN") THEN "Qianfan"
                ELSEIF STARTSWITH([OBJECT_NAME], "HULIANWANG") THEN "Guowang"
                ELSE "Other: " + [OWNER] END
```

`CIS` is CelesTrak's code for the USSR **and** Russia; `OWNER` is a registry-style code, not the operator — log both decisions.

11.3.3 3 · Replicate — 35 min

11.3.3.1 Sheet 1 · "Objects in Earth orbit, by type" — the target figure (data source A)

1. `Year` → **Columns**: right-click → *Convert to Dimension*, keep it **continuous** (green).
2. `SUM([Change])` → **Rows**. You see net additions per year — a jagged line.
3. Right-click the pill → *Quick Table Calculation* → **Running Total**. Now you see the **stock**.
4. `OBJECT_TYPE` → **Colour**. *Edit Table Calculation* → *Compute using* → *Specific Dimensions* → tick `Year` only, so the running total restarts for each type. Mark type **Area** (stacked).
5. Filter `Year` ≤ 2025 to match ESA's end point. **Never filter out early years** with a normal filter: it runs before the table calculation and the running total would restart from zero. (To hide early years, use a table-calculation filter such as `LOOKUP(MIN([Year]), 0)`.)
6. **Annotate** 2007, 2009 and 2021 (*right-click* → *Annotate* → *Point*). Look at the debris band. Where are the steps?
7. Title stating the finding; caption with source, snapshot date and "objects dated by launch date".

11.3.3.2 Sheet 2 · "Payloads launched per year, by power" (data source B)

8. `Launch year` → Columns (continuous dimension), `CNT([NORAD_CAT_ID])` → Rows, `Power` → Colour, mark type **Bar**. Where does the Cold War end, and where does the new race begin?

11.3.3.3 Sheet 3 · "How concentrated is the new space race?" — the HHI (data source B)

9. Create:

```
Owner share  =  {FIXED [Launch year], [OWNER] : COUNT([NORAD_CAT_ID])}
              / {FIXED [Launch year] : COUNT([NORAD_CAT_ID])}
HHI piece    =  10000 * [Owner share] / {FIXED [Launch year] : COUNT([NORAD_CAT_ID])}
```

Why the second line: each owner's squared share must be counted **once**, but the owner appears on as many rows as it has payloads. Spreading `share × 10,000` over the year's rows means that summing an owner's rows gives exactly `share² × 10,000`.

10. `Launch year` → Columns, `SUM([HHI piece])` → Rows, mark type **Line**. That sum **is** the HHI. Check: 1957 must give **10,000** (two Soviet payloads). Reference lines at **2,500** and **1,800**, labelled.
11. Why the data source filter in step 6: a `FIXED` expression is computed **before** ordinary filters. Had you filtered `OBJECT_TYPE` on the sheet, the shares would still include debris and rocket bodies. (Adding the filter to **context** also works.)

11.3.3.4 Sheet 4 · "Who launched that year?" (data source B)

12. `OWNER` → Rows, `CNT([NORAD_CAT_ID])` → Columns, sorted descending, `Operator` → Colour; a *Top 10 by count* filter on `OWNER`. This is the **target** of the action.

11.3.3.5 Sheet 5 · "Who operates the active satellites today?" (data source B)

13. Filter `Active` = True and `ORBIT_CENTER` = `EA`. `Operator` → Rows, `CNT([NORAD_CAT_ID])` → Columns, sorted; *Quick Table Calculation → Percent of Total*. Caption: "status as of the snapshot date; blank status excluded".

11.3.3.6 Dashboard · "Space infrastructure exposure"

14. *New Dashboard*. Sheet 1 top left, Sheet 3 top right, Sheet 4 bottom left, Sheet 5 bottom right.
15. **Action 1.** *Dashboard → Actions → Add Action → Filter*. Source: Sheet 1. Target: Sheet 4. Run on: *Select*. Clearing the selection: *Show all values*. Filter: **Selected fields** → source field `Year` → target data source *SATCAT objects*, target field `Launch year`. Because the two sheets use different data sources, map the fields explicitly.
16. **Action 2.** The same, with Sheet 3 as source (same data source as Sheet 4: field `Launch year` → `Launch year`).
17. Click 1965 on Sheet 3, then 2025. Sheet 4 should show who launched that year. If it shows nothing, the field mapping is wrong: fix it, and log it.

11.3.4 4 · Compare and break — 15 min

1. **Differ.** Put Sheet 1 beside ESA's Figure 2.1 and Table 3.1 (end of 2025: 44,964 objects, of which 16,946 payloads). Give your totals by type and **one** reason for each gap — which catalogues ESA adds, what it calls "unidentified", what the public catalogue leaves out.
2. **Break: when does debris appear?** Your debris band does not step up in 2007, 2009 or 2021. Re-date the four big events' debris to the date the event happened. In data source A:

```
Appearance year  =  IF [Event] = "LAUNCH_DATE" AND CONTAINS([OBJECT_NAME], "FENGYUN 1C DEB") THEN 2007
                    ELSEIF [Event] = "LAUNCH_DATE" AND (CONTAINS([OBJECT_NAME], "COSMOS 2251 DEB")
                         OR CONTAINS([OBJECT_NAME], "IRIDIUM 33 DEB")) THEN 2009
                    ELSEIF [Event] = "LAUNCH_DATE" AND CONTAINS([OBJECT_NAME], "COSMOS 1408 DEB") THEN 2021
                    ELSE [Year] END
```

   Duplicate Sheet 1 and replace `Year` by `Appearance year`. How big is each step now? What about the thousands of other fragments whose event you did not re-date?
3. **Break: flag or firm?** On Sheet 5, replace `Operator` by `OWNER`, and compute the HHI of active satellites both ways. Which number describes your firm's **dependence** better?

4. **Extend.** For your **team's firm**, name the satellite services it relies on (GNSS timing, connectivity at sea or in remote sites, imagery). Find their operators in the file — `NAVSTAR` (GPS), `GSAT0…` (Galileo, coded `ESA`), `BEIDOU`, `[GLONASS-M]` — and add the one that matters to Sheet 5. Optional, with GCAT: orbital launch attempts per year (`LaunchCode` starting `O` or `D`) by launch-provider state.

11.3.5 5 · The catch — 10 min

One slide, three panels:

Original	Ours	The catch
ESA's Figure 2.1	Your dashboard	What it does **not** license your firm to conclude

The model catch, if you need one to beat: *a count of objects is not a count of risk — the fragments able to disable a satellite are mostly too small to catalogue, and the public catalogue dates debris by its parent's launch, so the ASAT tests vanish from a chart built naively on it; and concentration by flag hides dependence on one firm.* Beat it — say something specific to your firm.

A member drawn at random presents in two minutes.

11.3.6 Hand in

Upload to ZoneCours before leaving: the `.twbx`, the provenance log, and the slide. Formative: commented, not marked; counts towards participation. (The SATCAT is freely available, so the workbook may also go to Tableau Public — credit CelesTrak in the caption.)

The one rule: the log records at least one place where the data disagreed with the paper — a total that does not reach ESA's, debris dated to the wrong year, an owner code that is not the operator.

11.3.7 Into the team dashboard

Before Session 12, re-point Sheet 5 at **your firm's** satellite services, keep the action, and add the dependence view to your **channel views** with a threshold — for example *more than half of a critical service from one operator → contract a second one*. A board member should be able to click a year and see who controlled access to orbit then, and see in one screen who the firm depends on now.

11.3.8 What loses credit

- A running total that restarts because early years were filtered out
- Objects around the Sun or the Moon counted as "in orbit"
- An HHI that is not 10,000 in 1957 — shares computed after a sheet filter, or squared on every row
- `OWNER` presented as the operator, or blank status counted as inactive
- A dashboard action that silently filters to nothing
- A catch about ESA rather than about your firm

Chapter 12

Team presentations

Session 12 · A geopolitical risk assessment for one multinational Presentation-day slides · Session folder

Can you make a board act on your assessment — and say what would change your mind?

12.0.1 A geopolitical risk assessment for one multinational

Can you make a board act on your assessment — and say what would change your mind?

What happens	Each team presents and defends its geopolitical risk dashboard for its multinational
Assessment	Presentation and defence: **5 of the 20 points** of the team work. Memo and published dashboard (15 points) due **one week later**
Closing readings	Kobrin, S. J. (1979). Political risk: A review and reconsideration. *Journal of International Business Studies*, 10(1), 67–80. doi:10.1057/palgrave.jibs.8490631 · Henisz, W. J. (2000). The institutional environment for economic growth. *Economics & Politics*, 12(1), 1–31. doi:10.1111/1468-0343.00066
Deck	`INTE60019A-S12-Presentations.qmd`
Brief	`assessments/team-project/`

There is no pre-session reading beyond your own work, no practice round and no Risk Brief.

Lecture notes: this session is also Chapter 12 of the course book.

12.1 Run of show

Each team has **20 minutes**: **12 minutes of presentation**, the dashboard driven live, then **8 minutes of defence** — questions from the instructor and from the room.

Time	
0:00–0:05	Order of presentation drawn at random; the two-team "panel" for each slot announced
0:05–2:50	Teams 1–8, 20 minutes each, back to back. A five-minute break after team 4
2:50–3:00	Closing address: *from indicators to decisions*

With more than eight teams the session runs in two rooms in parallel, or is extended by agreement; the order and room are posted on ZoneCours the week before.

The panel. For each presentation, two other teams act as the **board**. Each prepares one question in advance from the published dashboard link (sent 48 hours before) and asks it in the defence. The quality of a team's board question counts towards its members' participation.

Who answers. The instructor addresses at least one question to a member **drawn at random**. Every member must be able to defend every part of the dashboard and the method behind it.

12.2 What the twelve minutes must contain

Minutes	What the board must see
0–2	The firm and its **footprint**: where its revenue, assets, suppliers and people are
2–5	The **composite exposure index** for its main countries — indicators, normalisation, weights — and one live move of a weight parameter to show what the ranking is sensitive to
5–9	**Two channel views**, driven live: e.g. sanctions, critical inputs, bloc alignment, cyber
9–11	The **interaction**: how two risks compound or offset for *this* firm
11–12	**Decisions and the watchlist**: three indicators, three thresholds, three named decisions

Overrun is stopped at thirteen minutes.

12.3 Rubric (20 points — consistent with the team-work brief)

Criterion	Points	Assessed	Full marks look like
Conceptual grounding	3	memo + presentation	Risks framed with the course's concepts — order, anarchy, institutions, diplomacy, sanctions, interdependence — not newspaper vocabulary
Data and provenance	3	memo	Four or more datasets collected at source, from at least three families of indicators, every one logged; decisions named
Method of the composite index	4	memo	Indicators justified; normalisation and weighting explained; sensitivity to weights shown

Criterion	Points	Assessed	Full marks look like
Analysis of interactions	3	memo	How risks compound or offset — e.g. sanctions *and* input concentration *and* bloc alignment
Recommendations	2	memo	Decisions with triggers and owners, not "monitor the situation"
Presentation and defence	**5**	**today**	Twelve minutes; the dashboard driven live; questions answered by any member, including the one drawn at random

Presentation and defence (5 points), in detail:

Points	
2	Clarity and time: the five blocks above, in twelve minutes, the dashboard driven rather than screenshotted
2	Defence: answers that engage the question, concede what the data cannot show, and say what would change the recommendation
1	Random-member answer: the member drawn answers competently without help

Individual accountability (the peer-rating multiplier) is described in the team-work brief.

12.4 The closing address — from indicators to decisions

Ten minutes, after the last team. Its content is in the deck; in summary:

1. **Kobrin (1979) was right about the hard part.** Political risk is not the event — it is the *contingency* that the event affects *this* firm's operations. Every dashboard presented today is a claim about that contingency.
2. **Henisz (2000) on constraint.** The POLCON index measures how many independent veto players constrain a government. Low constraint means policy can change fast — in either direction. It is the structural complement to every event- and text-based indicator of the course.
3. **Indicators do not decide.** A watchlist is useful only if each indicator has a threshold and an owner. The best dashboards today were the ones that would make someone *do* something on a given Monday.
4. **What the twelve weeks established.** Geopolitical risk is measured through families of indicators that disagree; each figure in a famous work is a stack of decisions; and the analyst's job is to make those decisions visible to the person who must act.

12.5 After today

One week after Session 12	Memo (2,500 words + appendices) and the final published Tableau Public dashboard, on ZoneCours; confidential peer ratings
Final examination period	**Take-home examination** — 40%, individual, 3 hours of work in a 48-hour window. Brief and specimen: `assessments/take-home/`

Prepare for the take-home by re-reading the **catch** slides from your eleven practice rounds: together they are the course's list of what each indicator cannot tell a firm.

← Previous session · Syllabus

Appendix A

The replications

Each of Sessions 01–11 pairs one **big article or book** with the **data it is built on**. Before class you read the work and locate its source. In the 90-minute practice round your team:

1. **Collects** the data from the original producer and logs its provenance;
2. **Prepares** it in Tableau — pivot, join, recode — recording each decision;
3. **Replicates** the target figure as closely as the data allow;
4. **Compares** yours with the original, **breaks** one decision, and **extends** it in one direction;
5. Presents **one slide — original · ours · the catch** — in two minutes, by a member drawn at random.

Every source, its access conditions and its traps: `DATA-SOURCES.md`.

A.1 Article–data map

Session	Big work	Data you collect	Target figure	Compare · break · extend
01 · What is geopolitical risk?	Caldara & Iacoviello (2022), *Measuring Geopolitical Risk*, AER · doi:10.1257/aer.20191823	GPR index	The monthly GPR index since 1985, spikes annotated; the historical index since 1900	Threats vs acts · monthly vs yearly averages · one country index against the global
02 · The international economic order	Baldwin (2016), *The Great Convergence*, Belknap/Harvard UP	World Bank WDI: GDP (current US$ and PPP), trade % GDP	The **G7 share of world GDP**, 1970–today — the "great convergence" after 1990	Current dollars vs PPP · Baldwin's "Industrialising Six" · world trade/GDP and the "slowbalisation" after 2008
03 · Anarchy and the balance of power	Mearsheimer (2001), *The Tragedy of Great Power Politics*; Allison (2017), *Destined for War*	COW National Material Capabilities v7.0 (CINC); SIPRI military expenditure	**Shares of world capabilities** of the major powers, 1816–2022, and the US–China crossover	CINC vs military spending vs GDP — three measures, three different power transitions · a parameter to choose the challenger

Session	Big work	Data you collect	Target figure	Compare · break · extend
04 · Trade, conflict and fragmentation	Gopinath, Gourinchas, Presbitero & Topalova (2025), *Changing Global Linkages: A New Cold War?*, JIE · doi:10.1016/j.jinteco.2024.104042; Mansfield (2021)	UNGA ideal points (Voeten); IMF bilateral goods trade (IMTS)	**Trade within vs between geopolitical blocs**, indexed to before the 2022 invasion of Ukraine	Bloc definitions: thresholds on ideal-point distance · mirror trade (exports vs imports) · one "connector" country
05 · International institutions	Kuziemko & Werker (2006), *How Much Is a Seat on the Security Council Worth?*, JPE · doi:10.1086/507155	UNSC elected members; US foreign aid (ForeignAssistance.gov Country Summary — economic and military combined) or the AidData replication file	**US aid to elected members in event time**, years −3 to +3 around a Security Council term	Current vs constant dollars · mean vs median · years with and without salient Council business
06 · Private authority and standards	Prakash & Potoski (2006), *Racing to the Bottom? Trade, Environmental Governance, and ISO 14001*, AJPS · doi:10.1111/j.1540-5907.2006.00188.x; Abbott & Snidal (2009); Mattli & Büthe (2003)	ISO Survey (ISO 14001, ISO 9001); IMF bilateral exports; WDI GDP	**The diffusion of ISO 14001** across countries (map and small multiples), and adoption against the adoption level of each country's export markets	Counts vs per-$ of GDP · ISO 9001 as the placebo · trade-weighted vs unweighted export markets
07 · Measuring risk: text, coding, big data, indices	Hassan, Hollander, van Lent & Tahoun (2019), *Firm-Level Political Risk*, QJE · doi:10.1093/qje/qjz021; Baker, Bloom & Davis (2016), QJE; OECD/JRC (2008) *Handbook on Constructing Composite Indicators*	firmlevelrisk.com firm-quarter file; EPU; a corpus of news items from GDELT that **you code yourselves**	**Average firm-level political risk over time** against EPU — then **your own coded index**	Intercoder agreement on your corpus · normalisation choices · weights as parameters: how much does the ranking move?
08 · Cyber geopolitics	Valeriano & Maness (2015), *Cyber War versus Cyber Realities*, OUP · doi:10.1093/acprof:oso/9780190204792.001.0001; Farrell & Newman (2019)	DCID v2.0; EuRepoC v1.3	**Incidents by year, method and severity**, and the **initiator × target matrix** of rival dyads	DCID (state rivals) vs EuRepoC (all incidents) · severity distribution — evidence of "cyber restraint"? · incidents hitting the private sector

Session	Big work	Data you collect	Target figure	Compare · break · extend
09 · Economic diplomacy	Rose (2007), *The Foreign Service and Foreign Trade: Embassies as Export Promotion*, World Economy · doi:10.1111/j.1467-9701.2007.00870.x; Bjola & Manor (2018); Putnam (1988)	Lowy Global Diplomacy Index; IMF bilateral exports; WDI GDP	**Exports against the number of diplomatic posts** in the partner, log scale, with trend line	Raw vs GDP-adjusted · embassies vs consulates · reverse causality: posts follow trade?
10 · Economic sanctions	Hufbauer, Schott, Elliott & Oegg (2007), *Economic Sanctions Reconsidered*, PIIE; Felbermayr et al. (2020), *The Global Sanctions Data Base*, EER · doi:10.1016/j.euroecorev.2020.103561; Pape (1997)	GSDB Release 5 (by request); TIES v4.0	**Active sanctions cases per year by type**, 1950–today, and the **share judged successful** by objective	Threats (TIES) vs impositions · success by objective · a Gantt chart of one target's sanctions history
11 · Space geopolitics	ESA Space Debris Office (2026), *ESA's Annual Space Environment Report*, Issue 10.0 · PDF; Samson & Brett (Eds., 2026), *Global Counterspace Capabilities*, SWF; Pekkanen (2019)	CelesTrak SATCAT; GCAT launch log (optional)	**Objects in geocentric orbit by object class**, 1957–2025 (ESA Fig. 2.1), with the 2007, 2009 and 2021 events annotated; the **HHI of who launches payloads**	Debris dated by parent launch vs by event · concentration by flag (`OWNER`) vs by firm (operator) · public catalogue vs ESA's catalogued + asserted objects
12 · Team presentations	Kobrin (1979); Henisz (2000)	All of the above; POLCON; WGI	**Your team's dashboard** for one multinational	Defended live, in front of the room

A.2 Why replicate figures, not regressions

This is not an econometrics course. The figure at the heart of a big work is where its argument meets the reader — the image that ends up in board presentations and ministerial speeches. Rebuilding it forces every decision the authors made into the open: which countries, which years, which deflator, which coding rule, which normalisation. A multinational's risk team makes the same decisions every time it builds a dashboard. Learning to see them is the point.

A.3 Citing

Cite the work **and** the dataset, with version and access date. They are distinct contributions, often by different people, and the dataset has usually been revised since the work was published.

Appendix B

Data sources

Every dataset used in the course: who produces it, where to collect it, what it costs you to get it, and the trap it holds. **Collect at source, every time**: the course repository contains no data.

Access status was last checked on **16 October 2026**. Sources move: if a link has changed, the producer's landing page is canonical, and the change goes in your provenance log.

> **Lead times.** Two sources need action **days before** the session: the **Global Sanctions Data Base** (request form, data e-mailed within about 24 hours — request it after Session 08) and the **ISO Survey** (requires a free IAF CertSearch account — log in at iafcertsearch.org/services/iso-survey; create it after Session 04).

B.1 The registry

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
1	**Geopolitical Risk (GPR) index**	Caldara & Iacoviello	matteoiacoviello.com/gpr.htm	`gpr_webpage_data.xls` (one sheet, `Sheet1`; monthly `GPR`, `GPRT`, `GPRA`, historical `GPRH`; 44 country series `GPRC_XXX` — percent of articles, from 1985 — and historical `GPRHC_XXX`; variable notes in the `var_name`/`var_label` columns)	1900 – 1985 – last month	free	01, 07, 12

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
2	**World Development Indicators**	World Bank	DataBank WDI	`NY.GDP.MKTP.CD` (GDP, current US$), `NE.TRD.GNFS.ZS` (trade, % GDP), `NY.GDP.MKTP.PP.CD` (GDP, PPP) — per-indicator CSV: `api.worldbank.org/v2/en/indicator/{CODE}?downloadformat=csv` (4 lines above the header; `Indicator Code`, not `Series Code`; missing = blank)	1960–2025	free, CC BY 4.0	02, 03, 09
3	**National Material Capabilities v7.0**	Correlates of War	correlatesofwar.org/data-sets/national-material-capabilities	`wp-content/uploads/NMCv7.zip` (codebook `NMC_Documentation_v7.pdf` + nested `NMC-v7-abridged.zip` → `NMC-70-abridged.csv`)	1816–2022	free; cite Singer, Bremer & Stuckey (1972)	03
4	**Military Expenditure Database**	SIPRI	sipri.org/databases/milex	`SIPRI-Milex-data-1949-2025_v1.2.xlsx` (revised 27 April 2026)	1949–2025	free for non-commercial use	03
5	**UN General Assembly ideal points**	Bailey, Strezhnev & Voeten	Harvard Dataverse doi:10.7910/DVN/LEJQZ	`Idealpointestimates_2025_2026.csv` (country-year: `ccode`, `year`, `IdealPointFP`; `session` = `year` − 1945), `AgreementScores.csv`, codebook	1946–2025	free	04, 12

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
6	**International Trade in Goods by partner (IMTS, formerly DOTS)**	IMF	data.imf.org — IMF.STA:IMTS	exporter × partner × year: indicator `XG_FOB_USD`; columns `COUNTRY`, `COUNTER-PART_COUNTRY`, `TIME_PERIOD`, `OBS_VALUE` (US dollars); ISO3 codes except `KOS`, `WBG`; aggregates coded `G…/TX…` (Download button)	from 1948 for many countries	free	04, 06, 09
7	**Kuziemko & Werker replication data**	AidData	AidData replication page	"Download Data" link (a Google Drive file; required a Google sign-in when checked in October 2026 — contents unverified)	1946–2001	free	05
8	**UN Security Council elected members**	United Nations	main.un.org/securitycouncil — elected members	the table, copied into a sheet: country × term	1946–today	free	05

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
9	**US foreign aid (Greenbook)**	US State Dept / ForeignAssistance.gov	foreignassistance.gov/data · frozen mirror (December 2024; use its Dropbox zip): foreignassistance.andrewheiss.com	Country Summary `us_foreign_aid_country.csv` — ISO3 in `Country Code`; filter `Transaction Type Name` = Obligations; use `constant_amount`. Economic and military **combined** — the split is only in the Greenbook xlsx on the mirror (`greenbook.html`, FY 1946–2020)	official 1946–2026 · mirror 1946–2024	free; the official site has been unstable since 2025 — use the mirror if it fails, and log it	05
10	**ISO Survey of certifications**	ISO, via IAF CertSearch	iso.org/the-iso-survey.html · IAF CertSearch — ISO Survey	ISO 14001 and ISO 9001 certificates by country and year; from 2025 compiled from IAF CertSearch data (file format unverified)	1993–2024	free; IAF CertSearch account required	06
11	**Firm-level political risk**	Hassan, Hollander, van Lent & Tahoun	firmlevelrisk.com/download	`firm_level_risk_2022Q1.csv` (firm × quarter `PRisk`, `NPRisk`, sentiment, topic risks; **tab-separated despite `.csv`**; `hqcountrycode` is ISO **2**-letter)	2002Q1–2022Q1	free; no longer updated	07

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
12	**Economic Policy Uncertainty**	Baker, Bloom & Davis	policyuncertainty.com	`media/Global_Policy_Uncertainty_Data.xlsx` (`Year`, `Month`, `GEPU_current`, `GEPU_ppp` — no `GEPU` column), `media/All_Country_Policy_Uncertainty_Data.xlsx` (sheet `EPU`, one column per country, named in English)	1997–present, monthly (some countries from 1985; the US from 1900 in a separate file)	free, CC BY 4.0	07
13	**GDELT**	The GDELT Project	gdeltproject.org	news articles, via the DOC 2.0 API (CSV columns `URL`, `MobileURL`, `Date`, `Title`; one request every 5 seconds) or the web "Summary" tool	the DOC API searches a rolling last 3 months — use `timespan=12weeks`	free	07
14	**Dyadic Cyber Incident and Campaign Data (DCID) v2.0**	Maness, Valeriano, Hedgecock, Macias & Jensen	Harvard Dataverse doi:10.7910/DVN/GOQMYV	`Master_DCID_2.0_Release_public_Sep2022.tab` (Access File — original CSV); codebook at ryanmaness.com/cyber-conflict-dataset, not on Dataverse	2000–Jan 2021, 433 incidents	public, CC0	08
15	**EuRepoC Global Dataset of Cyber Incidents v1.3**	European Repository of Cyber Incidents	eurepoc.eu · Zenodo doi:10.5281/zenodo.14965305	`eurepoc_global_dataset_1_3.csv`, `eurepoc_codebook_1_3.pdf`	2000–2024, 3,414 rows in the CSV (the record says 3,416)	free, CC BY-NC 4.0	08
16	**CFR Cyber Operations Tracker**	Council on Foreign Relations	cfr.org/cyber-operations	no official download — browse and hand-code	2005–2024	free to read	08 (reading only)

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
17	**Global Diplomacy Index**	Lowy Institute	globaldiplomacyindex.lowyinstitute.com	direct download `Lowy Institute Global Diplomacy Index Data 2016, 2017, 2019, 2021, 2023.xlsx` — one sheet per data year (`2023`, `2021`, `2019`, `2017`, `2016`), one row per diplomatic post: `COUNTRY`, `POST CITY`, `POST COUNTRY`, `TYPE OF POST`; **the 2019 sheet is incomplete** (Venezuela only, 40 rows)	2016, 2017, 2019, 2021, 2023 (2024 edition)	free	09
18	**Global Sanctions Data Base, Release 5**	Felbermayr, Kirilakha, Syropoulos, Yalcin & Yotov	globalsanctionsdatabase.com	**request form** → files by e-mail (case-level and dyadic). Case file columns (from a public notebook, not yet checked against the codebook): `case_id`, `sanctioned_state`, `sanctioning_state`, `begin_year`, `end_year`, six 0/1 type columns, `objective`, `success`	1,794 cases; begin years 1949–2025 (the home page says 1950)	free on request; **no redistribution, no commercial use** — never publish raw data to Tableau Public	10

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
19	**TIES — Threat and Imposition of Economic Sanctions v4.0**	Morgan, Bapat & Kobayashi	sanctions.web.unt.edu	`TIESv4-1.xls` + `tiesusers-manu-alv4.pdf`; states as COW codes; threat-only cases have `imposition` = 0	1945–2005	free	10
20	**POLCON — Political Constraint Index**	Witold Henisz, Wharton	POLCON dataset page	`POLCON_2025_FINAL_PUBLIC.xlsx` (sheets *Data* and *Key*)	1800–2025	free	12
21	**Worldwide Governance Indicators**	World Bank	worldbank.org — WGI	`wgidataset_with_sourcedata-2025.xlsx`: one sheet per dimension (`pv` = political stability), `Economy (code)`, `Year`, `Governance score (0–100)` — not to be mixed with the estimate column. The 2026 update (1996–2025) is now the landing-page default and recalculates history again: use one edition and log it	1996–2024 (2025 ed.)	free	07, 12

#	Dataset	Producer	Landing page	File to collect	Coverage	Access	Session
22	**SATCAT — satellite catalogue**	CelesTrak (T. S. Kelso), from the US Space Force public catalogue	celestrak.org/satcat/ · format	`pub/satcat.csv` (one row per catalogued object, 17 columns: `OBJECT_TYPE` PAY/R/B/DEB/UNK, `OPS_STATUS_CODE` — active = `+` `P B S X`, `OWNER`, `LAUNCH_DATE`, `DECAY_DATE` empty = in space, `ORBIT_CENTER`)	1957 – day of download	free; usage policy: one download per update	11
23	**GCAT — General Catalog of Artificial Space Objects**	Jonathan C. McDowell	planet4589.org/space/gcat	`tsv/cat/launch/launch.tsv` + `tsv/tables/orgs.tsv` (tab-separated; line 2 a "# Updated" comment; orbital = `LaunchCode` starting `0` or `D`; state via `Agency` → `orgs.Code` → `StateCode`)	1942 – last week	free	11 (optional)

B.2 Traps, source by source

Dataset	The trap
GPR	Revised monthly; vintages exist (`data_gpr_export_YYYYMM.xls`). Country indices are the anglophone press's attention to a country, not that country's risk.
WDI	Aggregates ("World", "High income", "G7" is *not* provided — build it) sit in the same file as countries. Current-dollar shares move with exchange rates.

Dataset	The trap
NMC	Uses COW country codes, not ISO; states enter and exit the system (`GMY` absent 1946–1990, when `GFR`/`GDR` cover 1955/1954–1990). Missing components are coded `-9`. Yearly CINC sums run 1.000–1.075, not exactly 1. Ends in 2022.
SIPRI	Several sheets (`Constant (2024) US$`, `Current US$`, `Share of GDP`, `Per capita`, `Share of Govt. spending`, local-currency and `Regional totals`); `...` = data unavailable, `xxx` = the state did not exist — notes atop each sheet, plus a `Footnotes` sheet. Country strings: `United States of America`; `Russia` from 1992, `USSR` before.
UNGA ideal points	Carries both COW `ccode` and ISO `iso3c`; estimated by calendar year since the July 2025 releases (v36/37; current v39, 30 July 2026) — earlier versions by session, so the variable meaning changed between versions. Higher = closer to the US.
IMF IMTS	Exports reported by A ≠ imports reported by B ("mirror" asymmetry). Choose one side and say which.
UNSC members	Two-year terms starting 1 January; election happens the October before — the paper's "election year" matters.
US aid	Obligations vs disbursements; current vs constant dollars; the official site moved in 2025.
ISO Survey	Coverage of certification bodies changed over time: jumps in a country's count can be reporting, not adoption.
Firm-level risk	Listed firms holding earnings calls in English — mostly US. Stops in 2021.
DCID	State-to-state rivals only; `StateA`/`StateB` are the dyad as **names**, not initiator and target — derive the target from `initiator` (COW code) and `Dyadpair`; coded from open sources — attribution is by definition uncertain.
EuRepoC	Much wider than DCID (includes criminal and non-state); filter on `initiator_category` (`State` / `State affiliated actor`) before comparing.
Lowy GDI	66 countries or territories only (including Taiwan and the EU's EEAS); names, not codes, with typos (`Portual`, `Nigaragua`); counts posts, not staff or budget.
GSDB	Licence forbids redistribution: workbooks go to ZoneCours, never Tableau Public with the raw data. "Success" is the coders' judgement of objectives met.
TIES	Includes **threats** of sanctions — a case may never have been imposed.
POLCON / WGI	WGI's 2025 edition changed scale and recalculated history: never mix editions.
SATCAT	Debris carries its parent's `LAUNCH_DATE` (Fengyun-1C debris is dated 1999, not 2007); `OWNER` is not the operator (Starlink `US`, Galileo `ESA`, Russia `CIS`); filter `ORBIT_CENTER` = `EA`; a blank status is not "inactive"; one download per team.
GCAT	The state is the launch provider's, not the customer's (Rocket Lab = New Zealand); vague-date text, padded values.
UCS Satellite Database	Frozen: data current through 1 May 2023. Use SATCAT for anything later.

B.3 Country codes

Join on **ISO 3166 alpha-3** codes, never on names. NMC and DCID (in `initiator` and `Dyadpair`) use **Correlates of War** codes (official list): keep a concordance sheet (COW `ccode` ↔ ISO3) in your workbook. The `countrycode` concordance tables published by Arel-Bundock et al. are the standard reference. The UNGA ideal points carry `iso3c` too. Kosovo is `KOS` at the IMF and `XKX` at the World Bank; the West Bank and Gaza is `WBG` at the IMF.

Appendix C

The Tableau toolkit

Everything you need to install, the one new skill each session adds, the conventions every chart follows, and the traps that cost most marks.

C.1 1. Installation — before Session 01

Tool	How	Used for
Tableau Desktop	Free academic licence through **Tableau for Students** (Tableau Academic Programs). Request it with your HEC e-mail; the key arrives by e-mail.	every practice round
Tableau Prep Builder	Same licence, same key, where the academic programme includes it — optional	cleaning, reshaping and unioning files (from Session 02; essential in Session 07)
Tableau Public	Free account at public.tableau.com	publishing the team dashboard and Risk Briefs
A spreadsheet	Excel, LibreOffice Calc or Numbers	opening files at source, reading codebooks

Test. Open Tableau Desktop → *Sample – Superstore* → `Order Date` to Columns, `Sales` to Rows. A line appears: you are ready.

If the key has not arrived install the 14-day trial and switch to the key when it comes. The academic licence is renewable each year while you are enrolled.

> **If the academic programme no longer covers your case.** Tableau's academic terms have changed more than once (reports in 2025 indicated that free student licences for Desktop and Prep were being withdrawn in some programmes; check tableau.com/academic/students and HEC's own software page at the start of term). The fallback is the free **Tableau Public** desktop edition. Everything in the course works in it — pivots, joins, relationships, LOD, parameters, maps, dashboards — with two differences you must manage:
>
> 1. **It saves only to Tableau Public**, where workbooks are public. For data whose licence forbids redistribution (the Global Sanctions Data Base, possibly the ISO Survey), publish only aggregated views, keep "allow download" off, and hand in screenshots plus your calculated-field definitions instead of a `.twbx`.
> 2. **No Tableau Prep.** Do the Session 07 cleaning in the data source pane (pivot, split, union, calculated fields) — the practice brief gives both routes.

Mac and Windows behave identically for everything in this course, with one exception: on a Mac, right-drag is *Option*-drag.

C.2 2. The skills ladder

One new skill per session, always applied to the session's replication. Nothing is taught twice; everything is assumed afterwards.

Session	Replication	The new skill
01	GPR index	Connect to Excel · fix data types · continuous dates · reference lines · annotations · captions
02	G7 share of world GDP	**Pivot** wide years to long · filter on ISO codes, not names · area chart · **percent of total** (table calculation)
03	Distribution of power	**Calculated fields** · ratios · **parameters** to choose a country · dual axis
04	Trade between blocs	**Joins and relationships** on dyads (exporter × importer × year) · **groups** and **sets** for blocs · indexing to a base year
05	Aid and the Security Council	**LOD expressions** (`FIXED`) to compute "years relative to a term" · event-time charts · reference bands
06	ISO 14001 diffusion	**Maps** · geocoding by ISO code · fixing unmatched countries · the **Pages** shelf for animation · small multiples
07	Firm-level risk and your coded corpus	**Tableau Prep** flows · unions · cleaning text-coded data · **normalisation** (min–max, z-score) · **parameters as weights** in a composite index
08	Cyber incidents	**Highlight tables** · dyad matrices (initiator × target) · **filter and highlight actions**
09	Embassies and exports	**Log scales** · **trend lines** and what their R^2 and p-value do and do not say · clusters
10	Sanctions episodes	**Gantt charts** for episode durations · stacked bars of shares · **quick table calculations**
11	Objects in orbit and the new space race	**Running sum** (table calculation) on pivoted launch/decay events · **Herfindahl–Hirschman index** with LOD · **dashboard actions** across data sources
12	Your team dashboard	**Dashboards** · **stories** · publishing to Tableau Public · writing captions a board can read

C.3 3. Conventions every chart follows

1. **A title that states the finding**, not the variable. *"The G7's share of world GDP fell by a third after 1990"*, not *"GDP share"*.
2. **A source caption**: dataset, producer, version, access date. No caption, no credit.
3. **Units on every axis.** Current US dollars or constant? Index base year? Share or count?
4. **Direct labels** in place of legends wherever there are fewer than six series.
5. **One accent colour** for the series the argument is about; greys for the rest.
6. **Annotate the events** that explain the shape of the line.
7. **Name your decisions** in a footnote: countries dropped, years rebased, codes changed.

C.4 4. The traps that cost marks

Trap	Symptom	Fix
Joining on country names	Côte d'Ivoire, Türkiye, Korea, Rep. silently drop out	Always join on **ISO 3166 alpha-3** codes. Keep a concordance sheet for datasets that use COW or UN codes
Wide years	66 columns called `1960` … `2025`	**Pivot** them in the data source pane (select columns → right-click → *Pivot*)
SUM of an index	A line ten times too high	An index is averaged, not summed. Check the aggregation on every pill
Discrete vs continuous dates	A bar for "May" across all years	Use the *continuous* date (green pill) for time series
Duplicated rows after a join	Totals inflate	Check the row count before and after; use a **relationship** instead of a join when the grain differs
Revised data	Your numbers do not match last year's paper	That is a finding, not an error: log the release you used
Aggregates in country lists	"World", "High income" and "Euro area" counted as countries	Filter on a country list, not on "everything that is not blank"
Percent of total computed the wrong way	Shares that sum to 100% across years instead of within each year	*Edit Table Calculation* → *Compute using* the country dimension, restart every year

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C.5 5. The provenance log

One CSV per practice round, in the session's `data/` folder. Template: `templates/provenance-log.csv`.

Column	What goes in it
`file`	The file name exactly as saved in `raw/`
`source`	The producer: author, agency, organisation
`url`	The landing page you downloaded from — not a mirror
`version_or_release`	Release number, "last updated" date, or edition
`accessed`	Today's date, ISO format: `2026-09-02`

Column	What goes in it
`licence`	Free / registration / request / restricted redistribution
`notes`	Anything odd — and **at least one place where the data disagreed with the paper**

C.6 6. Handing in

- *File → Export Packaged Workbook* (`.twbx`) — it contains the data extract, so the workbook opens on any machine. Do not upload a `.twb`: it does not contain the data.
- For data whose licence forbids redistribution (several of them — see `DATA-SOURCES.md`), upload to **ZoneCours only**, never to Tableau Public.

Appendix D

Readings

The **required reading** of each session is its big work: the article or book whose central figure the practice round rebuilds. It is read and annotated **before** class. **Further readings** are for the Risk Brief, the team memo and the take-home; none is required.

Article–data map: `REPLICATIONS.md`. Datasets: `DATA-SOURCES.md`.

D.1 Core references of the course

These are the principal references listed in the course's official description. Each is the required or further reading of at least one session (in brackets).

- Abbott, K. W., & Snidal, D. (2009). The governance triangle: Regulatory standards institutions and the shadow of the state. In W. Mattli & N. Woods (Eds.), *The politics of global regulation* (pp. 44–88). Princeton University Press. https://doi.org/10.1515/9781400830732.44 *[06]*
- Bjola, C., & Manor, I. (2018). Revisiting Putnam's two-level game theory in the digital age: Domestic digital diplomacy and the Iran nuclear deal. *Cambridge Review of International Affairs*, 31(1), 3–32. https://doi.org/10.1080/09557571.2018.1476836 *[09]*
- Brunsson, N., Rasche, A., & Seidl, D. (2012). The dynamics of standardization: Three perspectives on standards in organization studies. *Organization Studies*, 33(5–6), 613–632. https://doi.org/10.1177/0170840612450120 *[06]*
- Green, J. F. (2013). *Rethinking private authority: Agents and entrepreneurs in global environmental governance*. Princeton University Press. *[06]*
- Kuziemko, I., & Werker, E. (2006). How much is a seat on the Security Council worth? Foreign aid and bribery at the United Nations. *Journal of Political Economy*, 114(5), 905–930. https://doi.org/10.1086/507155 *[05]*
- Le Billon, P., & Shykora, L. (2020). Conflicts, commodities and the environmental geopolitics of supply chains. In S. O'Lear (Ed.), *A research agenda for environmental geopolitics* (pp. 59–73). Edward Elgar. https://doi.org/10.4337/9781788971249.00012 *[04]*
- Mansfield, E. D. (2021). International trade and conflict. In J. C. W. Pevehouse & L. Seabrooke (Eds.), *The Oxford handbook of international political economy* (pp. 660–671). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198793 *[04]*
- Mattli, W., & Büthe, T. (2003). Setting international standards: Technological rationality or primacy of power? *World Politics*, 56(1), 1–42. https://doi.org/10.1353/wp.2004.0006 *[06]*

D.2 Session 01 — What is geopolitical risk?

Required. Caldara, D., & Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225. https://doi.org/10.1257/aer.20191823

Further. - Kobrin, S. J. (1979). Political risk: A review and reconsideration. *Journal of International Business Studies*, 10(1), 67–80. https://doi.org/10.1057/palgrave.jibs.8490631 - Knight, F. H. (1921). *Risk, uncertainty and profit*. Houghton Mifflin. - Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica*, 77(3), 623–685. https://doi.org/10.3982/ECTA6248 - Bremmer, I., & Keat, P. (2009). *The fat tail: The power of political knowledge for strategic investing*. Oxford University Press.

D.3 Session 02 — The international economic order

Required. Baldwin, R. (2016). *The great convergence: Information technology and the new globalization*. Belknap Press of Harvard University Press. https://doi.org/10.4159/9780674972667

Further. - Ikenberry, G. J. (2011). *Liberal leviathan: The origins, crisis, and transformation of the American world order*. Princeton University Press. - Rodrik, D. (2011). *The globalization paradox: Democracy and the future of the world economy*. W. W. Norton. - Keohane, R. O. (1984). *After hegemony: Cooperation and discord in the world political economy*. Princeton University Press.

D.4 Session 03 — Anarchy and the balance of power

Required. Mearsheimer, J. J. (2001). *The tragedy of great power politics*. W. W. Norton — chapters 1–3. And: Allison, G. (2017). *Destined for war: Can America and China escape Thucydides's trap?* Houghton Mifflin Harcourt — introduction and chapter 1.

Further. - Waltz, K. N. (1979). *Theory of international politics*. Addison-Wesley. - Organski, A. F. K. (1958). *World politics*. Knopf. - Singer, J. D., Bremer, S., & Stuckey, J. (1972). Capability distribution, uncertainty, and major power war, 1820–1965. In B. Russett (Ed.), *Peace, war, and numbers* (pp. 19–48). Sage.

D.5 Session 04 — Trade, conflict and geoeconomic fragmentation

Required. Gopinath, G., Gourinchas, P.-O., Presbitero, A. F., & Topalova, P. (2025). Changing global linkages: A new Cold War? *Journal of International Economics*, 153, 104042. https://doi.org/10.1016/j.jinteco.2024.104042 (IMF Working Paper WP/24/76, free). And: Mansfield (2021), above.

Further. - Bailey, M. A., Strezhnev, A., & Voeten, E. (2017). Estimating dynamic state preferences from United Nations voting data. *Journal of Conflict Resolution*, 61(2), 430–456. https://doi.org/10.1177/0022002715595700 - Martin, P., Mayer, T., & Thoenig, M. (2008). Make trade not war? *Review of Economic Studies*, 75(3), 865–900. https://doi.org/10.1111/j.1467-937X.2008.00492.x - Barbieri, K. (2002). *The liberal illusion: Does trade promote peace?* University of Michigan Press. - Le Billon & Shykora (2020), above — how supply chains carry conflict, through commodities.

D.6 Session 05 — International institutions

Required. Kuziemko & Werker (2006), above.

Further. - Vreeland, J. R., & Dreher, A. (2014). *The political economy of the United Nations Security Council: Money and influence*. Cambridge University Press. - Keohane (1984), above (Session 02). - Abbott & Snidal (2009), above — as the bridge to Session 06.

D.7 Session 06 — Private authority, standards and global regulation

Required. Prakash, A., & Potoski, M. (2006). Racing to the bottom? Trade, environmental governance, and ISO 14001. *American Journal of Political Science*, 50(2), 350–364. https://doi.org/10.1111/j.1540-5907.2006.00188.x. And: Abbott & Snidal (2009), above.

Further. - Mattli & Büthe (2003), above. - Brunsson, Rasche & Seidl (2012), above. - Green (2013), above. - Büthe, T., & Mattli, W. (2011). *The new global rulers: The privatization of regulation in the world economy*. Princeton University Press.

D.8 Session 07 — Measuring risk: content analysis, coding, big data and composite indices

Required. Hassan, T. A., Hollander, S., van Lent, L., & Tahoun, A. (2019). Firm-level political risk: Measurement and effects. *Quarterly Journal of Economics*, 134(4), 2135–2202. https://doi.org/10.1093/qje/qjz021

Further. - Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636. https://doi.org/10.1093/qje/qjw024 - OECD & JRC (2008). *Handbook on constructing composite indicators: Methodology and user guide*. OECD Publishing. https://doi.org/10.1787/9789264043466-en - Krippendorff, K. (2018). *Content analysis: An introduction to its methodology* (4th ed.). SAGE. - Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3), 267–297. https://doi.org/10.1093/pan/mps028

D.9 Session 08 — Cyber geopolitics and digital threats

Required. Valeriano, B., & Maness, R. C. (2015). *Cyber war versus cyber realities: Cyber conflict in the international system*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780190204792.001.0001 — chapters 1, 4 and 5.

Further. - Farrell, H., & Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1), 42–79. https://doi.org/10.1162/isec_a_00351 - Kello, L. (2017). *The virtual weapon and international order*. Yale University Press. - Rid, T. (2013). *Cyber war will not take place*. Oxford University Press.

D.10 Session 09 — The redefinition of economic diplomacy

Required. Rose, A. K. (2007). The foreign service and foreign trade: Embassies as export promotion. *The World Economy*, 30(1), 22–38. https://doi.org/10.1111/j.1467-9701.2007.00870.x. And: Bjola & Manor (2018), above.

Further. - Putnam, R. D. (1988). Diplomacy and domestic politics: The logic of two-level games. *International Organization*, 42(3), 427–460. https://doi.org/10.1017/S0020818300027697 - Blackwill, R. D., & Harris, J. M. (2016). *War by other means: Geoeconomics and statecraft*. Belknap Press of Harvard University Press.

D.11 Session 10 — Economic sanctions

Required. Felbermayr, G., Kirilakha, A., Syropoulos, C., Yalcin, E., & Yotov, Y. V. (2020). The Global Sanctions Data Base. *European Economic Review*, 129, 103561. https://doi.org/10.1016/j.euroecorev.2020.103561. And: Hufbauer, G. C., Schott, J. J., Elliott, K. A., & Oegg, B. (2007). *Economic sanctions reconsidered* (3rd ed.). Peterson Institute for International Economics — chapters 1–3.

Further. - Pape, R. A. (1997). Why economic sanctions do not work. *International Security*, 22(2), 90–136. https://doi.org/10.1162/isec.22.2.90 - Morgan, T. C., Bapat, N., & Kobayashi, Y. (2014). Threat and imposition of

economic sanctions 1945–2005: Updating the TIES dataset. *Conflict Management and Peace Science*, 31(5), 541–558. https://doi.org/10.1177/0738894213520379 - Drezner, D. W. (1999). *The sanctions paradox: Economic statecraft and international relations*. Cambridge University Press.

D.12 Session 11 — Space geopolitics: the orbital commons and the new space race

Required. ESA Space Debris Office. (2026). *ESA's annual space environment report* (Issue 10.0, 1 May 2026; GEN-DB-LOG-00288-OPS-SD). European Space Agency. https://www.sdo.esoc.esa.int/publications/Space_Environment_Report_I10R0_20260501 — executive summary, §1.1–1.2, §2.1, Table 3.1. And: Samson, V., & Brett, K. (Eds.). (2026). *Global counterspace capabilities: An open source assessment*. Secure World Foundation — executive summary. And: Pekkanen, S. M. (2019). Governing the new space race. *AJIL Unbound*, 113, 92–97. https://doi.org/10.1017/aju.2019.16

Further. - Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies (1967), 610 U.N.T.S. 205. https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/outerspacetreaty.ht - Dolman, E. C. (2002). *Astropolitik: Classical geopolitics in the space age*. Frank Cass. https://doi.org/10.4324/9780203016640 - Bowen, B. E. (2020). *War in space: Strategy, spacepower, geopolitics*. Edinburgh University Press. https://doi.org/10.3366/edinburgh/978 - Kessler, D. J., & Cour-Palais, B. G. (1978). Collision frequency of artificial satellites: The creation of a debris belt. *Journal of Geophysical Research*, 83(A6), 2637–2646. https://doi.org/10.1029/JA083iA06p02637 - Farrell & Newman (2019), above (Session 08).

D.13 Session 12 — Team presentations, and the closing address

Closing readings. - Kobrin (1979), above (Session 01). - Henisz, W. J. (2000). The institutional environment for economic growth. *Economics & Politics*, 12(1), 1–31. https://doi.org/10.1111/1468-0343.00066

Further, for the team memo and the take-home. - Jensen, N. M. (2003). Democratic governance and multinational corporations: Political regimes and inflows of foreign direct investment. *International Organization*, 57(3), 587–616. https://doi.org/10.1017/S0020818303573040 - Kaufmann, D., Kraay, A., & Mastruzzi, M. (2011). The Worldwide Governance Indicators: Methodology and analytical issues. *Hague Journal on the Rule of Law*, 3(2), 220–246.

Appendix E

Assessment

Component	Weight	Individual?	Form	When	Brief
Participation	10%	yes	active engagement, continuous	every session	below
Quizzes	15%	yes	3 × 5%, 20 min, in class, on paper, closed book	Sessions 04, 07, 10	`quizzes/`
Risk Brief	15%	yes	4-minute presentation + one question (a 5-minute slot)	one of Sessions 02–11	`risk-brief/`
Team work	20%	no	risk dashboard for one multinational + memo + presentation	presented Session 12 · memo one week later	`team-project/`
Take-home examination	40%	yes	3 hours of work within a 48-hour window	final examination period	`take-home/`

80% individual · 20% team. In HEC Montréal's terms: A07 Participation (10%), A03 Tests and quizzes (15%), A06 Individual presentation (15%), A09 Team work (20%), A11 Take-home examination (40%).

E.1 How each component maps to the learning outcomes

Outcome	Participation	Quizzes	Risk Brief	Team work	Take-home
Knowledge — define GPR analysis, identify indicators	·	●●	·	·	●
Comprehension — explain the international context of MNEs	●	●●	●	·	●

Outcome	Participation	Quizzes	Risk Brief	Team work	Take-home
Application — economic diplomacy and new risks	●	●	●●	●	●
Analysis — interactions between risks, implications for MNEs	●	·	●	●●	●●
Synthesis — integrate content analysis, coding, big data, statistics	·	·	●	●●	●●
Evaluation — design risk assessment strategies for MNEs	·	·	●	●●	●●

●● principal · ● contributes · · marginal

E.2 Participation — 10%

Marked against a published, absolute bar — never ranked:

Mark	What it looks like across the term
9–10	Regularly moves the discussion forward with reference to the reading; asks real questions after Risk Briefs; fully engaged in every practice round
7–8	Contributes often; engaged in practice rounds; occasional questions
5–6	Present and engaged when asked; rarely volunteers
< 5	Disengaged, or absent from practice rounds without reason

Practice rounds are **formative**: commented on, not marked. What is recorded is that you engaged and handed in the `.twbx` and provenance log.

E.3 Late work and missed assessments

- **Quiz** missed with a documented reason: its weight moves to the take-home. Without: zero.
- **Risk Brief** missed with a documented reason: rescheduled. Without: zero.
- **Team memo**: −5% of its mark per day late.
- **Take-home**: submissions after the window closes are not accepted, per HEC regulations.

E.4 Quizzes

Twenty minutes, at the start of Sessions **04, 07 and 10**, on paper, closed book, individual.

Quiz	Session	Covers
Quiz 1	04	Sessions 01–03: what geopolitical risk is; the international economic order; anarchy and the balance of power
Quiz 2	07	Sessions 04–06: trade, conflict and fragmentation; international institutions; private authority and standards
Quiz 3	10	Sessions 07–09: measuring risk; cyber geopolitics; economic diplomacy

E.4.1 The format — 10 points each

Part	Points	What it asks
A · Concepts	4	Four short definitions or distinctions, one or two sentences each
B · The big work	3	The research question, the data and the central finding of one of the session readings — and one limitation
C · Read the chart	3	An unseen Tableau chart: say what it shows, what decision it was built with, and what it does not license a firm to conclude

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E.4.2 Specimen — Quiz 1

A · Concepts (4 points, 1 each)

1. Distinguish *risk* from *uncertainty* in Knight's sense, and give one geopolitical example of each.
2. What is the difference between Kobrin's *macro* and *micro* political risk?
3. In the GPR index, why are counts of geopolitical articles divided by the total number of articles?
4. What does "anarchy" mean in the realist tradition — and what does it *not* mean?

B · The big work (3 points)

Caldara and Iacoviello (2022) distinguish *GPR Threats* from *GPR Acts*. (a) Which categories enter each? (b) Why does the distinction matter for a firm's investment decision? (c) Name one limitation of the index for a firm whose operations are mainly in sub-Saharan Africa.

C · Read the chart (3 points)

[A stacked area chart: shares of world GDP, current US dollars, 1970–2023, for the G7, China, and the rest of the world.]

(a) State the finding in one sentence. (b) The chart uses current US dollars. How would it change at purchasing-power parity, and why? (c) What does the chart *not* tell a Canadian manufacturer deciding where to build its next plant?

E.4.3 Specimen answers (Part A)

1. *Risk*: probabilities estimable from experience (piracy losses on a shipping route). *Uncertainty*: probabilities not estimable (whether a given strait will be blockaded within five years).
2. *Macro* risk affects all foreign firms in a country (a general expropriation, currency inconvertibility); *micro* risk affects one sector or firm (a mining-code change, a contract cancellation).
3. Because newspapers' total output changes over time; the share removes the effect of size, so the index measures *relative* attention to geopolitical tension.
4. The absence of a government above states — no authority to enforce agreements. It does not mean chaos or constant war.

E.5 The Risk Brief

> **Four minutes. One development. One view. One firm. One recommendation.**

Each student delivers one Risk Brief, at the start of one of Sessions 02–11. Sign-up opens at the end of Session 01; four briefs per session.

E.5.1 The brief

You are the geopolitical risk analyst of a named multinational. Something happened in the last **four weeks**. Brief your chief risk officer.

Minute	What you say
0:00–0:30	**The development.** What happened, where, who decided it. One sentence of context, no history lecture.
0:30–2:00	**The view.** One Tableau chart, built from data **you collected at source**, that puts the development in perspective — is it large, new, accelerating, concentrated?
2:00–3:00	**The exposure.** Through which channel does it reach *your* firm — market access, inputs, finance, assets and people, data and operations? How much of the firm sits in the path?
3:00–4:00	**The recommendation.** One decision, and the indicator and threshold that would make you change it.

One question follows (one minute), from the room or the instructor. Four briefs fill the first twenty minutes of the session.

The firm. Any real multinational with public information on its footprint (annual report, 10-K, sustainability report). It may — but need not — be your team's firm.

The data. Any source in `DATA-SOURCES.md` or any other public source, cited with version and access date. Ideally the concept or dataset of a session you have already had.

E.5.2 What you hand in

Before class on the day: the `.twbx` (or a Tableau Public link) and your provenance log, on ZoneCours.

E.5.3 Rubric (15 points)

Criterion	Points	Full marks look like
The development	2	Precisely dated, sourced, and genuinely geopolitical (between states or political actors)
The view	4	Data collected at source; correct aggregation; title states the finding; source caption; the chart *does* work in the argument
The exposure	4	A specific channel; evidence from the firm's own disclosures; an order of magnitude
The recommendation	3	One decision, an indicator and a threshold that would reverse it
Time and question	2	Four minutes, not six; answers the question asked

Automatic deductions: no source caption (−2); over five minutes (stopped at five); a development older than four weeks (−2).

E.6 Team work

> **Design the geopolitical risk assessment a real multinational should be running — build it, and defend the method behind it to its board.**

Teams of three or four, formed at the start of Session 02. Each team adopts **one named multinational** for the whole term — no two teams the same firm.

E.6.1 What you deliver

Deliverable	Points	When
The dashboard — a Tableau Public workbook	part of the 15 below	one week after Session 12
The memo to the board — 2,500 words, plus appendices	15 (dashboard + memo together)	one week after Session 12
The presentation and defence — 12 minutes + 8 minutes of questions	5	Session 12

E.6.1.1 The dashboard

At least **four datasets** collected in the practice rounds, from at least **three of the six families of indicators** (Session 01): event-based, text-based, expert-based, structural, market-based, firm-level. It must contain:

1. **The firm's footprint** — where its revenue, assets, suppliers and people are (from its own disclosures), as a map or a table.

2. **A composite exposure index** for the firm's main countries, built with the method of Session 07: indicators chosen and justified, normalised, weighted by **parameters** the board can move.
3. **At least two channel views** — e.g. sanctions exposure (S10), dependence on satellite services (S11), bloc alignment of key markets (S04), cyber-incident exposure of its sector (S08).
4. **A watchlist** — three indicators, each with a threshold that triggers a named decision.

E.6.1.2 The memo

Section	Words	Content
Executive summary	250	The three risks that matter, and what to do
The firm and its footprint	300	Where it is exposed, and through which channels
Method	700	Which indicators, why these, how combined, what each cannot see — the Session 07 discipline
Findings	800	What the dashboard shows, risk by risk, interaction by interaction
Recommendations and watchlist	450	Decisions, triggers, owners
Appendices	—	Provenance log for every dataset; data decisions; contribution statement

E.6.2 Rubric (20 points)

Criterion	Points	Full marks look like
Conceptual grounding	3	The risks are framed with the course's concepts — order, anarchy, institutions, diplomacy, sanctions, interdependence — not newspaper vocabulary
Data and provenance	3	Four or more datasets collected at source, every one logged; decisions named
Method of the composite index	4	Indicators justified; normalisation and weighting explained; sensitivity to weights shown
Analysis of interactions	3	Shows how risks compound or offset — e.g. sanctions *and* input concentration *and* bloc alignment
Recommendations	2	Decisions with triggers and owners, not "monitor the situation"
Presentation and defence	5	Twelve minutes, the dashboard driven live, questions answered by any member

Individual accountability. Each memo ends with a contribution statement signed by all members. At the end of term each student submits a confidential peer rating (four dimensions, 1–5). The team mark is multiplied by an individual factor (default 1.00, range 0.80–1.05) only when the peer ratings and the instructor's observation of the practice rounds independently agree.

E.6.3 Milestones

Session	Milestone
02	Team formed, firm chosen and registered on ZoneCours
04	One-paragraph footprint: the firm's five most important countries, with sources
07	Draft composite index — indicators and weights — reviewed in the practice round
10	Dashboard skeleton reviewed in the last twenty minutes of the practice round
12	Presentation and defence
12 + 1 week	Memo and published dashboard

E.7 Take-home examination

Three hours of work, within a 48-hour window, during the final examination period. Individual. Open book, open notes, open internet. No collaboration of any kind.

E.7.1 The format

At the opening of the window you receive, on ZoneCours:

- a **scenario** — a geopolitical development, real or plausible, dated;
- a **firm** — a real multinational, with an extract from its disclosures;
- a **data instruction** — one or two sources from `DATA-SOURCES.md` to collect from, with the variables to use.

You hand in, before the window closes:

Part	Points	What
A · Concepts in context	12	Three short essays (≈ 300 words each) applying the course's concepts to the scenario: e.g. what realism and liberal institutionalism each predict; whether sanctions in this case are likely to work; how the firm's standards exposure changes
B · The evidence	14	One packaged Tableau workbook (`.twbx`) built from data you collected, with its provenance log, and a 400-word reading of it
C · The assessment	14	A 900-word memo to the firm's chief risk officer: the exposure, the interactions between risks, the decision, the watchlist with thresholds

Length limits are strict. Words beyond the limit are not read.

E.7.2 Marking criteria

Criterion	Weight
Correct and precise use of the course's concepts	30%
Data collected at source, correctly prepared, honestly logged	25%
Analysis of interactions between risks and their implications for the firm	25%
A decision with triggers, defensible from the evidence	20%

E.7.3 On language models and integrity

You may use a language model to check syntax or to help you read a codebook. You may not use it to produce any text you submit. Every submission includes a one-paragraph statement of the tools you used and how. Undeclared use, or collaboration with another person, is a violation of HEC Montréal's academic integrity regulations.

E.7.4 Specimen scenario

Dated 15 March. The European Union and the United States announce coordinated export controls on advanced battery-manufacturing equipment to a major Asian economy; that economy responds within a week with an export licensing regime on refined graphite.

Firm: a European automotive manufacturer with an EV assembly plant in Central Europe and a joint venture in the affected Asian economy.

Data instruction: IMF International Trade in Goods by partner country (`XG_FOB_USD`); the Global Sanctions Data Base; UN General Assembly ideal points (Bailey, Strezhnev & Voeten); the firm's last annual report.

Part A (example): Using Farrell and Newman's concept of weaponised interdependence, explain which side holds the stronger *chokepoint* and why. *Part B:* Build a view of goods trade between the EU, the United States and the affected economy since 2000, set against the sanctions cases in force between them and the distance between their UN voting positions. *Part C:* Advise the chief risk officer on the joint venture and on sourcing for the European plant.